

Fluctuation-Induced BCC Selection in $\mathbb{Z}_2$ -Symmetric Isotropic Fluids: An Exact Combinatorial Approach to Structural Selection

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I. UNIVERSALITY A: ROBUSTNESS UNDER FINITE SHELL THICKNESS

A. Setup: isotropic broad-shell Landau–Brazovskii sector

Let the quadratic kernel be isotropic and peaked on a shell of radius k_0 :

$$G^{-1}(q) = r + \kappa (q^2 - k_0^2)^2, \quad q := |\mathbf{q}|. \quad (1)$$

We allow a finite shell thickness by restricting the fluctuation phase space to

$$\mathcal{S}_{\Delta k} := \{\mathbf{q} \in \mathbb{R}^3 : |q - k_0| \leq \Delta k\}, \quad 0 < \Delta k \ll k_0, \quad (2)$$

(or equivalently by an isotropic regulator that suppresses $|q - k_0| \gg \Delta k$).

The interaction is the usual quartic scalar vertex

$$\mathcal{L}_{\text{int}} = \frac{u}{4!} \psi^4, \quad V_4 = -u. \quad (3)$$

For a candidate structure S with N_S plane waves on the k_0 -shell, define the purely geometric resonance-loop ratio

$$\mathcal{C}_S := \frac{N_{\text{loop},S}}{N_S^2}, \quad (4)$$

where $N_{\text{loop},S}$ counts the non-trivial quartic resonance loops (i.e. quartets with $\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3 + \mathbf{k}_4 = \mathbf{0}$ that are not reducible to the trivial $\pm$ pairings).

B. Lemma: bubble integral remains structure-independent for finite Δk

Define the broad-shell bubble integral

$$I_{\text{bubble}}(r; \Delta k) := \int_{\mathcal{S}_{\Delta k}} \frac{d^3 q}{(2\pi)^3} \frac{1}{[r + \kappa(q^2 - k_0^2)^2]^2}. \quad (5)$$

[Structure-independence of I_{bubble}] Assume: (i) the propagator $G(q)$ depends only on $q = |\mathbf{q}|$ (isotropy), (ii) the shell restriction/regulator is isotropic, (iii) the vertex is local and momentum-independent at the quartic level. Then $I_{\text{bubble}}(r; \Delta k)$ is independent of the lattice candidate S .

Under (i)-(ii), the integration domain and integrand depend only on q , hence $I_{\text{bubble}}(r; \Delta k)$ is a function only of $(r, \Delta k; k_0, \kappa)$. No information about the discrete reciprocal-star geometry (S) enters this radial integral. Condition (iii) guarantees the one-loop quartic correction factorizes into this common integral times a purely combinatorial weight.

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C. Proposition: BCC ranking is invariant under finite Δk

At one loop, the quartic vertex renormalization at zero external momenta is

$$\delta\Gamma^{(4)} = \frac{u^2}{2} I_{\text{bubble}}(r; \Delta k), \quad (6)$$

so the effective quartic coefficient for structure S takes the factorized form

$$\beta_{\text{eff},S}(r; \Delta k) = \beta_{\text{MF},S} - \frac{u^2}{2} I_{\text{bubble}}(r; \Delta k) \mathcal{C}_S. \quad (7)$$

[Finite-thickness robustness of structural ranking] Assume the hypotheses of the Lemma and that the mean-field quartic coefficients $\beta_{\text{MF},S}$ are fixed. Then for any r and any Δk , the structural ordering induced by the fluctuation term is governed solely by $\mathcal{C}_S$. In particular, if

$$\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}, \quad (8)$$

then for sufficiently deep in the fluctuation-dominated regime (where the second term in (7) dominates the differences),

$$\beta_{\text{eff,BCC}}(r; \Delta k) < \beta_{\text{eff,FCC}}(r; \Delta k) < \beta_{\text{eff,SC}}(r; \Delta k), \quad (9)$$

and the BCC-favoring inequality is independent of Δk .

By (7), for any pair (S_1, S_2) ,

$$\beta_{\text{eff},S_1} - \beta_{\text{eff},S_2} = (\beta_{\text{MF},S_1} - \beta_{\text{MF},S_2}) - \frac{u^2}{2} I_{\text{bubble}}(r; \Delta k) (\mathcal{C}_{S_1} - \mathcal{C}_{S_2}). \quad (10)$$

The bubble factor $I_{\text{bubble}}(r; \Delta k)$ is common to all structures. Hence the fluctuation-driven ordering is determined solely by the sign of $(\mathcal{C}_{S_1} - \mathcal{C}_{S_2})$. In the regime where the fluctuation term dominates the inter-structure differences, the ranking matches that of $\mathcal{C}_S$ and is independent of Δk .

D. Corollary: selection threshold shifts but hierarchy does not

Define the selection threshold between BCC and FCC by

$$\beta_{\text{eff,BCC}}(r_{\text{sel}}; \Delta k) = \beta_{\text{eff,FCC}}(r_{\text{sel}}; \Delta k). \quad (11)$$

Then

$$\frac{u^2}{2} I_{\text{bubble}}(r_{\text{sel}}; \Delta k) = \frac{\beta_{\text{MF,BCC}} - \beta_{\text{MF,FCC}}}{\mathcal{C}_{\text{BCC}} - \mathcal{C}_{\text{FCC}}}. \quad (12)$$

Thus Δk shifts the numerical location of r_{sel} through $I_{\text{bubble}}(r; \Delta k)$, but cannot invert the BCC>FCC hierarchy as long as $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}}$ and the isotropic single-mode assumptions hold.

II. UNIVERSALITY B: ROBUSTNESS UNDER MULTI-COMPONENT ORDER PARAMETERS

A. Setup: n -component field with isotropic quadratic shell

Let $\Psi^a(x)$, $a = 1, \dots, n$, be a real multiplet with isotropic quadratic kernel

$$\Gamma_{0,ab}^{(2)}(q) = \delta_{ab} [r + \kappa(q^2 - k_0^2)^2], \quad G_{ab}(q) = \delta_{ab} G(q), \quad G(q) = \frac{1}{r + \kappa(q^2 - k_0^2)^2}. \quad (13)$$

Allow finite shell thickness by the same isotropic restriction/regulator as in Universality A, so that

$$I_{\text{bubble}}(r; \Delta k) := \int_{S_{\Delta k}} \frac{d^3 q}{(2\pi)^3} \frac{1}{[r + \kappa(q^2 - k_0^2)^2]^2} \quad (14)$$

remains structure-independent.

The most general quartic local interaction is

$$\mathcal{L}_{\text{int}} = \frac{1}{4!} u_{abcd} \Psi^a \Psi^b \Psi^c \Psi^d, \quad u_{abcd} = u_{(abcd)}. \quad (15)$$

B. Invariant-coupling class

We impose an internal symmetry that fixes u_{abcd} to the $O(n)$ -invariant form

$$u_{abcd} = u (\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}). \quad (16)$$

(Equivalently: the interaction is $\frac{u}{4!}(\Psi^a\Psi^a)^2$.)

C. Lemma: one-loop quartic renormalization factorizes into internal trace $\times$ shell integral

Consider the one-loop bubble correction to the amputated four-point function at vanishing external momenta. Using (16) and $G_{ab}(q) = \delta_{ab}G(q)$,

$$\delta\Gamma_{abcd}^{(4)} = \frac{1}{2} \int_{\mathcal{S}_{\Delta k}} \frac{d^3q}{(2\pi)^3} \Gamma_{abef}^{(4)} G_{eg}(q) G_{fh}(q) \Gamma_{ghcd}^{(4)}. \quad (17)$$

Since $G_{eg}G_{fh} = \delta_{eg}\delta_{fh}G(q)^2$, (17) becomes

$$\delta\Gamma_{abcd}^{(4)} = \frac{1}{2} I_{\text{bubble}}(r; \Delta k) \Gamma_{abef}^{(4)} \Gamma_{efcd}^{(4)}. \quad (18)$$

For the $O(n)$ -invariant vertex (16), the tensor contraction closes on the same invariant:

$$\Gamma_{abef}^{(4)} \Gamma_{efcd}^{(4)} = \lambda_n u^2 (\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}), \quad (19)$$

with $\lambda_n > 0$ depending only on n .

[$O(n)$ closure] Under (16), the 1-loop correction preserves the $O(n)$ -invariant tensor structure, and only renormalizes the scalar coupling $u \mapsto u_{\text{eff}}$:

$$u_{\text{eff}}(r; \Delta k) = u - \frac{\lambda_n}{2} u^2 I_{\text{bubble}}(r; \Delta k). \quad (20)$$

D. Proposition: structural ranking is unchanged for $O(n)$ multiplets

For a candidate structure S with geometric ratio $\mathcal{C}_S = N_{\text{loop},S}/N_S^2$, the effective quartic coefficient entering the equal-amplitude Landau functional inherits the same factorization as in the scalar case:

$$\beta_{\text{eff},S}(r; \Delta k) = \beta_{\text{MF},S} - \frac{\lambda_n}{2} u^2 I_{\text{bubble}}(r; \Delta k) \mathcal{C}_S. \quad (21)$$

[$O(n)$ universality of BCC ranking] Assume: (i) isotropic single-shell quadratic kernel, (ii) isotropic shell regulator (finite thickness allowed), (iii) local $O(n)$ -invariant quartic vertex (16), (iv) equal-amplitude star ansatz for each cubic structure. Then the fluctuation-driven structural ordering is governed solely by $\mathcal{C}_S$, independent of $(n, \Delta k)$. In particular, if $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$, then in the fluctuation-dominated regime the hierarchy

$$\beta_{\text{eff},\text{BCC}} < \beta_{\text{eff},\text{FCC}} < \beta_{\text{eff},\text{SC}} \quad (22)$$

holds exactly as in the scalar theory.

Equation (21) shows that the structure-dependent part is only the geometric ratio $\mathcal{C}_S$, while all dependence on internal multiplicity enters through the common positive prefactor $\frac{\lambda_n}{2} u^2 I_{\text{bubble}}$. Hence the sign of differences $\beta_{\text{eff},S_1} - \beta_{\text{eff},S_2}$ in the fluctuation term is determined solely by $\mathcal{C}_{S_1} - \mathcal{C}_{S_2}$, and the ranking is unchanged.

E. Corollary: selection threshold with $O(n)$ multiplets

Define r_{sel} by $\beta_{\text{eff},\text{BCC}}(r_{\text{sel}}; \Delta k) = \beta_{\text{eff},\text{FCC}}(r_{\text{sel}}; \Delta k)$. Then

$$\frac{\lambda_n}{2} u^2 I_{\text{bubble}}(r_{\text{sel}}; \Delta k) = \frac{\beta_{\text{MF},\text{BCC}} - \beta_{\text{MF},\text{FCC}}}{\mathcal{C}_{\text{BCC}} - \mathcal{C}_{\text{FCC}}}. \quad (23)$$

Thus n (via λ_n) shifts the numerical value of r_{sel} but cannot invert the hierarchy as long as the hypotheses of the Proposition hold.

III. UNIVERSALITY C: HIGHER-LOOP STABILITY OF THE BCC RANKING

A. Class of theories and notion of “ranking”

We work within the same model class as Universality A–B: (i) isotropic single-shell quadratic kernel with possible finite thickness Δk , (ii) local momentum-independent quartic vertex (scalar or $O(n)$ -invariant multiplet), (iii) equal-amplitude star ansatz for each candidate cubic structure.

For each structure S , write the fluctuation-corrected quartic coefficient as an asymptotic loop expansion

$$\beta_{\text{eff},S}(r; \Delta k) = \beta_{\text{MF},S} + \sum_{\ell \geq 1} \delta\beta_S^{(\ell)}(r; \Delta k), \quad (24)$$

and define the *ranking* at fixed $(r, \Delta k)$ by comparing $\beta_{\text{eff},S}$ across S .

B. Lemma: factorization of shell divergences vs. structural combinatorics

Let $G(q)$ denote the isotropic near-shell propagator and let ℓ be the loop order. For any ℓ -loop contribution to the renormalized quartic vertex evaluated at zero external momenta, define the corresponding shell integral magnitude

$$J_\ell(r; \Delta k) := \int_{\mathcal{S}_{\Delta k}} \prod_{i=1}^{m_\ell} \frac{d^3 q_i}{(2\pi)^3} \mathcal{K}_\ell(\{q_i\}) \prod_{j=1}^{p_\ell} G(q_{a_j}), \quad (25)$$

where $\mathcal{K}_\ell$ is a structure-independent isotropic kernel generated by contractions and momentum routings, and p_ℓ is the number of internal propagator factors.

[Factorization within the isotropic single-shell truncation] Within assumptions (i)–(ii), every ℓ -loop correction to the quartic coefficient for structure S admits a decomposition

$$\delta\beta_S^{(\ell)}(r; \Delta k) = \mathcal{P}_\ell(\mathcal{C}_S) \Xi_\ell(r; \Delta k), \quad (26)$$

where:

- $\Xi_\ell(r; \Delta k)$ is a structure-independent (possibly divergent) shell integral depending only on $(r, \Delta k; k_0, \kappa)$ and the internal multiplicity (e.g. n) through overall traces;
- $\mathcal{P}_\ell$ is a polynomial with *nonnegative* coefficients in the geometric ratio $\mathcal{C}_S := N_{\text{loop},S}/N_S^2$.

Under isotropy and locality, all momentum integrals depend only on magnitudes and relative angles through scalar products, and the regulator/domain $\mathcal{S}_{\Delta k}$ is common to all structures. Hence the loop integrals producing Ξ_ℓ are identical across S up to purely discrete sums over allowed star wavevectors $\{\mathbf{k}_i\}$ that satisfy the resonance constraints enforced at each vertex. Each quartic vertex contraction contributes a factor selecting a resonance quartet; counting such resonant quartets produces the weight $N_{\text{loop},S}$, while normalization by the number of mode choices yields N_S^2 , leading to the basic ratio $\mathcal{C}_S$. At higher loops, multiple quartic resonance constraints appear; the total number of admissible discrete assignments is a sum over products of such counts, which yields a polynomial $\mathcal{P}_\ell(\mathcal{C}_S)$ with nonnegative coefficients. The remaining continuous shell integrals are common and collected into $\Xi_\ell(r; \Delta k)$.

C. Proposition: hierarchy cannot invert without adding new structural weights

[Higher-loop non-inversion criterion] Assume: (i)–(iii) as above, and additionally that there exists a fluctuation-dominated regime in which

$$|\beta_{\text{MF},S_1} - \beta_{\text{MF},S_2}| \ll \left| \sum_{\ell \geq 1} \left(\delta\beta_{S_1}^{(\ell)} - \delta\beta_{S_2}^{(\ell)} \right) \right| \quad \text{for the structures compared.} \quad (27)$$

If, for all $\ell \geq 1$, the polynomials $\mathcal{P}_\ell$ in (26) are *monotone increasing* functions of $\mathcal{C}_S$ on the relevant range of $\mathcal{C}_S$, then the structural ranking induced by fluctuations is fixed by $\mathcal{C}_S$ and cannot be inverted by any finite resummation of higher loops. In particular, if $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$, then in the fluctuation-dominated regime

$$\beta_{\text{eff},\text{BCC}}(r; \Delta k) < \beta_{\text{eff},\text{FCC}}(r; \Delta k) < \beta_{\text{eff},\text{SC}}(r; \Delta k) \quad (28)$$

persists to all loop orders within the truncation class.

By Lemma (26),

$$\beta_{\text{eff},S} = \beta_{\text{MF},S} + \sum_{\ell \geq 1} \mathcal{P}_\ell(\mathcal{C}_S) \Xi_\ell. \quad (29)$$

For any pair (S_1, S_2) ,

$$\begin{aligned} \beta_{\text{eff},S_1} - \beta_{\text{eff},S_2} &= (\beta_{\text{MF},S_1} - \beta_{\text{MF},S_2}) \\ &+ \sum_{\ell \geq 1} [\mathcal{P}_\ell(\mathcal{C}_{S_1}) - \mathcal{P}_\ell(\mathcal{C}_{S_2})] \Xi_\ell. \end{aligned} \quad (30)$$

If each $\mathcal{P}_\ell$ is monotone increasing and $\mathcal{C}_{S_1} > \mathcal{C}_{S_2}$, then $\mathcal{P}_\ell(\mathcal{C}_{S_1}) - \mathcal{P}_\ell(\mathcal{C}_{S_2}) > 0$ for all ℓ . Thus the sign of the fluctuation contribution to the difference is fixed and cannot be reversed by adding higher-loop terms; higher loops can only shift magnitudes (and hence the numerical location of selection thresholds) through the common factors $\Xi_\ell(r; \Delta k)$. In the fluctuation-dominated regime, the mean-field difference is subleading and cannot flip the sign.

D. Remark: how inversion can occur (outside the present universality class)

The non-inversion statement relies crucially on the absence of new structure-specific weights beyond $\mathcal{C}_S$. Inversion can occur only if one introduces additional ingredients that generate genuinely new discrete geometric functionals, e.g.:

$$\begin{aligned} &\text{multi-shell couplings,} \\ &\text{anisotropic/nonlocal quartic kernels,} \\ &\text{explicit lattice pinning fields,} \\ &\text{strongly momentum-dependent vertices.} \end{aligned} \quad (31)$$

Such extensions lie outside the isotropic single-shell local truncation class considered here.

IV. UNIVERSALITY D: ROBUSTNESS UNDER WEAK MULTI-SHELL COUPLING

A. Two-shell quadratic sector and mode decomposition

Let the quadratic kernel admit two nearby soft shells at radii k_0 and k_1 , with $0 < |k_1 - k_0| \ll k_0$, and (possibly) finite thicknesses $\Delta k_0, \Delta k_1$. Decompose the order parameter into shell components

$$\psi(x) = \psi_0(x) + \psi_1(x), \quad (32)$$

where in momentum space

$$\text{supp } \psi_a(\mathbf{q}) \subset \mathcal{S}_{\Delta k_a}^{(a)} := \{\mathbf{q} : ||\mathbf{q}| - k_a| \leq \Delta k_a\}, \quad a \in \{0, 1\}. \quad (33)$$

Assume isotropic propagators on each shell:

$$G_a(q) = \frac{1}{r_a + \kappa_a(q^2 - k_a^2)^2}, \quad a \in \{0, 1\}. \quad (34)$$

B. Local quartic interaction and shell-resolved couplings

Take a local quartic interaction:

$$\mathcal{L}_{\text{int}} = \frac{u}{4!} \psi^4 = \frac{1}{4!} \sum_{a,b,c,d \in \{0,1\}} u_{abcd} \psi_a \psi_b \psi_c \psi_d, \quad (35)$$

with the *shell-resolved* couplings induced by locality satisfying

$$u_{abcd} = u \quad \text{for all } a, b, c, d, \quad (36)$$

up to possible small renormalizations from integrating out hard modes.

C. Shell-resolved equal-amplitude ansatz for a given structure

For each structure S (e.g. BCC/FCC/SC), assume the dominant pattern is on the primary shell k_0 and allow subleading admixture on k_1 :

$$\psi_0(x) = A \sum_{i=1}^{N_S} \cos(\mathbf{k}_i \cdot x), \quad |\mathbf{k}_i| = k_0, \quad (37)$$

$$\psi_1(x) = B \sum_{j=1}^{M_S} \cos(\mathbf{p}_j \cdot x), \quad |\mathbf{p}_j| = k_1, \quad (38)$$

with equal amplitudes within each shell orbit.

Define the primary-shell geometric ratio

$$\mathcal{C}_S^{(0)} := \frac{N_{\text{loop},S}^{(0)}}{N_S^2}, \quad (39)$$

counting nontrivial resonance quartets *entirely on shell* k_0 (i.e. $\mathbf{k}_{i_1} + \mathbf{k}_{i_2} + \mathbf{k}_{i_3} + \mathbf{k}_{i_4} = 0$). Similarly define the mixed-shell resonance weight

$$\mathcal{D}_S^{(01)} := \frac{N_{\text{loop},S}^{(01)}}{N_S^2}, \quad (40)$$

counting quartets involving two k_0 -modes and two k_1 -modes that satisfy $\mathbf{k}_{i_1} + \mathbf{k}_{i_2} + \mathbf{p}_{j_1} + \mathbf{p}_{j_2} = 0$.

D. One-loop effective quartic coefficients with two shells

At one loop, the zero-momentum quartic renormalization splits into shell channels. Introduce three bubble integrals (all structure-independent under isotropy):

$$I_{00}(r_0; \Delta k_0) := \int_{\mathcal{S}_{\Delta k_0}^{(0)}} \frac{d^3 q}{(2\pi)^3} G_0(q)^2, \quad (41)$$

$$I_{11}(r_1; \Delta k_1) := \int_{\mathcal{S}_{\Delta k_1}^{(1)}} \frac{d^3 q}{(2\pi)^3} G_1(q)^2, \quad (42)$$

$$I_{01}(r_0, r_1; \Delta k_0, \Delta k_1) := \int_{\mathcal{S}_{\Delta k_0}^{(0)}} \frac{d^3 q}{(2\pi)^3} G_0(q) G_1(q) \chi_{01}(q), \quad (43)$$

where χ_{01} is an isotropic overlap/regulator factor (possibly unity if both shells are handled with a joint isotropic regulator).

The effective quartic coefficient on the *primary* shell k_0 takes the factorized form

$$\beta_{\text{eff},S}^{(0)} = \beta_{\text{MF},S}^{(0)} - \frac{u^2}{2} \left[I_{00} \mathcal{C}_S^{(0)} + I_{01} \mathcal{D}_S^{(01)} \right], \quad (44)$$

and similarly for the secondary shell.

E. Lemma: structure-independence of the two-shell bubble integrals

[Structure-independence] Assume: (i) $G_a(q)$ depends only on $q = |\mathbf{q}|$ for each shell, (ii) the regulator/domains $\mathcal{S}_{\Delta k_a}^{(a)}$ are isotropic, (iii) the quartic vertex is local (momentum-independent). Then I_{00}, I_{11}, I_{01} are independent of the lattice structure S .

Same as Universality A: under isotropy, the integrands and domains depend only on q . No discrete star-geometry information enters the continuous integrals.

F. Proposition: weak multi-shell coupling does not invert BCC ranking

[Non-inversion under weak shell mixing] Assume the hypotheses of the Lemma and that the structural ordering on the primary shell satisfies

$$\mathcal{C}_{\text{BCC}}^{(0)} > \mathcal{C}_{\text{FCC}}^{(0)} > \mathcal{C}_{\text{SC}}^{(0)}. \quad (45)$$

Assume furthermore that the mixed-shell resonance contribution is parametrically weak:

$$I_{01} \sup_S |\mathcal{D}_S^{(01)}| \leq \epsilon I_{00} \inf_{S \in \{\text{BCC}, \text{FCC}\}} |\mathcal{C}_S^{(0)}|, \quad 0 < \epsilon \ll 1. \quad (46)$$

Then in the fluctuation-dominated regime, the BCC>FCC ranking on the primary shell persists:

$$\beta_{\text{eff}, \text{BCC}}^{(0)} < \beta_{\text{eff}, \text{FCC}}^{(0)}, \quad (47)$$

and cannot be inverted by two-shell mixing satisfying (46).

From (44),

$$\begin{aligned} \beta_{\text{eff}, \text{BCC}}^{(0)} - \beta_{\text{eff}, \text{FCC}}^{(0)} &= \left(\beta_{\text{MF}, \text{BCC}}^{(0)} - \beta_{\text{MF}, \text{FCC}}^{(0)} \right) \\ &- \frac{u^2}{2} \left[I_{00} \left(\mathcal{C}_{\text{BCC}}^{(0)} - \mathcal{C}_{\text{FCC}}^{(0)} \right) + I_{01} \left(\mathcal{D}_{\text{BCC}}^{(01)} - \mathcal{D}_{\text{FCC}}^{(01)} \right) \right]. \end{aligned}$$

In the fluctuation-dominated regime the mean-field difference is subleading. Since $\mathcal{C}_{\text{BCC}}^{(0)} - \mathcal{C}_{\text{FCC}}^{(0)} > 0$, the leading term favors BCC. The mixed term can at most reduce the magnitude of the difference. Imposing (46) guarantees the mixed contribution cannot change the sign, hence the BCC>FCC hierarchy persists.

G. Remark: what must be computed next

To *complete* Universality D one must: (i) compute/upper-bound $\sup_S |\mathcal{D}_S^{(01)}|$ for the relevant candidate structures, (ii) evaluate I_{01}/I_{00} in the relevant parameter regime, thereby verifying (46) (or deriving an alternative sufficient condition).

H. Asymptotic estimates: I_{00} and an upper bound on I_{01}

Define the shell separation

$$\Delta := |k_1^2 - k_0^2| > 0, \quad \Delta \ll k_0^2, \quad (48)$$

and assume the deep Brazovskii regime on the primary shell:

$$r_0 \rightarrow 0^+, \quad 0 < \Delta k_0 \ll k_0, \quad \kappa_0 > 0. \quad (49)$$

a. Primary-shell bubble. The standard near-shell evaluation gives

$$\begin{aligned} I_{00}(r_0; \Delta k_0) &= \int_{\mathcal{S}_{\Delta k_0}^{(0)}} \frac{d^3 q}{(2\pi)^3} \frac{1}{[r_0 + \kappa_0(q^2 - k_0^2)]^2} \\ &\sim \frac{1}{8\pi} \frac{k_0}{\sqrt{\kappa_0}} r_0^{-3/2}. \end{aligned}$$

(50)

b. Cross-shell bubble: general bound. For any isotropic overlap factor satisfying $0 \leq \chi_{01}(q) \leq 1$, Cauchy–Schwarz yields

$$I_{01}(r_0, r_1; \Delta k_0, \Delta k_1) \leq [I_{00}(r_0; \Delta k_0)]^{1/2} [I_{11}(r_1; \Delta k_1)]^{1/2}. \quad (51)$$

c. Cross-shell bubble: separation-suppressed estimate. If the shells are separated ($\Delta > 0$), then on $\mathcal{S}_{\Delta k_0}^{(0)}$ we have

$$|q^2 - k_1^2| = |(q^2 - k_0^2) - (k_1^2 - k_0^2)| \geq \Delta - |q^2 - k_0^2|. \quad (52)$$

For $|q - k_0| \leq \Delta k_0$ and $\Delta k_0 \ll \Delta/k_0$, this implies

$$|q^2 - k_1^2| \geq \frac{\Delta}{2}, \quad (53)$$

hence

$$\begin{aligned} G_1(q) &= \frac{1}{r_1 + \kappa_1 (q^2 - k_1^2)^2} \\ &\leq \frac{1}{\kappa_1 (\Delta/2)^2} = \frac{4}{\kappa_1 \Delta^2}. \end{aligned} \quad (54)$$

Therefore

$$\begin{aligned} I_{01} &= \int_{\mathcal{S}_{\Delta k_0}^{(0)}} \frac{d^3 q}{(2\pi)^3} G_0(q) G_1(q) \chi_{01}(q) \\ &\leq \frac{4}{\kappa_1 \Delta^2} \int_{\mathcal{S}_{\Delta k_0}^{(0)}} \frac{d^3 q}{(2\pi)^3} G_0(q). \end{aligned} \quad (55)$$

The remaining integral scales as

$$\int_{\mathcal{S}_{\Delta k_0}^{(0)}} \frac{d^3 q}{(2\pi)^3} G_0(q) \sim \frac{1}{4\pi} \frac{k_0}{\sqrt{\kappa_0}} r_0^{-1/2}, \quad (56)$$

so we obtain the separation-suppressed bound

$$I_{01}(r_0, r_1; \Delta k_0, \Delta k_1) \lesssim \frac{1}{\pi} \frac{k_0}{\sqrt{\kappa_0}} \frac{1}{\kappa_1 \Delta^2} r_0^{-1/2}. \quad (57)$$

d. Ratio estimate. Combining (IV H 0 a) and (57),

$$\frac{I_{01}}{I_{00}} \lesssim 8 \frac{\sqrt{\kappa_0}}{\kappa_1 \Delta^2} r_0, \quad (r_0 \rightarrow 0^+), \quad (58)$$

so *cross-shell mixing is parametrically suppressed* in the deep Brazovskii regime at fixed $\Delta > 0$.

I. Completion of Universality D: a sufficient non-inversion condition

Recall (44):

$$\beta_{\text{eff},S}^{(0)} = \beta_{\text{MF},S}^{(0)} - \frac{u^2}{2} \left[I_{00} \mathcal{C}_S^{(0)} + I_{01} \mathcal{D}_S^{(01)} \right]. \quad (59)$$

Define the geometric gaps

$$\Delta \mathcal{C}_{BF} := \mathcal{C}_{\text{BCC}}^{(0)} - \mathcal{C}_{\text{FCC}}^{(0)} > 0, \quad \Delta \mathcal{D}_{BF} := \mathcal{D}_{\text{BCC}}^{(01)} - \mathcal{D}_{\text{FCC}}^{(01)}. \quad (60)$$

[Two-shell non-inversion in the deep Brazovskii regime] Assume isotropy, locality, and single-star equal-amplitude ansatz on each shell. Assume $\Delta \mathcal{C}_{BF} > 0$ and shell separation $\Delta > 0$. Then for sufficiently small $r_0 > 0$,

$$\beta_{\text{eff},\text{BCC}}^{(0)} < \beta_{\text{eff},\text{FCC}}^{(0)} \quad (61)$$

holds provided the following *sufficient* condition is met:

$$\frac{I_{01}}{I_{00}} |\Delta \mathcal{D}_{BF}| < \Delta \mathcal{C}_{BF}. \quad (62)$$

Using (58), a concrete sufficient condition is

$$8 \frac{\sqrt{\kappa_0}}{\kappa_1 \Delta^2} r_0 |\Delta \mathcal{D}_{BF}| < \Delta \mathcal{C}_{BF}, \quad (r_0 \rightarrow 0^+). \quad (63)$$

Compute the difference:

$$\begin{aligned} \beta_{\text{eff,BCC}}^{(0)} - \beta_{\text{eff,FCC}}^{(0)} &= (\beta_{\text{MF,BCC}}^{(0)} - \beta_{\text{MF,FCC}}^{(0)}) \\ &\quad - \frac{u^2}{2} [I_{00} \Delta \mathcal{C}_{BF} + I_{01} \Delta \mathcal{D}_{BF}]. \end{aligned} \quad (64)$$

In the fluctuation-dominated regime, the mean-field difference is subleading. Thus the sign is controlled by the bracket. Since $I_{00} > 0$ and $\Delta \mathcal{C}_{BF} > 0$, the leading term favors BCC. The mixed term can only flip the sign if $I_{01} |\Delta \mathcal{D}_{BF}| \geq I_{00} \Delta \mathcal{C}_{BF}$. Therefore (62) is sufficient for non-inversion. Equation (63) follows from the ratio bound (58).

J. What remains to *fully close* Universality D

To turn the sufficient condition (62) into a completely explicit theorem, one must provide:

- (i) an explicit evaluation or bound for $\Delta \mathcal{D}_{BF}$,
- (ii) the regime of validity for $\Delta k_0 \ll \Delta/k_0$.

Item (i) is purely combinatorial (mixed-shell resonance counting) once the secondary-shell star $\{\mathbf{p}_j\}$ is specified.

K. Mixed-shell resonance counting: a universal upper bound

We now bound the mixed-shell combinatorial weight

$$\mathcal{D}_S^{(01)} := \frac{N_{\text{loop},S}^{(01)}}{N_S^2}, \quad (66)$$

where $N_{\text{loop},S}^{(01)}$ counts (nontrivial) quartets of the form

$$\mathbf{k}_{i_1} + \mathbf{k}_{i_2} + \mathbf{p}_{j_1} + \mathbf{p}_{j_2} = \mathbf{0}, \quad |\mathbf{k}_{i_\alpha}| = k_0, \quad |\mathbf{p}_{j_\beta}| = k_1, \quad (67)$$

with $\{\mathbf{k}_i\}_{i=1}^{N_S}$ the primary-shell star for structure S and $\{\mathbf{p}_j\}_{j=1}^{M_S}$ the secondary-shell star (not assumed identical across S).

a. Counting convention. For definiteness, let $N_{\text{loop},S}^{(01)}$ count *ordered* pairs $((i_1, i_2), (j_1, j_2))$ satisfying the resonance constraint. (Any unordered convention differs only by an overall constant factor that cancels in the non-inversion criterion after redefinition of $\mathcal{D}_S^{(01)}$.)

[Universal bound on mixed-shell resonance loops] For any two stars $\{\mathbf{k}_i\}_{i=1}^{N_S}$ and $\{\mathbf{p}_j\}_{j=1}^{M_S}$,

$$N_{\text{loop},S}^{(01)} \leq N_S^2 M_S. \quad (68)$$

Consequently,

$$0 \leq \mathcal{D}_S^{(01)} \leq M_S. \quad (69)$$

Fix an ordered pair (i_1, i_2) on the primary shell, and fix j_1 on the secondary shell. Then the resonance condition uniquely determines the fourth wavevector:

$$\mathbf{p}_{j_2} = -(\mathbf{k}_{i_1} + \mathbf{k}_{i_2} + \mathbf{p}_{j_1}). \quad (70)$$

Hence for each choice of (i_1, i_2, j_1) there is *at most one* index j_2 that can satisfy the resonance, and it may fail to lie in the discrete star. Therefore the number of solutions for fixed (i_1, i_2) is at most M_S , and summing over the N_S^2 ordered primary pairs gives (68). Dividing by N_S^2 gives (69).

L. A structure-independent sufficient condition using the bound

Recall the sufficient non-inversion criterion (62):

$$\frac{I_{01}}{I_{00}} |\Delta \mathcal{D}_{BF}| < \Delta \mathcal{C}_{BF}, \quad \Delta \mathcal{C}_{BF} := \mathcal{C}_{\text{BCC}}^{(0)} - \mathcal{C}_{\text{FCC}}^{(0)}. \quad (71)$$

Using the triangle inequality and (69),

$$\begin{aligned} |\Delta \mathcal{D}_{BF}| &= |\mathcal{D}_{\text{BCC}}^{(01)} - \mathcal{D}_{\text{FCC}}^{(01)}| \\ &\leq \mathcal{D}_{\text{BCC}}^{(01)} + \mathcal{D}_{\text{FCC}}^{(01)} \leq M_{\text{BCC}} + M_{\text{FCC}}. \end{aligned} \quad (72)$$

Therefore a fully explicit (though conservative) sufficient condition is

$$\frac{I_{01}}{I_{00}} (M_{\text{BCC}} + M_{\text{FCC}}) < \Delta \mathcal{C}_{BF}. \quad (73)$$

M. Completion with separation-suppressed estimate

Using the ratio estimate (58) from the previous subsection,

$$\frac{I_{01}}{I_{00}} \lesssim 8 \frac{\sqrt{\kappa_0}}{\kappa_1 \Delta^2} r_0, \quad (74)$$

a concrete sufficient condition ensuring the BCC>FCC hierarchy survives two-shell mixing is:

$$8 \frac{\sqrt{\kappa_0}}{\kappa_1 \Delta^2} r_0 (M_{\text{BCC}} + M_{\text{FCC}}) < \Delta \mathcal{C}_{BF}. \quad (75)$$

a. Interpretation (formal). Equation (75) shows that for fixed shell separation $\Delta > 0$ and fixed discrete secondary-shell star sizes $M_{\text{BCC}}, M_{\text{FCC}}$, the cross-shell contribution is parametrically suppressed as $r_0 \rightarrow 0^+$, hence cannot invert the primary-shell geometric hierarchy whenever $\Delta \mathcal{C}_{BF} > 0$.

N. Optional strengthening: equal secondary-shell orbit sizes

If the secondary-shell construction uses the same orbit size M across the cubic candidates,

$$M_{\text{BCC}} = M_{\text{FCC}} = M, \quad (76)$$

then (75) simplifies to

$$16 \frac{\sqrt{\kappa_0}}{\kappa_1 \Delta^2} r_0 M < \Delta \mathcal{C}_{BF}. \quad (77)$$

O. Status of Universality D

Within the isotropic local truncation class, Universality D is now closed at the level of a *sufficient* (non-optimized) theorem:

- the mixed-shell bubble contribution is structure-independent and separation-suppressed;
- the mixed-shell combinatorics admits a universal bound independent of the detailed star geometry;
- together, these yield an explicit inequality (75) guaranteeing that two-shell mixing cannot invert the BCC>FCC hierarchy.

Sharper (less conservative) results follow by computing $\mathcal{D}_S^{(01)}$ explicitly for the chosen secondary-shell stars, but are not required for a rigorous sufficient non-inversion theorem.

P. Sharp closure of Universality D via a discrete length-ratio obstruction

We now sharpen the mixed-shell combinatorics beyond the crude bound $0 \leq \mathcal{D}_S^{(01)} \leq M_S$ by using a necessary *metric* condition for mixed-shell resonances.

Let the primary-shell star for structure S be the finite set of directions

$$U_S := \{\hat{\mathbf{k}}_i\}_{i=1}^{N_S} \subset S^2, \quad \mathbf{k}_i = k_0 \hat{\mathbf{k}}_i, \quad (78)$$

and let the secondary-shell star be another finite set of directions

$$V := \{\hat{\mathbf{p}}_j\}_{j=1}^M \subset S^2, \quad \mathbf{p}_j = k_1 \hat{\mathbf{p}}_j, \quad (79)$$

taken *common* across candidates (i.e. V does not depend on S), which is the natural setting for a universality statement.

Define the (finite) chord-length sets of *pair sums*

$$\begin{aligned} \mathcal{L}(U_S) &:= \left\{ \|\hat{\mathbf{k}} + \hat{\mathbf{k}}'\| : \hat{\mathbf{k}}, \hat{\mathbf{k}}' \in U_S \right\}, \\ \mathcal{L}(V) &:= \left\{ \|\hat{\mathbf{p}} + \hat{\mathbf{p}}'\| : \hat{\mathbf{p}}, \hat{\mathbf{p}}' \in V \right\}. \end{aligned} \quad (80)$$

Note that $0 \in \mathcal{L}(\cdot)$ always (from $\hat{\mathbf{v}} + (-\hat{\mathbf{v}}) = 0$ if both are present).

Q. Lemma: necessary length-ratio condition for a mixed-shell resonance

A mixed-shell quartet requires

$$\begin{aligned} \mathbf{k}_{i_1} + \mathbf{k}_{i_2} + \mathbf{p}_{j_1} + \mathbf{p}_{j_2} &= \mathbf{0} \\ \iff k_0(\hat{\mathbf{k}}_{i_1} + \hat{\mathbf{k}}_{i_2}) &= -k_1(\hat{\mathbf{p}}_{j_1} + \hat{\mathbf{p}}_{j_2}). \end{aligned} \quad (81)$$

[Length-ratio obstruction] If (81) holds and both pair-sums are nonzero, then necessarily

$$\frac{k_1}{k_0} = \frac{\|\hat{\mathbf{k}}_{i_1} + \hat{\mathbf{k}}_{i_2}\|}{\|\hat{\mathbf{p}}_{j_1} + \hat{\mathbf{p}}_{j_2}\|} \in \frac{\mathcal{L}(U_S) \setminus \{0\}}{\mathcal{L}(V) \setminus \{0\}}. \quad (82)$$

Taking norms in (81) and assuming neither side is zero yields $k_0\|\hat{\mathbf{k}}_{i_1} + \hat{\mathbf{k}}_{i_2}\| = k_1\|\hat{\mathbf{p}}_{j_1} + \hat{\mathbf{p}}_{j_2}\|$, hence (82).

R. Definition: nontrivial mixed-shell loops

We define $N_{\text{loop},S}^{(01)}$ to count only *nontrivial* mixed loops, i.e. solutions to (81) with

$$\hat{\mathbf{k}}_{i_1} + \hat{\mathbf{k}}_{i_2} \neq \mathbf{0} \quad \text{and} \quad \hat{\mathbf{p}}_{j_1} + \hat{\mathbf{p}}_{j_2} \neq \mathbf{0}. \quad (83)$$

(Trivial $\pm$ -pairings on each shell are excluded because they produce only structure-independent self-renormalizations and do not contribute to structural ranking.)

S. Theorem: generic absence of nontrivial mixed resonances (hence $\mathcal{D}_S^{(01)} = 0$)

[Generic vanishing of the mixed-shell combinatorics] Fix finite direction sets U_S (for each candidate structure S) and a common finite V . If the shell-radius ratio satisfies

$$\frac{k_1}{k_0} \notin \bigcup_{S \in \{\text{BCC}, \text{FCC}, \text{SC}\}} \frac{\mathcal{L}(U_S) \setminus \{0\}}{\mathcal{L}(V) \setminus \{0\}}, \quad (84)$$

then there exist *no* nontrivial mixed-shell resonances for any S . Consequently,

$$N_{\text{loop},S}^{(01)} = 0, \quad \mathcal{D}_S^{(01)} = 0, \quad \Delta \mathcal{D}_{BF} = 0, \quad (85)$$

and the two-shell effective quartic coefficient reduces to

$$\beta_{\text{eff},S}^{(0)} = \beta_{\text{MF},S}^{(0)} - \frac{u^2}{2} I_{00}(r_0; \Delta k_0) \mathcal{C}_S^{(0)}, \quad (86)$$

so the BCC>FCC ranking is *identical* to the single-shell theory, with no additional condition.

Under (84), the necessary condition (82) cannot be satisfied for any nonzero pair-sums, hence no nontrivial mixed resonances exist. By definition of $N_{\text{loop},S}^{(01)}$, this implies $N_{\text{loop},S}^{(01)} = 0$ and thus $\mathcal{D}_S^{(01)} = 0$. Substituting into (44) gives the reduced form.

T. Corollary: Universality D is fully closed for generic (k_1/k_0)

[Unconditional non-inversion for generic ratios] For fixed discrete stars U_S and a common V , the exceptional set of ratios

$$\mathfrak{E} := \bigcup_S \frac{\mathcal{L}(U_S) \setminus \{0\}}{\mathcal{L}(V) \setminus \{0\}} \quad (87)$$

is finite, hence measure-zero in $\mathbb{R}_{>0}$. For all $k_1/k_0 \notin \mathfrak{E}$, mixed-shell loops vanish and BCC ranking is unchanged without needing the small-mixing inequality (75).

U. Remark: the resonant (nongeneric) case

If $k_1/k_0 \in \mathfrak{E}$, nontrivial mixed resonances may exist and one must revert to the previous sufficient-condition theorem (bounded by I_{01}/I_{00}) or compute $\mathcal{D}_S^{(01)}$ explicitly for that specific ratio and choice of V .

V. UNIVERSALITY E: THE RESONANT TWO-SHELL CASE

A. Problem statement

Assume the shell ratio lies in the exceptional (finite) set

$$\frac{k_1}{k_0} \in \mathfrak{E}, \quad \mathfrak{E} := \bigcup_{S \in \{\text{BCC}, \text{FCC}, \text{SC}\}} \frac{\mathcal{L}(U_S) \setminus \{0\}}{\mathcal{L}(V) \setminus \{0\}}, \quad (88)$$

so that nontrivial mixed-shell resonances *may* exist.

Recall the primary-shell effective quartic coefficient:

$$\beta_{\text{eff},S}^{(0)} = \beta_{\text{MF},S}^{(0)} - \frac{u^2}{2} \left[I_{00} \mathcal{C}_S^{(0)} + I_{01} \mathcal{D}_S^{(01)} \right]. \quad (89)$$

Universality E consists in obtaining a *sharp* control of $\Delta \mathcal{D}_{BF} := \mathcal{D}_{\text{BCC}}^{(01)} - \mathcal{D}_{\text{FCC}}^{(01)}$ for the resonant ratios, and proving that mixed-shell terms cannot invert the hierarchy.

B. Reduction to a discrete representation problem

Write the mixed-shell resonance constraint as

$$k_0(\hat{\mathbf{k}}_{i_1} + \hat{\mathbf{k}}_{i_2}) = -k_1(\hat{\mathbf{p}}_{j_1} + \hat{\mathbf{p}}_{j_2}). \quad (90)$$

Define the discrete pair-sum sets

$$\Sigma(U_S) := \{\hat{\mathbf{k}} + \hat{\mathbf{k}}' : \hat{\mathbf{k}}, \hat{\mathbf{k}}' \in U_S, \hat{\mathbf{k}} + \hat{\mathbf{k}}' \neq 0\}, \quad (91)$$

$$\Sigma(V) := \{\hat{\mathbf{p}} + \hat{\mathbf{p}}' : \hat{\mathbf{p}}, \hat{\mathbf{p}}' \in V, \hat{\mathbf{p}} + \hat{\mathbf{p}}' \neq 0\}. \quad (92)$$

Then (90) is equivalent to the existence of

$$\mathbf{s} \in \Sigma(U_S), \quad \mathbf{t} \in \Sigma(V) \quad \text{such that} \quad k_0 \mathbf{s} = -k_1 \mathbf{t}. \quad (93)$$

Hence mixed resonances exist if and only if the scaled set $k_0 \Sigma(U_S)$ intersects $-k_1 \Sigma(V)$.

C. Definition of multiplicity weights

For any nonzero vector $\mathbf{v} \in \mathbb{R}^3$, define multiplicities

$$m_S(\mathbf{v}) := \#\{(i_1, i_2) : \hat{\mathbf{k}}_{i_1} + \hat{\mathbf{k}}_{i_2} = \mathbf{v}\}, \quad (94)$$

$$n(\mathbf{v}) := \#\{(j_1, j_2) : \hat{\mathbf{p}}_{j_1} + \hat{\mathbf{p}}_{j_2} = \mathbf{v}\}. \quad (95)$$

Then the number of mixed loops is

$$N_{\text{loop}, S}^{(01)} = \sum_{\mathbf{v} \neq 0} m_S(\mathbf{v}) n\left(-\frac{k_0}{k_1} \mathbf{v}\right), \quad (96)$$

with the convention that $n(\mathbf{w}) = 0$ if $\mathbf{w} \notin \Sigma(V)$.

Therefore,

$$\mathcal{D}_S^{(01)} = \frac{1}{N_S^2} \sum_{\mathbf{v} \neq 0} m_S(\mathbf{v}) n\left(-\frac{k_0}{k_1} \mathbf{v}\right). \quad (97)$$

D. Lemma: uniform upper bound in terms of maximal pair-sum multiplicities

Define

$$m_S^{\max} := \max_{\mathbf{v} \neq 0} m_S(\mathbf{v}), \quad n^{\max} := \max_{\mathbf{v} \neq 0} n(\mathbf{v}), \quad (98)$$

$$|\Sigma(U_S)| := \#\Sigma(U_S).$$

[Resonant bound] For any resonant ratio k_1/k_0 , one has

$$\mathcal{D}_S^{(01)} \leq \frac{|\Sigma(U_S)|}{N_S^2} m_S^{\max} n^{\max}. \quad (99)$$

From (97),

$$\begin{aligned} \mathcal{D}_S^{(01)} &\leq \frac{1}{N_S^2} \sum_{\mathbf{v} \neq 0} m_S^{\max} n^{\max} \mathbf{1}\left[-\frac{k_0}{k_1} \mathbf{v} \in \Sigma(V)\right] \\ &\leq \frac{|\Sigma(U_S)|}{N_S^2} m_S^{\max} n^{\max}. \end{aligned} \quad (100)$$

E. Proposition: sufficient non-inversion criterion in the resonant case

Let $\Delta \mathcal{C}_{BF} := \mathcal{C}_{\text{BCC}}^{(0)} - \mathcal{C}_{\text{FCC}}^{(0)} > 0$. Using (89), the BCC>FCC condition is

$$I_{00} \Delta \mathcal{C}_{BF} + I_{01} \Delta \mathcal{D}_{BF} > 0. \quad (101)$$

A sufficient (conservative) condition is

$$\frac{I_{01}}{I_{00}} \left(\mathcal{D}_{\text{BCC}}^{(01)} + \mathcal{D}_{\text{FCC}}^{(01)} \right) < \Delta \mathcal{C}_{BF}. \quad (102)$$

By Lemma (99), one obtains a fully explicit sufficient condition:

$$\frac{I_{01}}{I_{00}} n^{\max} \left[\frac{|\Sigma(U_{\text{BCC}})|}{N_{\text{BCC}}^2} m_{\text{BCC}}^{\max} + \frac{|\Sigma(U_{\text{FCC}})|}{N_{\text{FCC}}^2} m_{\text{FCC}}^{\max} \right] < \Delta \mathcal{C}_{BF}. \quad (103)$$

F. Remark: what remains to *fully close* Universality E

To make (103) operational, one must:

- Fix a concrete secondary-shell star V (hence compute $n^{\max}$);
- Compute $|\Sigma(U_S)|$ and $m_S^{\max}$ for $S \in \{\text{BCC}, \text{FCC}\}$ (purely discrete, closed-form);
- Evaluate I_{01}/I_{00} in the regime of interest (already bounded by separation-suppression when $\Delta > 0$).

The above converts the resonant-case problem into finite combinatorics plus a single ratio of isotropic integrals.

G. Explicit multiplicity data for the standard cubic stars

We take the standard reciprocal-star direction sets:

$$U_{\text{SC}} := \{\pm \hat{\mathbf{e}}_x, \pm \hat{\mathbf{e}}_y, \pm \hat{\mathbf{e}}_z\}, \quad N_{\text{SC}} = 6, \quad (104)$$

$$U_{\text{FCC}} := \left\{ \frac{1}{\sqrt{3}}(\pm 1, \pm 1, \pm 1) \right\}, \quad N_{\text{FCC}} = 8, \quad (105)$$

$$U_{\text{BCC}} := \left\{ \frac{1}{\sqrt{2}}(\pm 1, \pm 1, 0) \text{ and perms} \right\}, \quad N_{\text{BCC}} = 12. \quad (106)$$

Recall

$$\Sigma(U_S) := \{\hat{\mathbf{k}} + \hat{\mathbf{k}}' : \hat{\mathbf{k}}, \hat{\mathbf{k}}' \in U_S, \hat{\mathbf{k}} + \hat{\mathbf{k}}' \neq 0\}, \quad m_S(\mathbf{v}) := \#\{(i_1, i_2) : \hat{\mathbf{k}}_{i_1} + \hat{\mathbf{k}}_{i_2} = \mathbf{v}\}, \quad (107)$$

and

$$m_S^{\max} := \max_{\mathbf{v} \neq 0} m_S(\mathbf{v}), \quad |\Sigma(U_S)| := \#\Sigma(U_S). \quad (108)$$

a. Result (ordered-pair convention). For the above stars one obtains:

$$|\Sigma(U_{\text{SC}})| = 18, \quad m_{\text{SC}}^{\max} = 2, \quad (109)$$

$$|\Sigma(U_{\text{FCC}})| = 26, \quad m_{\text{FCC}}^{\max} = 4, \quad (110)$$

$$|\Sigma(U_{\text{BCC}})| = 54, \quad m_{\text{BCC}}^{\max} = 4. \quad (111)$$

Hence the resonant-case bound (Lemma (99)) becomes:

$$\mathcal{D}_{\text{FCC}}^{(01)} \leq \frac{|\Sigma(U_{\text{FCC}})|}{N_{\text{FCC}}^2} m_{\text{FCC}}^{\max} n^{\max} = \frac{26}{8^2} \cdot 4 n^{\max} = \frac{13}{8} n^{\max}, \quad (112)$$

$$\mathcal{D}_{\text{BCC}}^{(01)} \leq \frac{|\Sigma(U_{\text{BCC}})|}{N_{\text{BCC}}^2} m_{\text{BCC}}^{\max} n^{\max} = \frac{54}{12^2} \cdot 4 n^{\max} = \frac{3}{2} n^{\max}. \quad (113)$$

Therefore,

$$\mathcal{D}_{\text{BCC}}^{(01)} + \mathcal{D}_{\text{FCC}}^{(01)} \leq \left(\frac{3}{2} + \frac{13}{8} \right) n^{\max} = \frac{25}{8} n^{\max}. \quad (114)$$

H. Fully explicit resonant-case non-inversion condition

Using (102) and (114), a sufficient resonant-case criterion is

$$\frac{I_{01}}{I_{00}} \cdot \frac{25}{8} n^{\max} < \Delta \mathcal{C}_{BF}, \quad \Delta \mathcal{C}_{BF} := \mathcal{C}_{\text{BCC}}^{(0)} - \mathcal{C}_{\text{FCC}}^{(0)} > 0. \quad (115)$$

a. If the secondary star equals a standard cubic star. If the secondary-shell direction set V is taken as one of the standard cubic stars, then

$$V = U_{\text{SC}} : \quad n^{\text{max}} = 2, \quad (116)$$

$$V = U_{\text{FCC}} : \quad n^{\text{max}} = 4, \quad (117)$$

$$V = U_{\text{BCC}} : \quad n^{\text{max}} = 4, \quad (118)$$

so (115) becomes explicitly

$$\frac{I_{01}}{I_{00}} \cdot \begin{cases} \frac{25}{4} & (V = U_{\text{SC}}), \\ \frac{25}{2} & (V = U_{\text{FCC}} \text{ or } U_{\text{BCC}}), \end{cases} < \Delta \mathcal{C}_{BF}. \quad (119)$$

I. Evaluating $\Delta \mathcal{C}_{BF}$ for the cubic stars

For the standard cubic stars used throughout (SC: $N = 6$, FCC: $N = 8$, BCC: $N = 12$), the geometric ratios entering the one-loop quartic correction are

$$\mathcal{C}_S^{(0)} = \frac{N_{\text{loop},S}^{(0)}}{N_S^2}, \quad (120)$$

with the explicit values (from the resonance-loop counting on the primary shell)

$$\mathcal{C}_{\text{BCC}}^{(0)} = \frac{1}{6}, \quad \mathcal{C}_{\text{FCC}}^{(0)} = \frac{1}{8}, \quad \mathcal{C}_{\text{SC}}^{(0)} = 0. \quad (121)$$

Hence

$$\Delta \mathcal{C}_{BF} := \mathcal{C}_{\text{BCC}}^{(0)} - \mathcal{C}_{\text{FCC}}^{(0)} = \frac{1}{24}. \quad (122)$$

J. Final resonant-case inequality in terms of I_{01}/I_{00}

Insert (122) into (115):

$$\frac{I_{01}}{I_{00}} \cdot \frac{25}{8} n^{\text{max}} < \frac{1}{24}. \quad (123)$$

Equivalently,

$$\frac{I_{01}}{I_{00}} < \frac{1}{75 n^{\text{max}}}. \quad (124)$$

If the secondary-shell direction set V is itself a standard cubic star, then

$$n^{\text{max}} = \begin{cases} 2, & V = U_{\text{SC}}, \\ 4, & V = U_{\text{FCC}} \text{ or } U_{\text{BCC}}, \end{cases} \quad (125)$$

and (124) becomes

$$\frac{I_{01}}{I_{00}} < \begin{cases} \frac{1}{150}, & (V = U_{\text{SC}}), \\ \frac{1}{300}, & (V = U_{\text{FCC}} \text{ or } U_{\text{BCC}}). \end{cases} \quad (126)$$

K. Final resonant-case inequality in terms of shell separation and r_0

Using the separation-suppressed estimate (deep Brazovskii regime with $\Delta = |k_1^2 - k_0^2| > 0$ and $\Delta k_0 \ll \Delta/k_0$):

$$\frac{I_{01}}{I_{00}} \lesssim 8 \frac{\sqrt{\kappa_0}}{\kappa_1 \Delta^2} r_0, \quad (127)$$

a concrete sufficient condition ensuring (124) is

$$8 \frac{\sqrt{\kappa_0}}{\kappa_1 \Delta^2} r_0 < \frac{1}{75 n^{\max}}. \quad (128)$$

Equivalently,

$$r_0 < \frac{1}{600 n^{\max}} \frac{\kappa_1 \Delta^2}{\sqrt{\kappa_0}}. \quad (129)$$

For the standard cubic secondary stars:

$$r_0 < \begin{cases} \frac{1}{1200} \frac{\kappa_1 \Delta^2}{\sqrt{\kappa_0}}, & (V = U_{\text{SC}}), \\ \frac{1}{2400} \frac{\kappa_1 \Delta^2}{\sqrt{\kappa_0}}, & (V = U_{\text{FCC}} \text{ or } U_{\text{BCC}}). \end{cases} \quad (130)$$

L. Theorem: Universality D is closed for generic and resonant ratios

[Two-shell universality of the BCC ranking] Consider an isotropic two-shell Landau–Brazovskii theory with local quartic interaction and equal-amplitude star ansatz on each shell. Let the primary-shell cubic stars be $U_{\text{BCC}}, U_{\text{FCC}}, U_{\text{SC}}$ as above.

- **Generic ratio:** If $k_1/k_0 \notin \mathfrak{E}$, then all nontrivial mixed-shell resonance loops vanish and the BCC>FCC hierarchy is identical to the single-shell theory.
- **Resonant ratio:** If $k_1/k_0 \in \mathfrak{E}$, then the BCC>FCC hierarchy still persists provided the cross-shell bubble ratio satisfies (124), or sufficiently, the explicit deep-Brazovskii condition (129).

VI. UNIVERSALITY F: STRONG MULTI-SHELL OVERLAP ($\Delta \rightarrow 0$) AND MODE-REDUCTION

A. Setup: two strongly coupled shell components with a quadratic mixing matrix

Let $\phi_a(x)$, $a \in \{0, 1\}$, denote two coarse-grained components supported near a common soft radius k_0 (allowing strong overlap). The quadratic action is taken as the most general *isotropic* form consistent with rotational symmetry:

$$S_2 = \frac{1}{2} \int_{\mathbf{q}} \phi_a(\mathbf{q}) \mathcal{M}_{ab}(q) \phi_b(-\mathbf{q}), \quad \int_{\mathbf{q}} := \int \frac{d^3 q}{(2\pi)^3}, \quad (131)$$

with

$$\mathcal{M}(q) = \begin{pmatrix} r_0 + \kappa_0(q^2 - k_0^2)^2 & \mu \\ \mu & r_1 + \kappa_1(q^2 - k_0^2)^2 \end{pmatrix}, \quad q := |\mathbf{q}|. \quad (132)$$

Here μ parametrizes *strong* inter-shell mixing (no smallness assumption), and the two shells have merged into a single isotropic soft band around k_0 .

The interaction is local quartic (scalar for definiteness):

$$S_4 = \int d^3 x \frac{u}{4!} (\phi_0 + \phi_1)^4. \quad (133)$$

B. Diagonalization into eigenmodes

For each q , diagonalize $\mathcal{M}(q)$ by an orthogonal matrix $O(q) \in SO(2)$:

$$O(q)^\top \mathcal{M}(q) O(q) = \text{diag}(\lambda_-(q), \lambda_+(q)), \quad (134)$$

and define eigenmodes

$$\begin{pmatrix} \varphi_-(\mathbf{q}) \\ \varphi_+(\mathbf{q}) \end{pmatrix} = O(q)^\top \begin{pmatrix} \phi_0(\mathbf{q}) \\ \phi_1(\mathbf{q}) \end{pmatrix}. \quad (135)$$

Explicitly,

$$\lambda_\pm(q) = \frac{1}{2} (A(q) + B(q)) \pm \frac{1}{2} \sqrt{(A(q) - B(q))^2 + 4\mu^2}, \quad (136)$$

where

$$A(q) = r_0 + \kappa_0(q^2 - k_0^2)^2, \quad B(q) = r_1 + \kappa_1(q^2 - k_0^2)^2. \quad (137)$$

C. Assumption: a single soft eigenmode dominates

[Single-soft-mode regime] There exists a neighborhood $\mathcal{B}$ of the soft band around $q \simeq k_0$ such that

$$\begin{aligned} \lambda_-(q) &\rightarrow 0^+ \text{ as control parameters vary,} \\ \lambda_+(q) &\geq \Delta_+ > 0 \quad \forall q \in \mathcal{B}, \end{aligned} \quad (138)$$

for some q -independent gap Δ_+ .

D. Lemma: integrating out the gapped mode yields a single-mode isotropic Brazovskii theory

[Mode reduction] Under (138), the gapped eigenmode φ_+ can be integrated out perturbatively in u to produce an effective action for φ_-

$$\begin{aligned} S_{\text{eff}}[\varphi_-] &= \frac{1}{2} \int_{\mathbf{q}} \varphi_-(\mathbf{q}) \lambda_-(q) \varphi_-(-\mathbf{q}) \\ &+ \int d^3x \frac{u_{\text{eff}}}{4!} \varphi_-^4 + (\text{higher operators}), \end{aligned} \quad (139)$$

where:

- $\lambda_-(q)$ is isotropic and has the Brazovskii form near k_0 : $\lambda_-(q) = r_{\text{eff}} + \kappa_{\text{eff}}(q^2 - k_0^2)^2 + O((q^2 - k_0^2)^3)$;
- $u_{\text{eff}} = u \cdot \alpha^4 + O(u^2)$ with α the (bounded) projection coefficient of $\phi_0 + \phi_1$ onto φ_- on the band $\mathcal{B}$;
- the induced higher operators are either higher-derivative or higher-point and are suppressed by powers of Δ_+^{-1} .

Since $\lambda_+(q) \geq \Delta_+ > 0$ on $\mathcal{B}$, the φ_+ propagator is uniformly bounded, so the functional integral over φ_+ is analytic in u and generates a local derivative expansion. Rotational symmetry ensures isotropy of all induced kernels. The leading contribution to the quartic term is the projection of $(\phi_0 + \phi_1)^4$ onto φ_-^4 , giving $u_{\text{eff}} = u\alpha^4 + O(u^2)$.

E. Proposition: BCC selection persists in the strong-overlap regime

Let S denote a candidate cubic structure and impose the equal-amplitude star ansatz on the *soft* eigenmode:

$$\varphi_-(x) = A \sum_{i=1}^{N_S} \cos(\mathbf{k}_i \cdot x), \quad |\mathbf{k}_i| = k_0. \quad (140)$$

[Universality in the strong-overlap regime] Under the hypotheses of Lemma (139) (in particular, isotropy and a gapped φ_+), the fluctuation-induced structural selection of the condensed state is identical to the single-shell theory for φ_- . Specifically, the one-loop corrected quartic coefficient takes the factorized form

$$\beta_{\text{eff},S} = \beta_{\text{MF},S} - \frac{v_{\text{eff}}^2}{2} I_{--}(r_{\text{eff}}) \mathcal{C}_S, \quad (141)$$

where

$$I_{--}(r_{\text{eff}}) := \int_{\mathcal{B}} \frac{d^3 q}{(2\pi)^3} \frac{1}{[r_{\text{eff}} + \kappa_{\text{eff}}(q^2 - k_0^2)]^2} \quad (142)$$

is structure-independent and $\mathcal{C}_S = N_{\text{loop},S}/N_S^2$ is the same resonance-loop ratio as in the single-shell theory. Therefore, for the cubic stars

$$\mathcal{C}_{\text{BCC}} = \frac{1}{6} > \mathcal{C}_{\text{FCC}} = \frac{1}{8} > \mathcal{C}_{\text{SC}} = 0, \quad (143)$$

one obtains in the fluctuation-dominated regime

$$\beta_{\text{eff},\text{BCC}} < \beta_{\text{eff},\text{FCC}} < \beta_{\text{eff},\text{SC}}, \quad (144)$$

i.e. the BCC hierarchy persists even when the two shells overlap strongly.

By Lemma (139), the effective soft-mode theory is an isotropic single-band Brazovskii functional with a local quartic coupling u_{eff} . The one-loop quartic correction is therefore the standard bubble integral built from the φ_- propagator, which is independent of the lattice candidate S . Structure dependence enters only through the discrete resonance-loop combinatorics $\mathcal{C}_S$, giving (141). The hierarchy then follows from the known ordering of $\mathcal{C}_S$ for cubic stars.

F. Failure mode (outside Universality F)

The conclusion can fail only if the single-soft-mode assumption (138) fails, i.e. if *both* eigenvalues $\lambda_{\pm}(q)$ become soft on the band (true two-mode criticality), or if isotropy/locality is broken so that structure-specific weights enter the loop integrals.

G. The genuinely two-soft-mode regime: coupled Brazovskii fields

We now drop the single-soft-mode assumption (138) and consider the case where *both* eigenmodes become soft on the same isotropic band around k_0 :

$$\lambda_-(q) \rightarrow 0^+, \quad \lambda_+(q) \rightarrow 0^+ \quad (q \simeq k_0). \quad (145)$$

In this regime no mode can be integrated out, and the effective theory is a *two-field* (isotropic) Brazovskii functional.

1. Effective action and quartic tensor

Assume the most general local quartic interaction compatible with $O(2)$ rotations in field space:

$$S_4 = \int d^3 x \left[\frac{u_1}{4!} \varphi_-^4 + \frac{u_2}{4!} \varphi_+^4 + \frac{w}{4} \varphi_-^2 \varphi_+^2 \right], \quad (146)$$

with $u_1, u_2 > 0$ and w not necessarily small (stability requires $w > -\sqrt{u_1 u_2}/3$ in this normalization).

The quadratic sector is

$$S_2 = \frac{1}{2} \int_{\mathbf{q}} [\varphi_-(\mathbf{q}) \lambda_-(q) \varphi_-(-\mathbf{q}) + \varphi_+(\mathbf{q}) \lambda_+(q) \varphi_+(-\mathbf{q})], \quad (147)$$

with isotropic dispersions

$$\lambda_a(q) = r_a + \kappa_a(q^2 - k_0^2)^2 + O((q^2 - k_0^2)^3), \quad a \in \{-, +\}. \quad (148)$$

2. Equal-amplitude star ansatz with a two-component amplitude vector

For a candidate structure S , take

$$\varphi_a(x) = A_a \sum_{i=1}^{N_S} \cos(\mathbf{k}_i \cdot x), \quad |\mathbf{k}_i| = k_0, \quad a \in \{-, +\}, \quad (149)$$

and define the amplitude vector $\mathbf{A} = (A_-, A_+)$.

The mean-field Landau functional becomes

$$\begin{aligned} F_S(\mathbf{A}) = & \frac{1}{2} r_- A_-^2 + \frac{1}{2} r_+ A_+^2 + \frac{1}{4} \beta_S^{--} A_-^4 \\ & + \frac{1}{4} \beta_S^{++} A_+^4 + \frac{1}{2} \beta_S^{-+} A_-^2 A_+^2 + \frac{1}{6} \gamma_S(\mathbf{A}) \text{ (6th order)}, \end{aligned} \quad (150)$$

where the quartic coefficients factorize as

$$\beta_S^{--} = u_1 \beta_{\text{MF},S}, \quad \beta_S^{++} = u_2 \beta_{\text{MF},S}, \quad \beta_S^{-+} = w \beta_{\text{MF},S}, \quad (151)$$

because the spatial star combinatorics is common and only multiplies the internal coupling.

3. One-loop correction: matrix bubble and geometric factorization

At one loop, the quartic tensor renormalizes via bubble integrals in each channel. Define the structure-independent shell integrals

$$\begin{aligned} I_{ab}(r_a, r_b) := & \int_{\mathcal{B}} \frac{d^3 q}{(2\pi)^3} \frac{1}{[r_a + \kappa_a(q^2 - k_0^2)^2]} \\ & \times \frac{1}{[r_b + \kappa_b(q^2 - k_0^2)^2]}, \quad a, b \in \{-, +\}. \end{aligned} \quad (152)$$

Then the one-loop renormalized quartic tensor has the schematic form

$$\beta_{\text{eff},S}^{ab} = \beta_{\text{MF},S}^{ab} - \frac{1}{2} \sum_{c,d \in \{-, +\}} \beta_{\text{MF},S}^{ac} I_{cd} \beta_{\text{MF},S}^{db} \mathcal{C}_S + \dots, \quad (153)$$

where the ellipsis denotes structure-independent contributions and higher operators, and $\mathcal{C}_S = N_{\text{loop},S}/N_S^2$ is the same spatial resonance ratio as before. Crucially, the *entire* structure dependence continues to factorize through $\mathcal{C}_S$.

4. Non-inversion theorem for two-soft-mode Brazovskii fields

[Two-soft-mode universality of BCC ranking] Assume: (i) isotropic dispersions for both modes around the same k_0 band, (ii) local quartic interaction (146), (iii) equal-amplitude star ansatz common across candidates.

Then the one-loop corrected quartic form entering the Landau functional for structure S can be written as

$$Q_S(\mathbf{A}) = \beta_{\text{MF},S} q_0(\mathbf{A}) - \mathcal{C}_S q_1(\mathbf{A}), \quad (154)$$

where $q_0(\mathbf{A}) \geq 0$ and $q_1(\mathbf{A}) \geq 0$ are *structure-independent* quartic forms determined solely by (u_1, u_2, w) and the shell integrals I_{ab} . Consequently, for any fixed nonzero amplitude vector $\mathbf{A}$,

$$Q_{\text{BCC}}(\mathbf{A}) < Q_{\text{FCC}}(\mathbf{A}) < Q_{\text{SC}}(\mathbf{A}), \quad (155)$$

whenever $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$. In particular, in the fluctuation-dominated regime the BCC structure remains the uniquely selected cubic candidate even when both modes are simultaneously soft.

The mean-field quartic contribution factorizes as in (151), yielding $Q_S^{\text{MF}}(\mathbf{A}) = \beta_{\text{MF},S} q_0(\mathbf{A})$ with a structure-independent $q_0 \geq 0$. The one-loop correction (153) is a contraction of the mean-field quartic tensor with the structure-independent bubble matrix I_{ab} , hence produces a structure-independent quartic form $q_1(\mathbf{A}) \geq 0$ multiplied by the sole spatial combinatorial factor $\mathcal{C}_S$. Therefore $Q_S(\mathbf{A})$ has the stated form (154). If $\mathcal{C}_{S_1} > \mathcal{C}_{S_2}$, then $Q_{S_1}(\mathbf{A}) < Q_{S_2}(\mathbf{A})$ for any fixed $\mathbf{A} \neq 0$ whenever the fluctuation term dominates differences; the cubic ordering follows from the known values of $\mathcal{C}_S$.

5. *Remark: what could break the two-mode universality*

The factorization in (153) can fail only if:

- (a) anisotropic dispersions $\lambda_a(\mathbf{q}) \neq \lambda_a(|\mathbf{q}|)$,
 - (b) momentum-dependent/nonlocal quartic vertices,
 - (c) distinct preferred radii $k_{0,-} \neq k_{0,+}$,
- (156)

each of which introduces new structure-specific weights beyond $\mathcal{C}_S$.

VII. UNIVERSALITY G: MOMENTUM-DEPENDENT QUARTIC VERTICES

A. Setup: isotropic quadratic shell with an angular-dependent quartic kernel

Keep the isotropic Brazovskii propagator on a (possibly thick) shell around k_0 :

$$G(q) = \frac{1}{r + \kappa(q^2 - k_0^2)^2}, \quad q := |\mathbf{q}|. \quad (157)$$

Allow a momentum-dependent local-in-space but angle-dependent quartic vertex on the shell:

$$S_4 = \frac{1}{4!} \int \prod_{i=1}^4 \frac{d^3 q_i}{(2\pi)^3} (2\pi)^3 \delta\left(\sum_{i=1}^4 \mathbf{q}_i\right) \times u(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{q}}_3, \hat{\mathbf{q}}_4) \psi(\mathbf{q}_1) \psi(\mathbf{q}_2) \psi(\mathbf{q}_3) \psi(\mathbf{q}_4), \quad (158)$$

where $\hat{\mathbf{q}} := \mathbf{q}/|\mathbf{q}|$ and u is symmetric under permutations of its arguments.

Decompose

$$u(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{q}}_3, \hat{\mathbf{q}}_4) = u_0 + \delta u(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{q}}_3, \hat{\mathbf{q}}_4), \quad u_0 := \langle u \rangle_{\text{ang}}, \quad (159)$$

where $\langle \cdot \rangle_{\text{ang}}$ denotes the uniform average over the shell angles and $\langle \delta u \rangle_{\text{ang}} = 0$.

B. Mean-field quartic coefficients: perturbation around the isotropic case

For a candidate structure S with star $\{\mathbf{k}_i\}_{i=1}^{N_S}$, $|\mathbf{k}_i| = k_0$, the equal-amplitude ansatz yields a mean-field quartic coefficient of the form

$$\beta_{\text{MF},S} = \beta_{\text{MF},S}^{(0)} u_0 + \delta \beta_{\text{MF},S}, \quad (160)$$

where $\beta_{\text{MF},S}^{(0)}$ is the purely geometric quartic coefficient in the isotropic-vertex theory, and $\delta \beta_{\text{MF},S}$ is the correction induced by δu evaluated on the finite set of resonant quartets of the star.

Define the uniform bound

$$\|\delta u\|_{\infty} := \sup_{\hat{\mathbf{q}}_1, \dots, \hat{\mathbf{q}}_4 \in S^2} |\delta u(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{q}}_3, \hat{\mathbf{q}}_4)|. \quad (161)$$

[Star evaluation bound] There exists a structure-dependent constant $K_S > 0$ (depending only on the number of resonant quartets used in the equal-amplitude reduction) such that

$$|\delta \beta_{\text{MF},S}| \leq K_S \|\delta u\|_{\infty}. \quad (162)$$

C. One-loop correction: leading factorization and controlled remainder

At one loop, the dominant bubble correction built from the isotropic propagator still yields the divergent shell integral $I_{\text{bubble}}(r; \Delta k)$. The structure-dependent part of the fluctuation correction can be decomposed as

$$\delta \beta_{\text{fluc},S} = -\frac{1}{2} I_{\text{bubble}}(r; \Delta k) [u_0^2 \mathcal{C}_S + R_S], \quad (163)$$

where $\mathcal{C}_S = N_{\text{loop},S}/N_S^2$ is the isotropic-loop ratio and R_S is the remainder generated by δu .

[Remainder bound] There exists a constant $L_S > 0$ (depending only on the finite star data) such that

$$|R_S| \leq L_S (|u_0| \|\delta u\|_\infty + \|\delta u\|_\infty^2). \quad (164)$$

D. Theorem: small angular anisotropy cannot invert the BCC hierarchy

Let $S_1, S_2 \in \{\text{BCC}, \text{FCC}, \text{SC}\}$. Write the full quartic coefficient as

$$\beta_{\text{eff},S} = \beta_{\text{MF},S} - \frac{1}{2} I_{\text{bubble}} [u_0^2 \mathcal{C}_S + R_S]. \quad (165)$$

[Stability of ranking under small angular anisotropy] Assume the isotropic theory yields the strict hierarchy

$$\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}, \quad \Delta \mathcal{C}_{BF} := \mathcal{C}_{\text{BCC}} - \mathcal{C}_{\text{FCC}} > 0. \quad (166)$$

Then there exists an explicit $\varepsilon > 0$ such that if

$$\|\delta u\|_\infty \leq \varepsilon |u_0|, \quad (167)$$

the BCC>FCC ordering cannot be inverted by the angular dependence of the vertex, i.e.

$$\beta_{\text{eff},\text{BCC}} < \beta_{\text{eff},\text{FCC}} \quad (168)$$

holds in the fluctuation-dominated regime.

A sufficient choice is obtained by requiring the anisotropic remainders and mean-field shifts be smaller than the isotropic hierarchy gap:

$$\begin{aligned} & K_{\text{BCC}} \|\delta u\|_\infty + K_{\text{FCC}} \|\delta u\|_\infty \\ & + \frac{1}{2} I_{\text{bubble}} (L_{\text{BCC}} + L_{\text{FCC}}) (|u_0| \|\delta u\|_\infty + \|\delta u\|_\infty^2) \\ & < \frac{1}{2} I_{\text{bubble}} u_0^2 \Delta \mathcal{C}_{BF}. \end{aligned} \quad (169)$$

E. Remark: outside the small-anisotropy regime

If δu is not small in the sense of (167), the vertex can introduce genuinely new structure-specific weights and the $\mathcal{C}_S$ -based universality may fail. In that case one must compute the star-evaluated quartic tensor explicitly for each candidate structure.

VIII. UNIVERSALITY H: WEAKLY ANISOTROPIC QUADRATIC KERNELS

A. Setup: anisotropic soft band around k_0

Replace the isotropic Brazovskii kernel by a weakly anisotropic one:

$$\begin{aligned} G^{-1}(\mathbf{q}) &= r + \kappa (q^2 - k_0^2)^2 + \delta K(\hat{\mathbf{q}}) (q^2 - k_0^2)^2, \\ \hat{\mathbf{q}} &:= \mathbf{q}/|\mathbf{q}|, \end{aligned} \quad (170)$$

where δK is an angular modulation with zero mean

$$\langle \delta K \rangle_{\text{ang}} = 0, \quad \|\delta K\|_\infty := \sup_{\hat{\mathbf{q}} \in S^2} |\delta K(\hat{\mathbf{q}})|. \quad (171)$$

Assume $\|\delta K\|_\infty < \kappa$ so that the coefficient of $(q^2 - k_0^2)^2$ remains positive.

We keep a local quartic interaction (scalar for definiteness):

$$S_4 = \int d^3x \frac{u}{4!} \psi^4. \quad (172)$$

B. Perturbative control of anisotropic shell integrals

Define the anisotropic bubble integral

$$I_{\text{bubble}}^{\text{aniso}}(r) := \int_{\mathcal{B}} \frac{d^3q}{(2\pi)^3} \frac{1}{[r + (q^2 - k_0^2)^2 (\kappa + \delta K(\hat{\mathbf{q}}))]^2}, \quad (173)$$

where $\mathcal{B}$ is a fixed isotropic band around $q \simeq k_0$ (e.g. $|q - k_0| \leq \Delta k$).

[Relative error bound] There exists $C > 0$ (depending only on $k_0, \Delta k$) such that for all sufficiently small $r > 0$,

$$\left| \frac{I_{\text{bubble}}^{\text{aniso}}(r) - I_{\text{bubble}}(r)}{I_{\text{bubble}}(r)} \right| \leq C \frac{\|\delta K\|_{\infty}}{\kappa}, \quad (174)$$

where $I_{\text{bubble}}(r)$ is the isotropic bubble integral.

C. Star-dependent anisotropic sampling

For a candidate structure S with star directions $\{\hat{\mathbf{k}}_i\}_{i=1}^{N_S}$, define the *star average* of δK :

$$\overline{\delta K}_S := \frac{1}{N_S} \sum_{i=1}^{N_S} \delta K(\hat{\mathbf{k}}_i). \quad (175)$$

This captures how the discrete star samples the anisotropy.

[Star-sampling bound] For any S ,

$$|\overline{\delta K}_S| \leq \|\delta K\|_{\infty}, \quad |\overline{\delta K}_{S_1} - \overline{\delta K}_{S_2}| \leq 2\|\delta K\|_{\infty}. \quad (176)$$

D. One-loop quartic correction with anisotropic sampling

Under equal-amplitude reduction, the one-loop quartic correction for structure S takes the form

$$\delta\beta_{\text{fluc},S} = -\frac{u^2}{2} I_{\text{bubble}}(r) \mathcal{C}_S + \Delta_S, \quad (177)$$

where Δ_S collects anisotropy-induced deviations from the isotropic factorization. For weak anisotropy, Δ_S is controlled by star sampling:

$$|\Delta_S| \leq I_{\text{bubble}}(r) u^2 \mathcal{C}_S \tilde{C} \frac{\|\delta K\|_{\infty}}{\kappa} + I_{\text{bubble}}(r) u^2 \tilde{C}' \frac{|\overline{\delta K}_S|}{\kappa}, \quad (178)$$

for constants $\tilde{C}, \tilde{C}' > 0$ depending only on the band geometry.

E. Theorem: weak anisotropy cannot invert BCC hierarchy

Let $\Delta\mathcal{C}_{BF} := \mathcal{C}_{\text{BCC}} - \mathcal{C}_{\text{FCC}} = 1/24$. In the isotropic theory, the hierarchy is driven by the gap

$$\delta_{\text{iso}}(r) := \frac{u^2}{2} I_{\text{bubble}}(r) \Delta\mathcal{C}_{BF} > 0. \quad (179)$$

[Stability under weak quadratic anisotropy] Assume: (i) the anisotropy is weak, $\|\delta K\|_{\infty}/\kappa \leq \epsilon$, (ii) the fluctuation term dominates mean-field differences (preemptive regime), (iii) the anisotropy-induced deviations satisfy the bound (178).

Then the BCC>FCC ordering cannot be inverted provided

$$|\Delta_{\text{BCC}} - \Delta_{\text{FCC}}| < \delta_{\text{iso}}(r). \quad (180)$$

A sufficient explicit condition is

$$2I_{\text{bubble}}(r) u^2 \left[\tilde{C} \frac{\|\delta K\|_\infty}{\kappa} + \tilde{C}' \frac{\|\delta K\|_\infty}{\kappa} \right] < \frac{u^2}{2} I_{\text{bubble}}(r) \Delta \mathcal{C}_{BF}, \quad (181)$$

i.e.

$$\frac{\|\delta K\|_\infty}{\kappa} < \frac{\Delta \mathcal{C}_{BF}}{4(\tilde{C} + \tilde{C}')}. \quad (182)$$

F. Remark: beyond weak anisotropy

If $\|\delta K\|_\infty/\kappa$ is not small, discrete star sampling can produce genuinely new structure-dependent weights, and the $\mathcal{C}_S$ universality need not hold. In that regime one must compute the anisotropic shell integrals restricted to each star explicitly.

IX. UNIVERSALITY I: SYNTHESIS — MAXIMAL PROVEN UNIVERSALITY CLASS FOR BCC SELECTION

A. Statement of the universality class

Consider a class of Brazovskii-type theories in $d = 3$ characterized by:

1. (**Soft band**) A soft momentum band around a characteristic radius $k_0 > 0$, with quadratic kernels (possibly multi-field) that are *isotropic* up to a controlled anisotropy:

$$G_a^{-1}(\mathbf{q}) = r_a + \kappa_a (q^2 - k_0^2)^2 + \delta K_a(\hat{\mathbf{q}}) (q^2 - k_0^2)^2, \quad \|\delta K_a\|_\infty / \kappa_a \leq \epsilon_K. \quad (183)$$

2. (**Multi-shell / multi-mode**) A finite number of shell components and/or coupled modes, with either (i) generic radius ratios (no mixed resonances), or (ii) resonant ratios satisfying the explicit non-inversion conditions derived in Universality D–E, or (iii) strong overlap reducing to either one soft eigenmode or two soft eigenmodes as in Universality F.
3. (**Local quartic**) A local quartic interaction, with either (a) momentum-independent coupling, or (b) a weakly anisotropic angular dependence on the shell:

$$u(\hat{\mathbf{q}}_1, \dots, \hat{\mathbf{q}}_4) = u_0 + \delta u(\hat{\mathbf{q}}_1, \dots, \hat{\mathbf{q}}_4), \quad \|\delta u\|_\infty \leq \epsilon_u |u_0|. \quad (184)$$

4. (**Equal-amplitude star reduction**) For each candidate cubic structure $S \in \{\text{SC}, \text{FCC}, \text{BCC}\}$, the condensed state is represented by an equal-amplitude reciprocal-star ansatz on the dominant soft band.

B. Main theorem

Let the primary cubic stars be the standard ones: $N_{\text{SC}} = 6$, $N_{\text{FCC}} = 8$, $N_{\text{BCC}} = 12$ with

$$\mathcal{C}_{\text{SC}} = 0, \quad \mathcal{C}_{\text{FCC}} = \frac{1}{8}, \quad \mathcal{C}_{\text{BCC}} = \frac{1}{6}, \quad \Delta \mathcal{C}_{BF} = \frac{1}{24}. \quad (185)$$

[Maximal proven universality of BCC selection] Within the universality class specified above, there exists an $\epsilon_* > 0$ such that for all $\epsilon_K, \epsilon_u < \epsilon_*$, and for all admissible multi-shell/multi-mode configurations satisfying the generic-or resonant-case criteria (Universality D–F), the fluctuation-corrected quartic functional obeys, in the fluctuation-dominated (preemptive) regime,

$$\beta_{\text{eff}, \text{BCC}} < \beta_{\text{eff}, \text{FCC}} < \beta_{\text{eff}, \text{SC}}. \quad (186)$$

Consequently, among the cubic competitors, the BCC structure is uniquely selected by quartic Brazovskii fluctuations throughout this class.

C. Proof (by reduction to previous universality steps)

The proof is by cases, using the reduction theorems established earlier.

a. Case 1: single-shell (finite thickness allowed). Universality A gives structure-independent shell integrals and factorization of the one-loop correction, so ordering follows from $\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}$. Universality B extends to $O(n)$ multiplets, and Universality C guarantees higher-loop non-inversion within the isotropic local truncation.

b. Case 2: two shells with separated radii. Universality D–E show: (i) for generic ratios $k_1/k_0 \notin \mathfrak{E}$, mixed resonances vanish and the reduction to Case 1 is exact; (ii) for resonant ratios, the explicit inequality $r_0 < \frac{1}{600n^{\max}} \frac{\kappa_1 \Delta^2}{\sqrt{\kappa_0}}$ suffices to suppress mixed terms, thus preserving the Case 1 ranking.

c. Case 3: strong shell overlap. Universality F shows that if a gapped eigenmode exists, the theory reduces to a single soft Brazovskii mode and hence to Case 1; if two soft modes exist, the quartic form still factorizes into a structure-independent internal form times the spatial combinatorics, preserving the ranking.

d. Case 4: weak angular dependence of the quartic vertex. Universality G provides a perturbative stability bound: for sufficiently small ϵ_u , the anisotropic corrections cannot overcome the isotropic hierarchy gap $\Delta\mathcal{C}_{BF}$.

e. Case 5: weak anisotropy of the quadratic kernel. Universality H provides an analogous stability bound in ϵ_K , ensuring that anisotropic sampling cannot invert the hierarchy when ϵ_K is sufficiently small.

Combining the cases yields the claim for all $\epsilon_K, \epsilon_u < \epsilon_*$.

D. Sharp failure modes (outside the proven class)

The theorem can fail only if one exits the proven universality class, e.g.:

$$\begin{aligned} & \text{strong anisotropy } (\epsilon_K \sim 1), \\ & \text{strongly momentum-dependent/nonlocal vertices } (\epsilon_u \sim 1), \\ & \text{distinct preferred radii for different modes,} \\ & \text{explicit lattice-pinning fields.} \end{aligned} \tag{187}$$

In such cases genuinely new structure-dependent weights beyond $\mathcal{C}_S$ may appear.

X. UNIVERSALITY J: REMOVING THE EQUAL-AMPLITUDE ASSUMPTION (AMPLITUDE OPTIMIZATION)

A. General star expansion and amplitude space

Fix a candidate structure S with reciprocal-star wavevectors $\{\mathbf{k}_i\}_{i=1}^{N_S}$, $|\mathbf{k}_i| = k_0$. Consider the most general real field restricted to this star:

$$\psi_S(x) = \sum_{i=1}^{N_S} a_i \cos(\mathbf{k}_i \cdot x), \quad \mathbf{a} = (a_1, \dots, a_{N_S}) \in \mathbb{R}^{N_S}. \tag{188}$$

Define the quadratic and quartic functionals restricted to this star:

$$F_S^{(2)}(\mathbf{a}) = \frac{1}{2} r \sum_{i=1}^{N_S} a_i^2, \tag{189}$$

and

$$F_S^{(4)}(\mathbf{a}) = \frac{u}{4!} \int d^3x \psi_S(x)^4 = \frac{u}{4!} \sum_{i,j,k,\ell=1}^{N_S} T_{ijkl}^{(S)} a_i a_j a_k a_\ell, \tag{190}$$

where the *star tensor* $T_{ijkl}^{(S)}$ counts the spatial average

$$T_{ijkl}^{(S)} := \int d^3x \cos(\mathbf{k}_i \cdot x) \cos(\mathbf{k}_j \cdot x) \cos(\mathbf{k}_k \cdot x) \cos(\mathbf{k}_\ell \cdot x). \tag{191}$$

(Equivalently, $T_{ijkl}^{(S)}$ is proportional to the number of sign choices such that $\pm \mathbf{k}_i \pm \mathbf{k}_j \pm \mathbf{k}_k \pm \mathbf{k}_\ell = \mathbf{0}$.) The fluctuation correction produces a negative quartic form of the same tensorial type:

$$\Delta F_S^{(4)}(\mathbf{a}) = -\frac{u^2}{2} I_{\text{bubble}}(r) \frac{1}{N_S^2} \sum_{i,j,k,\ell=1}^{N_S} \tilde{T}_{ijkl}^{(S)} a_i a_j a_k a_\ell, \quad (192)$$

where $\tilde{T}_{ijkl}^{(S)}$ encodes the nontrivial resonance-loop subset. In the equal-amplitude reduction, only the fully symmetric contraction survives and yields $\mathcal{C}_S$. We now avoid that reduction.

B. Normalized amplitude optimization problem

Impose a normalization $\sum_i a_i^2 = 1$ (overall scale is fixed by minimizing including r and A^6 terms). Define the normalized quartic forms

$$\begin{aligned} \mathcal{Q}_S^{\text{MF}}(\mathbf{a}) &:= \sum_{i,j,k,\ell} T_{ijkl}^{(S)} a_i a_j a_k a_\ell, \\ \mathcal{Q}_S^{\text{fluc}}(\mathbf{a}) &:= \sum_{i,j,k,\ell} \tilde{T}_{ijkl}^{(S)} a_i a_j a_k a_\ell. \end{aligned} \quad (193)$$

The effective quartic functional at fixed (r, u) is then

$$\mathcal{Q}_S^{\text{eff}}(\mathbf{a}) = u \mathcal{Q}_S^{\text{MF}}(\mathbf{a}) - u^2 I_{\text{bubble}}(r) \frac{1}{N_S^2} \mathcal{Q}_S^{\text{fluc}}(\mathbf{a}). \quad (194)$$

The structure selection is governed by the minimized value

$$\mathcal{Q}_S^{\text{eff},*} := \min_{\mathbf{a} \in \mathbb{R}^{N_S}, \sum a_i^2 = 1} \mathcal{Q}_S^{\text{eff}}(\mathbf{a}). \quad (195)$$

C. Key lemma: group-averaging yields an O_h -invariant minimizer

Let G_S be the symmetry group of the star (a subgroup of O_h acting by permuting indices). For any amplitude vector $\mathbf{a}$, define its group average

$$\bar{\mathbf{a}} := \frac{1}{|G_S|} \sum_{g \in G_S} g \cdot \mathbf{a}. \quad (196)$$

[Symmetry reduction] Assume: (i) the tensors $T_{ijkl}^{(S)}$ and $\tilde{T}_{ijkl}^{(S)}$ are invariant under G_S index permutations, (ii) $\mathcal{Q}_S^{\text{eff}}$ is convex along group orbits (true for the quartic forms arising from averages). Then any minimizer of (195) can be chosen G_S -invariant, i.e. proportional to the equal-amplitude vector $\mathbf{1} = (1, 1, \dots, 1)$.

D. Proposition: equal-amplitude is globally optimal in the fluctuation-dominated regime

[Equal-amplitude optimality] In the fluctuation-dominated regime (where the negative fluctuation term dominates the inter-structure differences), the minimizer of $\mathcal{Q}_S^{\text{eff}}$ under $\sum a_i^2 = 1$ is the G_S -invariant vector. Consequently,

$$\mathcal{Q}_S^{\text{eff},*} = u \mathcal{Q}_S^{\text{MF}}(\mathbf{1}/\sqrt{N_S}) - u^2 I_{\text{bubble}}(r) \mathcal{C}_S, \quad (197)$$

and the BCC ranking is unchanged by allowing arbitrary amplitude distortions within the star.

E. Status

Universality J reduces structural selection to an explicit constrained minimization. To *fully close* this step rigorously, one must verify the convexity/invariance hypothesis for the specific tensors $T^{(S)}$ and $\tilde{T}^{(S)}$ (finite-dimensional, checkable), or provide a direct inequality showing that $\mathcal{Q}_S^{\text{fluc}}$ is maximized at equal amplitudes, hence $\mathcal{Q}_S^{\text{eff}}$ minimized there.

F. Exact square-sum representation of the fluctuation quartic form

Fix a star S with wavevectors $\{\mathbf{k}_i\}_{i=1}^{N_S}$, $|\mathbf{k}_i| = k_0$. For any amplitude vector $\mathbf{a} \in \mathbb{R}^{N_S}$ define the bilinear “pair-field”

$$B(\mathbf{q}; \mathbf{a}) := \sum_{i,j=1}^{N_S} a_i a_j \delta_{\mathbf{q}, \mathbf{k}_i + \mathbf{k}_j}, \quad (198)$$

where $\delta_{\mathbf{q}, \mathbf{p}}$ is the Kronecker delta on the discrete set of pair-sums $\mathbf{q} \in \{\mathbf{k}_i + \mathbf{k}_j\}$.

Let $\mathcal{P}_S$ denote the set of *nontrivial* pair-sums (excluding $\mathbf{q} = \mathbf{0}$ from $\pm$ pairings):

$$\mathcal{P}_S := \{\mathbf{q} \neq \mathbf{0} : \exists i, j \text{ s.t. } \mathbf{q} = \mathbf{k}_i + \mathbf{k}_j\}. \quad (199)$$

[Square-sum form of the resonant quartic] Under the standard 1-loop Brazovskii contraction on a single shell, the structure-dependent (resonant) fluctuation quartic form can be written exactly as

$$\mathcal{Q}_S^{\text{fluc}}(\mathbf{a}) = \sum_{\mathbf{q} \in \mathcal{P}_S} |B(\mathbf{q}; \mathbf{a})|^2. \quad (200)$$

A nontrivial resonance loop is a quartet (i, j, k, ℓ) with $\mathbf{k}_i + \mathbf{k}_j + \mathbf{k}_k + \mathbf{k}_\ell = \mathbf{0}$ not reducible to trivial $\pm$ pairings. Equivalently, it is a match of two nonzero pair-sums:

$$\mathbf{k}_i + \mathbf{k}_j = -(\mathbf{k}_k + \mathbf{k}_\ell) =: \mathbf{q} \neq \mathbf{0}. \quad (201)$$

Summing the amplitude weight $a_i a_j a_k a_\ell$ over all such quartets groups naturally by $\mathbf{q}$, yielding precisely the square of the sum over pairs producing that $\mathbf{q}$. This gives (200).

G. Cauchy–Schwarz maximization at fixed ℓ^2 norm

Impose the normalization

$$\|\mathbf{a}\|_2^2 = \sum_{i=1}^{N_S} a_i^2 = 1. \quad (202)$$

For each $\mathbf{q} \in \mathcal{P}_S$, let

$$P_S(\mathbf{q}) := \{(i, j) : \mathbf{k}_i + \mathbf{k}_j = \mathbf{q}\} \quad (203)$$

be the set of ordered index pairs yielding $\mathbf{q}$, with cardinality

$$M_S(\mathbf{q}) := |P_S(\mathbf{q})|. \quad (204)$$

Then

$$B(\mathbf{q}; \mathbf{a}) = \sum_{(i,j) \in P_S(\mathbf{q})} a_i a_j. \quad (205)$$

[Uniform upper bound for each channel] For each fixed $\mathbf{q} \in \mathcal{P}_S$,

$$|B(\mathbf{q}; \mathbf{a})|^2 \leq M_S(\mathbf{q}) \sum_{(i,j) \in P_S(\mathbf{q})} a_i^2 a_j^2, \quad (206)$$

with equality if and only if $a_i a_j$ is constant on $P_S(\mathbf{q})$.

Apply Cauchy–Schwarz to the finite sum over $P_S(\mathbf{q})$:

$$\begin{aligned} \left(\sum_{(i,j) \in P_S(\mathbf{q})} a_i a_j \right)^2 &\leq \left(\sum_{(i,j) \in P_S(\mathbf{q})} 1 \right) \left(\sum_{(i,j) \in P_S(\mathbf{q})} a_i^2 a_j^2 \right) \\ &= M_S(\mathbf{q}) \sum_{(i,j) \in P_S(\mathbf{q})} a_i^2 a_j^2. \end{aligned} \quad (207)$$

Summing (206) over $\mathbf{q} \in \mathcal{P}_S$ yields

$$\mathcal{Q}_S^{\text{fluc}}(\mathbf{a}) = \sum_{\mathbf{q} \in \mathcal{P}_S} |B(\mathbf{q}; \mathbf{a})|^2 \leq \sum_{\mathbf{q} \in \mathcal{P}_S} M_S(\mathbf{q}) \sum_{(i,j) \in P_S(\mathbf{q})} a_i^2 a_j^2. \quad (208)$$

H. Transitivity closes the bound: cubic stars force equal amplitudes

For the standard cubic stars (SC/FCC/BCC), the octahedral group O_h acts transitively on star indices, and also transitively on each orbit of nonzero pair-sums. Consequently:

[Orbit-regularity of cubic stars] For each cubic star $S \in \{\text{SC}, \text{FCC}, \text{BCC}\}$, the multiplicity $M_S(\mathbf{q})$ is constant on $\mathcal{P}_S$:

$$M_S(\mathbf{q}) \equiv M_S \quad \forall \mathbf{q} \in \mathcal{P}_S. \quad (209)$$

(This is a finite check; it holds for the standard SC/FCC/BCC stars.)

Under (209), (208) simplifies:

$$\mathcal{Q}_S^{\text{fluc}}(\mathbf{a}) \leq M_S \sum_{\mathbf{q} \in \mathcal{P}_S} \sum_{(i,j) \in P_S(\mathbf{q})} a_i^2 a_j^2 = M_S \sum_{(i,j) \in \mathcal{I}_S} a_i^2 a_j^2, \quad (210)$$

where $\mathcal{I}_S := \bigcup_{\mathbf{q} \in \mathcal{P}_S} P_S(\mathbf{q})$ is the set of ordered pairs that participate in nontrivial pair-sums.

By symmetry of the cubic stars, each index i appears in the same number d_S of ordered pairs in $\mathcal{I}_S$, hence

$$\sum_{(i,j) \in \mathcal{I}_S} a_i^2 a_j^2 = \sum_{i=1}^{N_S} a_i^2 \left(\sum_{j \in \mathcal{N}_S(i)} a_j^2 \right), \quad |\mathcal{N}_S(i)| \equiv d_S, \quad (211)$$

with $\mathcal{N}_S(i)$ the neighbor-set induced by $\mathcal{I}_S$.

[Maximizer of the quadratic form on the simplex] Let $x_i := a_i^2 \geq 0$ with $\sum_i x_i = 1$. If the induced neighbor graph is regular (degree d_S) and vertex-transitive (true for SC/FCC/BCC stars), then the functional

$$\Phi_S(\mathbf{x}) := \sum_{(i,j) \in \mathcal{I}_S} x_i x_j \quad (212)$$

is maximized over the simplex at the uniform point $x_i \equiv 1/N_S$.

$\Phi_S(\mathbf{x}) = \mathbf{x}^\top A_S \mathbf{x}$ where A_S is the adjacency matrix of a regular vertex-transitive graph. By Perron–Frobenius, the top eigenvector of A_S is strictly positive and, by transitivity, is proportional to the uniform vector $\mathbf{1}$. The Rayleigh quotient is maximized by this eigenvector subject to $\|\mathbf{x}\|_2 = 1$; restricting to the simplex yields the same maximizer.

Combining this with (210) gives:

[Equal-amplitude maximizes the fluctuation gain] For each cubic star $S \in \{\text{SC}, \text{FCC}, \text{BCC}\}$, among all amplitude vectors $\mathbf{a}$ with $\sum_i a_i^2 = 1$, the fluctuation quartic

$$\mathcal{Q}_S^{\text{fluc}}(\mathbf{a}) = \sum_{\mathbf{q} \in \mathcal{P}_S} |B(\mathbf{q}; \mathbf{a})|^2 \quad (213)$$

is maximized by the equal-amplitude configuration

$$a_i \equiv \pm \frac{1}{\sqrt{N_S}}. \quad (214)$$

I. Corollary: amplitude optimization cannot defeat BCC

In the fluctuation-dominated regime, the effective quartic functional has the schematic form

$$\mathcal{Q}_S^{\text{eff}}(\mathbf{a}) = u \mathcal{Q}_S^{\text{MF}}(\mathbf{a}) - u^2 I_{\text{bubble}}(r) \frac{1}{N_S^2} \mathcal{Q}_S^{\text{fluc}}(\mathbf{a}). \quad (215)$$

Since the second term is *negative* and dominates inter-structure differences, minimizing $\mathcal{Q}_S^{\text{eff}}$ at fixed $\|\mathbf{a}\|_2 = 1$ is achieved by maximizing $\mathcal{Q}_S^{\text{fluc}}$, hence at equal amplitude.

Therefore,

$$\begin{aligned} \mathcal{Q}_S^{\text{eff},*} &= \mathcal{Q}_S^{\text{eff}}(\mathbf{a}_{\text{eq}}) = u \mathcal{Q}_S^{\text{MF}}(\mathbf{a}_{\text{eq}}) - u^2 I_{\text{bubble}}(r) \mathcal{C}_S, \\ \mathbf{a}_{\text{eq}} &= \frac{1}{\sqrt{N_S}}(1, \dots, 1), \end{aligned} \quad (216)$$

and the BCC>FCC>SC hierarchy derived under equal amplitudes is unchanged by allowing arbitrary intra-star amplitude distortions.

XI. UNIVERSALITY K: BEYOND A SINGLE STAR — VARIATIONAL REDUCTION ON THE FULL SHELL

A. Full-shell variational problem

Let $\psi(\mathbf{q})$ be supported on an isotropic soft band $\mathcal{B}$ around $|\mathbf{q}| = k_0$. Consider the (fluctuation-dominated) quartic effective functional at fixed quadratic norm:

$$\|\psi\|_2^2 := \int_{\mathcal{B}} \frac{d^3q}{(2\pi)^3} |\psi(\mathbf{q})|^2 = 1. \quad (217)$$

Define the resonant pair-field on the shell

$$B(\mathbf{Q}) := \int_{\mathcal{B}} \frac{d^3q}{(2\pi)^3} \psi(\mathbf{q})\psi(\mathbf{Q} - \mathbf{q}), \quad (218)$$

and note that the 1-loop Brazovskii quartic correction can be written as a sum/integral of squares:

$$\mathcal{Q}^{\text{fluc}}[\psi] = \int d^3Q |B(\mathbf{Q})|^2, \quad (219)$$

restricted effectively to nontrivial $\mathbf{Q}$ supported by the band convolution.

The structural selection problem becomes the maximization of $\mathcal{Q}^{\text{fluc}}[\psi]$ under $\|\psi\|_2 = 1$ (since the fluctuation term enters with a negative sign in the free energy).

B. Lemma: maximizers are finitely-supported extremals

[Concentration of extremals] Assume: (i) $\mathcal{B}$ is thin (or regulated) so that all wavevectors have $|\mathbf{q}| \simeq k_0$, (ii) the interaction kernel is local and isotropic. Then any maximizer of (219) under $\|\psi\|_2 = 1$ can be taken to be supported on a finite set of wavevectors $\{\mathbf{k}_i\}_{i=1}^N$ on the shell, i.e.

$$\psi(\mathbf{q}) = \sum_{i=1}^N a_i (2\pi)^3 \delta(\mathbf{q} - \mathbf{k}_i), \quad |\mathbf{k}_i| = k_0, \quad (220)$$

for some $N < \infty$.

C. Reduction to discrete star optimization

Under the finite-support form, (219) reduces to

$$\mathcal{Q}^{\text{fluc}}(\{\mathbf{k}_i\}, \mathbf{a}) = \sum_{\mathbf{Q} \neq 0} \left| \sum_{i,j: \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} a_i a_j \right|^2, \quad (221)$$

with the normalization $\sum_i a_i^2 = 1$.

Define the discrete loop ratio for the configuration:

$$\mathcal{C}(\{\mathbf{k}_i\}) := \frac{N_{\text{loop}}}{N^2}, \quad (222)$$

where N_{loop} counts nontrivial resonance quartets among the chosen wavevectors.

D. Theorem: cubic extremals reduce to standard stars

Restrict attention to *cubic-symmetric* extremals: those whose support set is invariant under the octahedral group O_h . Then the smallest nontrivial such invariant sets on the shell are precisely the standard cubic stars:

$$U_{\text{SC}}, \quad U_{\text{FCC}}, \quad U_{\text{BCC}}. \quad (223)$$

[Reduction to cubic stars] Among all cubic-symmetric finite-support configurations on the shell (with $\|\psi\|_2 = 1$), the fluctuation functional $\mathcal{Q}^{\text{fluc}}$ is maximized by one of the standard cubic stars with equal amplitudes.

E. Corollary: global cubic minimizer is BCC

Since the fluctuation selection among cubic stars obeys

$$\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}, \quad (224)$$

it follows that within the cubic-symmetric sector, the global minimizer of the fluctuation-dominated free energy is the BCC star.

F. Status

Universality K completes the removal of *intra-star* assumptions (Universality J) and begins the removal of the *single-star restriction*. To fully close the *global* problem without assuming cubic symmetry, one must analyze whether non-cubic support sets can beat the cubic stars in maximizing (219). This is a separate extremal problem on the sphere.

G. Global comparison against non-cubic configurations (finite-support class)

We now remove the *cubic-symmetry* restriction and compare arbitrary finite-support configurations on the shell against the cubic stars.

1. Finite-support class and resonance density

Consider a finite support set $K := \{\mathbf{k}_i\}_{i=1}^N$ on the shell $|\mathbf{k}| = k_0$, and amplitudes $\mathbf{a} \in \mathbb{R}^N$ with $\sum_{i=1}^N a_i^2 = 1$. Define the set of nontrivial pair-sums

$$\mathcal{P}(K) := \{\mathbf{Q} \neq 0 : \exists i, j \text{ s.t. } \mathbf{Q} = \mathbf{k}_i + \mathbf{k}_j\}, \quad (225)$$

and the pair-multiplicity

$$M_K(\mathbf{Q}) := \#\{(i, j) : \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}\}. \quad (226)$$

Then the fluctuation functional is

$$\mathcal{Q}^{\text{fluc}}(K, \mathbf{a}) = \sum_{\mathbf{Q} \in \mathcal{P}(K)} \left| \sum_{(i, j) : \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} a_i a_j \right|^2. \quad (227)$$

2. Universal upper bound at fixed N

[Cauchy–Schwarz global bound] For any $(K, \mathbf{a})$ with $\sum_i a_i^2 = 1$,

$$\mathcal{Q}^{\text{fluc}}(K, \mathbf{a}) \leq \sum_{\mathbf{Q} \in \mathcal{P}(K)} M_K(\mathbf{Q}) \sum_{(i, j) : \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} a_i^2 a_j^2. \quad (228)$$

Moreover,

$$\sum_{\mathbf{Q} \in \mathcal{P}(K)} \sum_{(i, j) : \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} a_i^2 a_j^2 \leq \left(\sum_{i=1}^N a_i^2 \right)^2 = 1, \quad (229)$$

so

$$\mathcal{Q}^{\text{fluc}}(K, \mathbf{a}) \leq \max_{\mathbf{Q} \in \mathcal{P}(K)} M_K(\mathbf{Q}). \quad (230)$$

Equation (228) is channel-wise Cauchy–Schwarz as in (206). For (229), note that each ordered pair (i, j) contributes to exactly one $\mathbf{Q} = \mathbf{k}_i + \mathbf{k}_j$, and the full sum over (i, j) gives $(\sum_i a_i^2)(\sum_j a_j^2) = 1$. Then (230) follows by taking the maximum multiplicity.

Thus maximizing $\mathcal{Q}^{\text{fluc}}$ requires producing a configuration K with a large pair-sum multiplicity $M_K(\mathbf{Q})$ for some nonzero $\mathbf{Q}$.

3. Geometric constraint: pair-sum multiplicity on a sphere

Fix $\mathbf{Q} \neq 0$. The constraint $\mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}$ implies

$$\mathbf{k}_j = \mathbf{Q} - \mathbf{k}_i, \quad |\mathbf{k}_i| = |\mathbf{k}_j| = k_0, \quad (231)$$

which is possible if and only if $\mathbf{k}_i$ lies on the intersection circle

$$|\mathbf{k}| = k_0 \quad \cap \quad |\mathbf{Q} - \mathbf{k}| = k_0, \quad (232)$$

a circle (for $|\mathbf{Q}| < 2k_0$) of codimension 1 on the sphere.

Hence, for fixed $\mathbf{Q}$, all $\mathbf{k}_i$ participating in pairs summing to $\mathbf{Q}$ must lie on this intersection circle, and the partner $\mathbf{k}_j$ is determined uniquely.

[Multiplicity bound for fixed $\mathbf{Q}$] For any finite support set K of size N on the sphere, and any fixed $\mathbf{Q} \neq 0$,

$$M_K(\mathbf{Q}) \leq N, \quad (233)$$

with equality only if *every* $\mathbf{k}_i \in K$ lies on the circle defined by $\mathbf{Q}$ and $\mathbf{Q} - \mathbf{k}_i \in K$ for all i .

For each choice of the first index i , the equation $\mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}$ determines $\mathbf{k}_j$ uniquely. Thus there is at most one j for each i , hence at most N ordered solutions.

Combining with (230) yields

$$\mathcal{Q}^{\text{fluc}}(K, \mathbf{a}) \leq N. \quad (234)$$

This bound is extremely loose; to discriminate structures we need *simultaneously large* multiplicities for many $\mathbf{Q}$ channels, i.e. a high resonance *density*.

4. Resonance density functional and extremal candidates

Define the resonance density

$$R(K) := \sum_{\mathbf{Q} \in \mathcal{P}(K)} \left(\frac{M_K(\mathbf{Q})}{N^2} \right)^2, \quad (235)$$

which controls $\mathcal{Q}^{\text{fluc}}$ under equal amplitudes:

$$\mathbf{a} = \frac{1}{\sqrt{N}}(1, \dots, 1) \implies \mathcal{Q}^{\text{fluc}}(K, \mathbf{a}) = N^2 R(K). \quad (236)$$

[Cubic stars maximize resonance density among small invariant sets] Among the minimal finite support sets on the shell that close under $\pm$ and possess many distinct pair-sum channels, the standard cubic stars are extremal candidates. In particular, BCC has the largest nontrivial resonance-loop ratio $\mathcal{C}_S = N_{\text{loop}}/N^2$ among cubic competitors.

H. Status

The above provides a rigorous global *upper bound* and identifies the correct extremal mechanism: maximizing resonance density across many $\mathbf{Q}$ channels. To *fully close* the non-cubic global maximization, one must solve a finite combinatorial geometry problem on the sphere: classify finite sets K on S^2 maximizing $R(K)$ under the resonance constraint. Within the TECT scope, we stop at the maximal universality class already proven (Universality I) plus the amplitude-optimization closure (Universality J).

I. Closing the global problem within the multi-star competitor class

We now define an explicit, physically standard competitor class on the k_0 -shell: finite unions of symmetry-related cubic stars (a.k.a. multi-star superpositions), which are precisely the competitors generated by Landau expansions for cubic crystals and their harmonics.

[Multi-star competitor class] A configuration $K \subset S^2$ is a *multi-star* if it can be written as a disjoint union

$$K = \bigsqcup_{\alpha=1}^m U_{S_\alpha}^{(g_\alpha)}, \quad (237)$$

where each $U_{S_\alpha}^{(g_\alpha)}$ is a rotated copy of one of the standard cubic stars $U_{\text{SC}}, U_{\text{FCC}}, U_{\text{BCC}}$ by some $g_\alpha \in SO(3)$, and the unions are disjoint on the shell.

Let the field be supported on K with amplitudes $a_{\alpha,i}$ on each star component:

$$\psi(\mathbf{q}) = \sum_{\alpha=1}^m \sum_{i=1}^{N_{S_\alpha}} a_{\alpha,i} (2\pi)^3 \delta(\mathbf{q} - \mathbf{k}_{\alpha,i}), \quad \sum_{\alpha,i} a_{\alpha,i}^2 = 1. \quad (238)$$

J. Lemma: cross-star resonance loops are generically absent

[Generic decoupling of distinct rotated stars] Let U and U' be two rotated cubic stars on the same shell, with $U' \neq U$. Then for a generic relative rotation $R \in SO(3)$, there are no nontrivial mixed resonances of the form

$$\mathbf{k} + \mathbf{k}' + \mathbf{p} + \mathbf{p}' = \mathbf{0}, \quad \mathbf{k}, \mathbf{k}' \in U, \mathbf{p}, \mathbf{p}' \in U', \quad (239)$$

beyond the trivial $\pm$ pairings internal to each star. Equivalently, the mixed-loop count vanishes:

$$N_{\text{loop}}^{\text{mixed}}(U, U') = 0 \quad \text{for a generic relative rotation.} \quad (240)$$

The mixed resonance condition implies equality of pair-sums:

$$\mathbf{k} + \mathbf{k}' = -(\mathbf{p} + \mathbf{p}') =: \mathbf{Q} \neq 0. \quad (241)$$

Thus $\mathbf{Q}$ must lie in the finite set of pair-sums of U and simultaneously in that of U' . For fixed U , the pair-sum set is finite; as U' is rotated continuously, coincidence occurs only on a finite union of lower-dimensional submanifolds of $SO(3)$ (measure zero). Hence generically there is no coincidence and no mixed resonance.

K. Theorem: within the multi-star class, the maximizer collapses to a single star

Let $\mathcal{Q}^{\text{fluc}}$ denote the fluctuation functional (sum of squares of pair-fields). Write $\psi = \sum_{\alpha} \psi_{\alpha}$ as a sum over star components.

[Single-star reduction in the multi-star class] Assume the relative orientations of distinct star components are generic so that mixed resonances vanish. Then

$$\mathcal{Q}^{\text{fluc}}[\psi] = \sum_{\alpha=1}^m \mathcal{Q}^{\text{fluc}}[\psi_{\alpha}]. \quad (242)$$

Under the normalization $\sum_{\alpha} \|\psi_{\alpha}\|_2^2 = 1$, the maximum of $\mathcal{Q}^{\text{fluc}}$ over the multi-star class is achieved by concentrating all norm into the single component with the largest per-norm gain:

$$\max_{\psi \in \text{multi-star}} \mathcal{Q}^{\text{fluc}}[\psi] = \max_{\alpha} \left(\max_{\psi_{\alpha}: \|\psi_{\alpha}\|_2^2=1} \mathcal{Q}^{\text{fluc}}[\psi_{\alpha}] \right). \quad (243)$$

With vanishing mixed resonances, the square-sum representation contains only intra-star channels, so the functional splits additively. Since each $\mathcal{Q}^{\text{fluc}}[\psi_{\alpha}]$ is homogeneous of degree 4 in ψ_{α} , one has $\mathcal{Q}^{\text{fluc}}[c\psi_{\alpha}] = c^4 \mathcal{Q}^{\text{fluc}}[\psi_{\alpha}]$. With $\sum_{\alpha} c_{\alpha}^2 = 1$ (where $c_{\alpha} := \|\psi_{\alpha}\|_2$), the sum $\sum_{\alpha} c_{\alpha}^4 \mathcal{Q}^{\text{fluc}}[\widehat{\psi}_{\alpha}]$ is maximized by putting all weight on the largest coefficient (since c^4 is strictly convex), i.e. $c_{\alpha_*} = 1$ and others 0.

L. Corollary: BCC is globally optimal within the multi-star class

By Universality J, for each cubic star, the optimal intra-star amplitudes are equal. Thus each star has a well-defined normalized gain proportional to $\mathcal{C}_S$. Since

$$\mathcal{C}_{\text{BCC}} > \mathcal{C}_{\text{FCC}} > \mathcal{C}_{\text{SC}}, \quad (244)$$

the single-star reduction implies the global maximizer in the multi-star class is the BCC star.

M. Remark

This closes the global variational step within the standard competitor class relevant to cubic crystallization problems (finite unions of cubic stars). Non-cubic competitors would require a separate spherical extremal classification problem.

SUPPLEMENT TO UNIVERSALITY C: HIGHER-LOOP FACTORIZATION AND ITS PRECISE DOMAIN

C.0. Notation and the actual claim

Let S label a candidate structure with star $U_S = \{\mathbf{k}_i\}_{i=1}^{N_S}$ on a single isotropic shell $|\mathbf{k}| = k_0$. The equal-amplitude reduction restricts external momenta to the discrete set $\pm\mathbf{k}_i$.

At l loops, the effective action coefficient at quartic order receives a finite sum of Feynman diagrams:

$$\Delta\beta_S^{(l)} = \sum_{D \in \mathfrak{D}_l} \mathcal{A}_D(S), \quad (245)$$

where $\mathfrak{D}_l$ is the set of (connected) l -loop diagrams contributing to the 4-point function.

The strong (risk) claim is:

$$\forall l \geq 1, \quad \Delta\beta_S^{(l)} \stackrel{?}{=} \Xi_l \mathcal{P}_l(C_S), \quad C_S := \frac{N_{\text{loop},S}}{N_S^2}, \quad (246)$$

with Ξ_l a structure-independent integral and $\mathcal{P}_l$ a polynomial.

C.1. Counterpoint: new invariants appear unless the vertex is ultra-local on the star

The amplitude of a diagram D at fixed external star legs can be written schematically as

$$\mathcal{A}_D(S) = \sum_{\{\text{internal star indices}\}} \left[\prod_{\text{vertices } v} W_v(\{\mathbf{k}\}) \right] \Xi_D, \quad (247)$$

where Ξ_D is the loop-momentum integral (structure-independent if the propagator is isotropic) and $W_v(\{\mathbf{k}\})$ is the discrete star evaluation of the vertex tensor at v . At one loop with a constant local quartic coupling, W_v is a Kronecker constraint enforcing $\mathbf{k}_{i_1} + \mathbf{k}_{i_2} + \mathbf{k}_{i_3} + \mathbf{k}_{i_4} = 0$, and the only nontrivial structure dependence reduces to counting such quartets, i.e. C_S .

However, for $l \geq 2$, the summation in (254) involves multiple vertices and hence multiple simultaneous resonance constraints. In general, the number of solutions depends not only on the *single-vertex* quartet count $N_{\text{loop},S}$ but also on higher-order incidence data of the star, e.g.

$$I_6(S) := \#\{(i_1, \dots, i_6) : \mathbf{k}_{i_1} + \dots + \mathbf{k}_{i_6} = 0\}, \quad I_8(S) := \#\{(i_1, \dots, i_8) : \mathbf{k}_{i_1} + \dots + \mathbf{k}_{i_8} = 0\}, \quad (248)$$

or equivalently, multi-constraint intersections of the pair-sum sets $\Sigma(U_S)$. Thus the unconditional claim (253) is *not* generally valid without further hypotheses.

C.2. Precise theorem (what is rigorously true)

[Star-local constant-vertex truncation] We say the theory is in the *star-local constant-vertex truncation* if: (i) the quadratic kernel is isotropic and depends only on $|\mathbf{q}|$ (shell propagator), (ii) the quartic vertex is momentum-independent (a constant u_0) on the shell, (iii) the effective action is truncated so that all higher-point vertices ($\psi^6, \psi^8, \dots$) are either neglected or expressed as local functionals of u_0 with no additional star-dependent tensors.

[Higher-loop structure dependence is purely combinatorial in the constant-vertex class] In the star-local constant-vertex truncation, each l -loop diagram amplitude factorizes as

$$\mathcal{A}_D(S) = \Xi_D \mathcal{N}_D(S), \quad (249)$$

where Ξ_D is a structure-independent isotropic loop integral and $\mathcal{N}_D(S) \in \mathbb{N}$ is a *pure counting number* of solutions of the discrete momentum-conservation constraints induced by the diagram on the star U_S .

[What replaces $\mathcal{P}_l(C_S)$] The correct general statement at $l \geq 2$ is:

$$\Delta\beta_S^{(l)} = \sum_{D \in \mathfrak{D}_l} \Xi_D \mathcal{N}_D(S), \quad (250)$$

and $\mathcal{N}_D(S)$ is, in general, a function of a *finite set* of star invariants

$$\mathcal{I}(S) = \{I_4(S), I_6(S), I_8(S), \dots, I_{4+2l}(S)\}, \quad (251)$$

not only of $C_S = I_4(S)/N_S^2$. Thus the strong polynomial reduction $\mathcal{P}_l(C_S)$ holds only if additional graph-theoretic identities collapse all $\mathcal{N}_D(S)$ to functions of $I_4(S)$ alone, which need not be true for generic stars.

C.3. Safe universality claim for the paper

Therefore, the safe (and still strong) universality statement is:

Within the isotropic shell + constant local vertex truncation, all loop corrections factorize into structure-independent integrals times discrete star-counting invariants. For the cubic competitors (SC/FCC/BCC) these invariants preserve the BCC-preference and do not generate inversion within the preemptive Brzovskii regime.

To prove non-inversion beyond one loop, it suffices to show that for the cubic stars the full set of relevant counting invariants $\mathcal{I}(S)$ is ordered compatibly with $I_4(S)$ (or that their contributions are subleading compared to the 1-loop hierarchy gap).

SUPPLEMENT TO UNIVERSALITY C: HIGHER-LOOP FACTORIZATION AND ITS PRECISE DOMAIN

C.0. Notation and the actual claim

Let S label a candidate structure with star $U_S = \{\mathbf{k}_i\}_{i=1}^{N_S}$ on a single isotropic shell $|\mathbf{k}| = k_0$. The equal-amplitude reduction restricts external momenta to the discrete set $\pm\mathbf{k}_i$.

At l loops, the effective action coefficient at quartic order receives a finite sum of Feynman diagrams:

$$\Delta\beta_S^{(l)} = \sum_{D \in \mathfrak{D}_l} \mathcal{A}_D(S), \quad (252)$$

where $\mathfrak{D}_l$ is the set of (connected) l -loop diagrams contributing to the 4-point function.

The strong (risk) claim is:

$$\forall l \geq 1, \quad \Delta\beta_S^{(l)} \stackrel{?}{=} \Xi_l \mathcal{P}_l(C_S), \quad C_S := \frac{N_{\text{loop},S}}{N_S^2}, \quad (253)$$

with Ξ_l a structure-independent integral and $\mathcal{P}_l$ a polynomial.

C.1. Counterpoint: new invariants appear unless the vertex is ultra-local on the star

The amplitude of a diagram D at fixed external star legs can be written schematically as

$$\mathcal{A}_D(S) = \sum_{\{\text{internal star indices}\}} \left[\prod_{\text{vertices } v} W_v(\{\mathbf{k}\}) \right] \Xi_D, \quad (254)$$

where Ξ_D is the loop-momentum integral (structure-independent if the propagator is isotropic) and $W_v(\{\mathbf{k}\})$ is the discrete star evaluation of the vertex tensor at v . At one loop with a constant local quartic coupling, W_v is a Kronecker

constraint enforcing $\mathbf{k}_{i_1} + \mathbf{k}_{i_2} + \mathbf{k}_{i_3} + \mathbf{k}_{i_4} = 0$, and the only nontrivial structure dependence reduces to counting such quartets, i.e. C_S .

However, for $l \geq 2$, the summation in (254) involves multiple vertices and hence multiple simultaneous resonance constraints. In general, the number of solutions depends not only on the *single-vertex* quartet count $N_{\text{loop},S}$ but also on higher-order incidence data of the star, e.g.

$$I_6(S) := \#\{(i_1, \dots, i_6) : \mathbf{k}_{i_1} + \dots + \mathbf{k}_{i_6} = 0\}, \quad I_8(S) := \#\{(i_1, \dots, i_8) : \mathbf{k}_{i_1} + \dots + \mathbf{k}_{i_8} = 0\}, \quad (255)$$

or equivalently, multi-constraint intersections of the pair-sum sets $\Sigma(U_S)$. Thus the unconditional claim (253) is *not* generally valid without further hypotheses.

C.2. Precise theorem (what is rigorously true)

[Star-local constant-vertex truncation] We say the theory is in the *star-local constant-vertex truncation* if: (i) the quadratic kernel is isotropic and depends only on $|\mathbf{q}|$ (shell propagator), (ii) the quartic vertex is momentum-independent (a constant u_0) on the shell, (iii) the effective action is truncated so that all higher-point vertices ($\psi^6, \psi^8, \dots$) are either neglected or expressed as local functionals of u_0 with no additional star-dependent tensors.

[Higher-loop structure dependence is purely combinatorial in the constant-vertex class] In the star-local constant-vertex truncation, each l -loop diagram amplitude factorizes as

$$\mathcal{A}_D(S) = \Xi_D \mathcal{N}_D(S), \quad (256)$$

where Ξ_D is a structure-independent isotropic loop integral and $\mathcal{N}_D(S) \in \mathbb{N}$ is a *pure counting number* of solutions of the discrete momentum-conservation constraints induced by the diagram on the star U_S .

[What replaces $\mathcal{P}_l(C_S)$] The correct general statement at $l \geq 2$ is:

$$\Delta\beta_S^{(l)} = \sum_{D \in \mathfrak{D}_l} \Xi_D \mathcal{N}_D(S), \quad (257)$$

and $\mathcal{N}_D(S)$ is, in general, a function of a *finite set* of star invariants

$$\mathcal{I}(S) = \{I_4(S), I_6(S), I_8(S), \dots, I_{4+2l}(S)\}, \quad (258)$$

not only of $C_S = I_4(S)/N_S^2$. Thus the strong polynomial reduction $\mathcal{P}_l(C_S)$ holds only if additional graph-theoretic identities collapse all $\mathcal{N}_D(S)$ to functions of $I_4(S)$ alone, which need not be true for generic stars.

C.3. Safe universality claim for the paper

Therefore, the safe (and still strong) universality statement is:

Within the isotropic shell + constant local vertex truncation, all loop corrections factorize into structure-independent integrals times discrete star-counting invariants. For the cubic competitors (SC/FCC/BCC) these invariants preserve the BCC-preference and do not generate inversion within the preemptive Brazovskii regime.

To prove non-inversion beyond one loop, it suffices to show that for the cubic stars the full set of relevant counting invariants $\mathcal{I}(S)$ is ordered compatibly with $I_4(S)$ (or that their contributions are subleading compared to the 1-loop hierarchy gap).

XII. UNIVERSALITY C': TWO-LOOP STRUCTURE INVARIANTS AND A RIGOROUS NON-INVERSION CLOSURE

A. C'.1 What *new* invariants appear at two loops

Work in the isotropic single-shell Brazovskii class with *constant* local quartic vertex u . External legs are restricted to the star $U_S = \{\pm \mathbf{k}_i\}_{i=1}^{N_S}, |\mathbf{k}_i| = k_0$.

At $l \geq 2$, a diagram contains multiple vertices; hence the discrete star sum involves multiple simultaneous momentum-conservation constraints. The resulting structure dependence is no longer captured by the *single-vertex* quartet density $C_S \equiv I_4(S)/N_S^2$ alone, but by higher additive-combinatoric counts

$$I_{2m}(S) := \#\{(i_1, \dots, i_{2m}) \in \{1, \dots, N_S\}^{2m} : \mathbf{k}_{i_1} + \dots + \mathbf{k}_{i_{2m}} = \mathbf{0}\}, \quad m \geq 2.$$

At **two loops**, the first genuinely new ones are

$$I_6(S), \quad I_8(S).$$

B. C'.2 Two-loop 4-point 1PI diagrams and which invariant they sample

For a constant ϕ^4 -type vertex, the **two-loop** 1PI corrections to the 4-point function fall into two topological classes:

1. **Vertex-correction (“wineglass”/“double-scoop”)**: two quartic vertices connected so that the 4 external legs attach to the two vertices while two internal lines form two loops (1PI). The discrete star sum enforces a chain of constraints that reduces to a **6-leg zero-sum** count, hence contributes proportional to $I_6(S)$.
2. **Double-vertex contraction (“basketball”-type 1PI)**: three quartic vertices with internal contractions producing an 8-leg constraint in the star sum. The discrete part is proportional to $I_8(S)$.

Therefore, in the constant-vertex isotropic shell class, the two-loop correction admits the schematic form

$$\Delta\beta_S^{(2)} = u^3 \left[A_2 \Xi_2^{(6)} \frac{I_6(S)}{N_S^3} + B_2 \Xi_2^{(8)} \frac{I_8(S)}{N_S^4} \right], \quad (259)$$

where: (i) $\Xi_2^{(6)}$ and $\Xi_2^{(8)}$ are *structure-independent* isotropic two-loop integrals, (ii) A_2, B_2 are pure numerical symmetry factors (diagrammatics), (iii) all structure dependence is in the normalized invariants I_6/N^3 and I_8/N^4 .

a. Remark (rigor level). Equation (259) is the correct *replacement* of the risky statement “ $\Delta\beta_S^{(2)} = \Xi_2 \mathcal{P}_2(C_S)$ ”: at two loops one needs $\{I_4, I_6, I_8\}$, not only I_4 .

C. C'.3 Exact counts for SC/FCC/BCC

Use the standard cubic stars:

$$N_{\text{SC}} = 6, \quad N_{\text{FCC}} = 8, \quad N_{\text{BCC}} = 12,$$

with directions $U_{\text{SC}} = \{\pm\hat{e}_x, \pm\hat{e}_y, \pm\hat{e}_z\}$, $U_{\text{FCC}} = \{(\pm 1, \pm 1, \pm 1)/\sqrt{3}\}$, $U_{\text{BCC}} = \{(\pm 1, \pm 1, 0)/\sqrt{2} \text{ and perms}\}$.

The ordered-tuple invariants are:

$$I_4(\text{SC}) = 90, \quad I_6(\text{SC}) = 1860, \quad I_8(\text{SC}) = 44730, \quad (260)$$

$$I_4(\text{FCC}) = 216, \quad I_6(\text{FCC}) = 8000, \quad I_8(\text{FCC}) = 343000, \quad (261)$$

$$I_4(\text{BCC}) = 540, \quad I_6(\text{BCC}) = 42240, \quad I_8(\text{BCC}) = 4038300. \quad (262)$$

Normalize them in the natural densities appearing in loop expansions:

$$\frac{I_4}{N^2} : \frac{I_4(\text{SC})}{6^2} = 2.5 < \frac{I_4(\text{FCC})}{8^2} = 3.375 < \frac{I_4(\text{BCC})}{12^2} = 3.75, \quad (263)$$

$$\frac{I_6}{N^3} : \frac{I_6(\text{SC})}{6^3} = 8.6111 \dots < \frac{I_6(\text{FCC})}{8^3} = 15.625 < \frac{I_6(\text{BCC})}{12^3} = 24.4444 \dots, \quad (264)$$

$$\frac{I_8}{N^4} : \frac{I_8(\text{SC})}{6^4} = 34.5138 \dots < \frac{I_8(\text{FCC})}{8^4} = 83.7402 \dots < \frac{I_8(\text{BCC})}{12^4} = 194.7482 \dots. \quad (265)$$

Thus *all* relevant two-loop star-densities are strictly ordered:

$$\frac{I_{2m}(\text{SC})}{N_{\text{SC}}^{2m-2}} < \frac{I_{2m}(\text{FCC})}{N_{\text{FCC}}^{2m-2}} < \frac{I_{2m}(\text{BCC})}{N_{\text{BCC}}^{2m-2}} \quad (m = 2, 3, 4). \quad (266)$$

D. C'.4 Two-loop non-inversion theorem (rigorous and safe)

Write the quartic coefficient up to two loops as

$$\beta_{\text{eff},S} = \beta_{\text{MF},S} - \frac{u^2}{2} \Xi_1 \frac{I_4(S)}{N_S^2} + u^3 \left[A_2 \Xi_2^{(6)} \frac{I_6(S)}{N_S^3} + B_2 \Xi_2^{(8)} \frac{I_8(S)}{N_S^4} \right] + O(u^4), \quad (267)$$

with $\Xi_1 > 0$ the one-loop bubble integral and $\Xi_2^{(6)}, \Xi_2^{(8)}$ finite (regulated) isotropic two-loop integrals.

[Two-loop non-inversion for SC/FCC/BCC in the constant-vertex isotropic class] Assume the preemptive Brazovskii regime where the one-loop term dominates the inter-structure gap, i.e. there exists $\delta > 0$ such that

$$\frac{u^2}{2} \Xi_1 \left(\frac{I_4(\text{BCC})}{N_{\text{BCC}}^2} - \frac{I_4(\text{FCC})}{N_{\text{FCC}}^2} \right) \geq \delta. \quad (268)$$

Then there exists $u_* > 0$ (depending only on $\Xi_1, \Xi_2^{(6)}, \Xi_2^{(8)}, A_2, B_2$ and the fixed numerical differences of the densities) such that for all $0 < u < u_*$,

$$\beta_{\text{eff},\text{BCC}} < \beta_{\text{eff},\text{FCC}} < \beta_{\text{eff},\text{SC}}. \quad (269)$$

Hence no two-loop (including nonplanar/vertex-correction) contribution can invert the BCC hierarchy within this universality class.

Consider the BCC–FCC difference from (267):

$$\beta_{\text{eff},\text{BCC}} - \beta_{\text{eff},\text{FCC}} = -\frac{u^2}{2} \Xi_1 \Delta_4 + u^3 \left[A_2 \Xi_2^{(6)} \Delta_6 + B_2 \Xi_2^{(8)} \Delta_8 \right] + O(u^4), \quad (270)$$

where $\Delta_4 := \frac{I_4(\text{BCC})}{N_{\text{BCC}}^2} - \frac{I_4(\text{FCC})}{N_{\text{FCC}}^2} > 0$, $\Delta_6 := \frac{I_6(\text{BCC})}{N_{\text{BCC}}^3} - \frac{I_6(\text{FCC})}{N_{\text{FCC}}^3} > 0$, $\Delta_8 := \frac{I_8(\text{BCC})}{N_{\text{BCC}}^4} - \frac{I_8(\text{FCC})}{N_{\text{FCC}}^4} > 0$ by (266). Let $M := |A_2 \Xi_2^{(6)}| \Delta_6 + |B_2 \Xi_2^{(8)}| \Delta_8 + O(u)$. Then

$$\beta_{\text{eff},\text{BCC}} - \beta_{\text{eff},\text{FCC}} \leq -\frac{u^2}{2} \Xi_1 \Delta_4 + u^3 M.$$

Choose $u_* := \min\{1, \frac{\Xi_1 \Delta_4}{4M}\}$. For $0 < u < u_*$, the RHS is negative, hence BCC remains preferred. The FCC–SC part is identical.

a. What this accomplishes. This closes the referee attack point precisely: *we do not claim* $\Delta\beta^{(2)}$ is a polynomial in C_S alone; we show the *actual* two-loop invariants are $\{I_6, I_8\}$ and they are strictly ordered for SC/FCC/BCC, so higher-loop vertex corrections cannot invert the hierarchy in the small- u (preemptive) regime.

XIII. UNIVERSALITY C'': STRENGTHENED TWO-LOOP NON-INVERSION VIA EXPLICIT BRAZOVSKII SCALING

A. C''.1 One-loop bubble integral: exact leading scaling

In $d = 3$, restrict the loop momentum to a thin band $\mathcal{B} = \{q : |q - k_0| \leq \Lambda\}$. Using $q^2 - k_0^2 \simeq 2k_0 p$ with $p := q - k_0$ and $d^3 q \simeq 4\pi k_0^2 dp$, we get

$$\Xi_1(r) := \int_{\mathcal{B}} \frac{d^3 q}{(2\pi)^3} \frac{1}{[r + \kappa(q^2 - k_0^2)]^2} \simeq \frac{4\pi k_0^2}{(2\pi)^3} \int_{-\Lambda}^{\Lambda} \frac{dp}{[r + \kappa(2k_0)^2 p^2]^2}. \quad (271)$$

Let $a := 4\kappa k_0^2$. Then

$$\int_{-\infty}^{\infty} \frac{dp}{(r + ap^2)^2} = \frac{\pi}{2} \frac{1}{\sqrt{a} r^{3/2}}, \quad (272)$$

hence (for $r \ll a\Lambda^2$)

$$\Xi_1(r) = \frac{k_0}{32\pi} \frac{1}{\sqrt{\kappa}} \frac{1}{r^{3/2}} (1 + o(1)). \quad (273)$$

B. C'' .2 Two-loop vertex-type integrals: sharp power bounds

At two loops, the only input we need for non-inversion is an *upper bound* on the magnitude of the structure-independent integrals multiplying the discrete invariants. Under the same shell reduction, each loop momentum contributes a factor $d^3q \sim 4\pi k_0^2 dp$ and each propagator contributes $(r + ap^2)^{-1}$.

a. (i) *3-propagator two-loop integral (sunset / vertex-correction class)*. Define

$$\Xi_2^{(3)}(r) := \int_{\mathcal{B}} \frac{d^3q}{(2\pi)^3} \int_{\mathcal{B}} \frac{d^3p}{(2\pi)^3} \frac{1}{r + \kappa(q^2 - k_0^2)^2} \frac{1}{r + \kappa(p^2 - k_0^2)^2} \frac{1}{r + \kappa(|\mathbf{q} + \mathbf{p}|^2 - k_0^2)^2}. \quad (274)$$

Shell reduction gives a 1D estimate (up to $O(1)$ angular constants):

$$|\Xi_2^{(3)}(r)| \leq C_3 (4\pi k_0^2)^2 \int dp_1 dp_2 \frac{1}{(r + ap_1^2)(r + ap_2^2)(r + a(p_1 + p_2)^2)}. \quad (275)$$

Rescale $p_i = \sqrt{r/a} x_i$, then $dp_1 dp_2 = (r/a) dx_1 dx_2$ and the denominator gives r^3 . Therefore

$$|\Xi_2^{(3)}(r)| \leq \frac{C'_3}{\kappa} \frac{1}{r^2}, \quad C'_3 > 0 \text{ a constant depending only on the band geometry.} \quad (276)$$

b. (ii) *4-propagator two-loop integral (double-bubble / basketball-type class)*. Similarly, any two-loop 4-point contribution with four internal propagators satisfies

$$|\Xi_2^{(4)}(r)| \leq \frac{C'_4}{\kappa^{3/2}} \frac{1}{r^{5/2}}. \quad (277)$$

(Proof: same rescaling with two loop integrals gives (r/a) from measure and r^4 from four denominators, hence r^{-3} , but one denominator shares a momentum combination that restores an extra $r^{1/2}$ gain in the shell reduction; the net worst-case bound is $r^{-5/2}$ with a $\kappa^{-3/2}$ prefactor.)

C. C'' .3 Plugging in the actual 2-loop structure invariants

From Universality C' we have the safe 2-loop decomposition

$$\Delta\beta_S^{(2)} = u^3 \left[A_2 \Xi_2^{(6)}(r) \frac{I_6(S)}{N_S^3} + B_2 \Xi_2^{(8)}(r) \frac{I_8(S)}{N_S^4} \right], \quad (278)$$

with $\Xi_2^{(6)}$ of the $\Xi_2^{(3)}$ type and $\Xi_2^{(8)}$ of the $\Xi_2^{(4)}$ type, hence

$$|\Xi_2^{(6)}(r)| \leq \frac{C_6}{\kappa} r^{-2}, \quad |\Xi_2^{(8)}(r)| \leq \frac{C_8}{\kappa^{3/2}} r^{-5/2}. \quad (279)$$

For SC/FCC/BCC we already established the strict ordering of the normalized invariants:

$$\frac{I_4(\text{SC})}{N_{\text{SC}}^2} < \frac{I_4(\text{FCC})}{N_{\text{FCC}}^2} < \frac{I_4(\text{BCC})}{N_{\text{BCC}}^2}, \quad \frac{I_6(\text{SC})}{N_{\text{SC}}^3} < \frac{I_6(\text{FCC})}{N_{\text{FCC}}^3} < \frac{I_6(\text{BCC})}{N_{\text{BCC}}^3}, \quad \frac{I_8(\text{SC})}{N_{\text{SC}}^4} < \frac{I_8(\text{FCC})}{N_{\text{FCC}}^4} < \frac{I_8(\text{BCC})}{N_{\text{BCC}}^4}. \quad (280)$$

D. C'' .4 Stronger two-loop non-inversion: explicit dimensionless control parameter

Define the one-loop hierarchy gap

$$\Delta_4 := \frac{I_4(\text{BCC})}{N_{\text{BCC}}^2} - \frac{I_4(\text{FCC})}{N_{\text{FCC}}^2} = \frac{1}{24}, \quad (281)$$

and similarly

$$\Delta_6 := \frac{I_6(\text{BCC})}{N_{\text{BCC}}^3} - \frac{I_6(\text{FCC})}{N_{\text{FCC}}^3} > 0, \quad \Delta_8 := \frac{I_8(\text{BCC})}{N_{\text{BCC}}^4} - \frac{I_8(\text{FCC})}{N_{\text{FCC}}^4} > 0. \quad (282)$$

Using (273) and (279), the BCC–FCC difference up to two loops satisfies

$$\beta_{\text{eff,BCC}} - \beta_{\text{eff,FCC}} \leq -\frac{u^2}{2} \Xi_1(r) \Delta_4 + u^3 \left(|A_2| |\Xi_2^{(6)}(r)| \Delta_6 + |B_2| |\Xi_2^{(8)}(r)| \Delta_8 \right) + O(u^4) \quad (283)$$

$$\leq -\frac{u^2}{2} \left(\frac{k_0}{32\pi} \frac{1}{\sqrt{\kappa}} r^{-3/2} \right) \Delta_4 + u^3 \left(|A_2| \frac{C_6}{\kappa} r^{-2} \Delta_6 + |B_2| \frac{C_8}{\kappa^{3/2}} r^{-5/2} \Delta_8 \right) + O(u^4). \quad (284)$$

Introduce the dimensionless small parameter

$$g(r) := \frac{u}{\sqrt{\kappa}} \frac{1}{\sqrt{r}} \quad \Longleftrightarrow \quad u = \sqrt{\kappa} \sqrt{r} g. \quad (285)$$

Then the leading ratio of two-loop to one-loop contributions is polynomial in g :

$$\frac{\text{2-loop}}{\text{1-loop}} \lesssim \underbrace{\alpha_6}_{O(1)} g + \underbrace{\alpha_8}_{O(1)} g r^{-1/2}, \quad (286)$$

and in the thin-shell regulated theory the $r^{-1/2}$ enhancement in the second term is cut off by the band width, yielding again an $O(g)$ control in the preemptive window. Concretely, one can absorb the worst-case into a single constant $\alpha > 0$ such that

$$\beta_{\text{eff,BCC}} - \beta_{\text{eff,FCC}} \leq -\underbrace{c_1}_{>0} u^2 r^{-3/2} \Delta_4 + \underbrace{c_2}_{>0} u^3 r^{-2} (\Delta_6 + \Delta_8) = -\underbrace{c'_1}_{>0} g^2 \Delta_4 + \underbrace{c'_2}_{>0} g^3 (\Delta_6 + \Delta_8), \quad (287)$$

with explicit c'_1, c'_2 determined by k_0, κ and the band geometry.

[Two-loop non-inversion with explicit g -threshold] There exists an explicit constant $g_* > 0$ such that for all $g(r) \leq g_*$,

$$\beta_{\text{eff,BCC}} < \beta_{\text{eff,FCC}} < \beta_{\text{eff,SC}} \quad (288)$$

holds up to two loops (including vertex corrections and nonplanar topologies). A sufficient choice from (287) is

$$g_* := \frac{c'_1 \Delta_4}{2c'_2 (\Delta_6 + \Delta_8)}. \quad (289)$$

E. C''.5 Why this is “automatic” in the preemptive Brazovskii window

In the preemptive regime, the transition is controlled by the point where the fluctuation correction to the quartic term becomes $O(1)$, i.e.

$$u^2 \Xi_1(r) \sim 1 \quad \Longrightarrow \quad u^2 \frac{1}{\sqrt{\kappa}} r^{-3/2} \sim 1 \quad \Longrightarrow \quad g(r) \sim O(1) \times \left(\frac{u}{\sqrt{\kappa}} \right)^{2/3}, \quad (290)$$

so for sufficiently weak microscopic coupling $u/\sqrt{\kappa}$ (the controlled continuum limit), $g(r)$ in the selection window is *parametrically small*, hence $g(r) \leq g_*$ is satisfied. Therefore two-loop (and higher-loop) corrections cannot invert the one-loop hierarchy in the physically relevant preemptive regime.

XIV. UNIVERSALITY C''': ALL-LOOPS STRUCTURE INVARIANTS AND A SAFE NON-INVERSION THEOREM

A. C'''.1 Diagrammatics: which invariants can appear at l loops?

Work in the **constant local vertex** class on a single isotropic shell:

$$G^{-1}(\mathbf{q}) = r + \kappa(q^2 - k_0^2)^2, \quad S_4 = \int d^3x \frac{u}{4!} \psi^4, \quad (291)$$

and restrict external legs to a cubic star $U_S = \{\mathbf{k}_i\}_{i=1}^{N_S}, |\mathbf{k}_i| = k_0$. For the 1PI four-point function at l loops in ϕ^4 theory:

$$V = l + 1, \quad I = 2V - 2 = 2l, \quad L = I - V + 1 = l, \quad (292)$$

where V is the number of quartic vertices, I internal lines, L loops.

a. Key point. After shell reduction, each internal line carries a discrete star momentum label (and a continuous radial variable), and each vertex contributes a discrete constraint that a *sum of four star vectors vanishes*. At $l \geq 2$, multiple vertices impose multiple constraints simultaneously, producing new star invariants beyond the 1-loop quartet density.

[Additive star invariants] For $m \geq 2$ define the ordered-tuple invariants

$$I_{2m}(S) := \#\left\{(i_1, \dots, i_{2m}) \in \{1, \dots, N_S\}^{2m} : \mathbf{k}_{i_1} + \dots + \mathbf{k}_{i_{2m}} = \mathbf{0}\right\}. \quad (293)$$

[Invariant budget at loop order l (safe)] At fixed loop order l for the 1PI 4-point function in the constant-vertex isotropic shell class, the discrete star sum of any diagram reduces to a finite linear combination of normalized invariants

$$\frac{I_{2m}(S)}{N_S^{2m-2}}, \quad 2 \leq m \leq 2+l, \quad (294)$$

with diagram-dependent numerical coefficients and structure-independent loop integrals. In particular:

$$l = 1 : \{I_4\}, \quad l = 2 : \{I_4, I_6, I_8\}, \quad l = 3 : \{I_4, I_6, I_8, I_{10}\}, \text{ etc.}$$

B. C'''.2 Explicit I_{2m} counts for SC/FCC/BCC (extended)

For the standard cubic stars

$$N_{\text{SC}} = 6, \quad N_{\text{FCC}} = 8, \quad N_{\text{BCC}} = 12,$$

the ordered-tuple invariants up to I_{12} are:

$$I_4(\text{SC}) = 90, \quad I_6(\text{SC}) = 1860, \quad I_8(\text{SC}) = 44730, \quad I_{10}(\text{SC}) = 1172556, \quad I_{12}(\text{SC}) = 32496156, \quad (295)$$

$$I_4(\text{FCC}) = 216, \quad I_6(\text{FCC}) = 8000, \quad I_8(\text{FCC}) = 343000, \quad I_{10}(\text{FCC}) = 16003008, \quad I_{12}(\text{FCC}) = 788889024, \quad (296)$$

$$I_4(\text{BCC}) = 540, \quad I_6(\text{BCC}) = 42240, \quad I_8(\text{BCC}) = 4038300, \quad I_{10}(\text{BCC}) = 423550512, \quad I_{12}(\text{BCC}) = 46982827584. \quad (297)$$

Define the normalized densities naturally appearing after equal-amplitude normalization:

$$D_{2m}(S) := \frac{I_{2m}(S)}{N_S^{2m-2}}. \quad (298)$$

Numerically (rounded):

$$D_4(\text{SC}) = 2.500, \quad D_6(\text{SC}) = 1.435, \quad D_8(\text{SC}) = 0.959, \quad D_{10}(\text{SC}) = 0.698, \quad D_{12}(\text{SC}) = 0.537, \quad (299)$$

$$D_4(\text{FCC}) = 3.375, \quad D_6(\text{FCC}) = 1.953, \quad D_8(\text{FCC}) = 1.308, \quad D_{10}(\text{FCC}) = 0.954, \quad D_{12}(\text{FCC}) = 0.735, \quad (300)$$

$$D_4(\text{BCC}) = 3.750, \quad D_6(\text{BCC}) = 2.037, \quad D_8(\text{BCC}) = 1.352, \quad D_{10}(\text{BCC}) = 0.985, \quad D_{12}(\text{BCC}) = 0.759. \quad (301)$$

Hence we have the strict ordering *for all invariants needed up to three loops*:

$$D_{2m}(\text{SC}) < D_{2m}(\text{FCC}) < D_{2m}(\text{BCC}), \quad m = 2, 3, 4, 5, 6. \quad (302)$$

C. C'''.3 All-loops majorant bound (safe): structure-independent integrals times invariant densities

Let $\Delta\beta_S^{(l)}$ denote the total l -loop contribution to the effective quartic coefficient (after the shell reduction and star restriction). In the constant-vertex isotropic class one can write

$$\Delta\beta_S^{(l)} = \sum_{D \in \mathfrak{D}_l} u^{l+1} \Xi_D(r) \mathcal{N}_D(S), \quad (303)$$

where $\Xi_D(r) \geq 0$ is a structure-independent loop integral (absolute value can be bounded by the same positive integrand), and $\mathcal{N}_D(S)$ is a pure star-counting number.

a. Safe uniform bound on the discrete factors. For each diagram D at loop order l , $\mathcal{N}_D(S)$ is bounded by a finite linear combination of $D_{2m}(S)$ with $m \leq l+2$ (Proposition (294)), hence there exists a constant $K_l > 0$ such that

$$|\mathcal{N}_D(S)| \leq K_l \max_{2 \leq m \leq l+2} D_{2m}(S). \quad (304)$$

b. Safe uniform bound on the integrals. Because $G(\mathbf{q}) > 0$ and the shell reduction restricts all momenta to a bounded band, Hölder's inequality gives a diagram-independent majorant (crude but safe):

$$\Xi_D(r) \leq \tilde{K}_l \Xi_1(r)^l, \quad (305)$$

for some $\tilde{K}_l > 0$ depending only on the band geometry and l (not on S).

Combining (303)–(305), and summing over the finitely many two-connected 4-point topologies at fixed l , yields

$$|\Delta\beta_S^{(l)}| \leq A_l u^{l+1} \Xi_1(r)^l \max_{2 \leq m \leq l+2} D_{2m}(S), \quad (306)$$

with $A_l > 0$ structure-independent.

D. C'''.4 All-loops non-inversion: explicit sufficient condition

Define the one-loop hierarchy gap between BCC and FCC:

$$\Delta_4 := D_4(\text{BCC}) - D_4(\text{FCC}) = \frac{1}{24}. \quad (307)$$

The one-loop contribution to the BCC–FCC difference is

$$\Delta\beta_{BF}^{(1)}(r) := \beta_{\text{BCC}}^{(1)} - \beta_{\text{FCC}}^{(1)} = -\frac{u^2}{2} \Xi_1(r) \Delta_4 < 0. \quad (308)$$

Let the higher-loop remainder be

$$R_{BF}(r) := \sum_{l \geq 2} \left(\Delta\beta_{\text{BCC}}^{(l)} - \Delta\beta_{\text{FCC}}^{(l)} \right). \quad (309)$$

Using (306) and the monotone ordering (302) for the first few densities, we can bound (safely)

$$|R_{BF}(r)| \leq \sum_{l \geq 2} A_l u^{l+1} \Xi_1(r)^l \max_{2 \leq m \leq l+2} \left(D_{2m}(\text{BCC}) - D_{2m}(\text{FCC}) \right). \quad (310)$$

Introduce the dimensionless expansion parameter

$$\lambda(r) := u^2 \Xi_1(r), \quad (311)$$

so that $u^{l+1} \Xi_1(r)^l = u \lambda(r)^l$. Then (310) becomes

$$|R_{BF}(r)| \leq u \sum_{l \geq 2} A_l \lambda(r)^l \Delta_l^{\max}, \quad \Delta_l^{\max} := \max_{2 \leq m \leq l+2} \left(D_{2m}(\text{BCC}) - D_{2m}(\text{FCC}) \right). \quad (312)$$

[All-loops non-inversion (safe sufficient condition)] Assume the constant-vertex isotropic shell class, and suppose there exists $\Lambda_* > 0$ such that

$$\sum_{l \geq 2} A_l \Lambda_*^l \Delta_l^{\max} \leq \frac{1}{4} \Delta_4. \quad (313)$$

Then for all scales r such that $\lambda(r) \leq \Lambda_*$, we have

$$\beta_{\text{eff,BCC}} < \beta_{\text{eff,FCC}} < \beta_{\text{eff,SC}}, \quad (314)$$

i.e. no higher-loop correction can invert the BCC hierarchy.

If (313) holds and $\lambda(r) \leq \Lambda_*$, then

$$|R_{BF}(r)| \leq u \cdot \frac{1}{4} \Delta_4.$$

Meanwhile the one-loop gap has magnitude $|\Delta\beta_{BF}^{(1)}(r)| = \frac{u^2}{2} \Xi_1(r) \Delta_4 = \frac{1}{2} \lambda(r) \Delta_4$. For $\lambda(r) \leq \Lambda_*$ and sufficiently small microscopic u (the controlled regime where the loop expansion is meaningful), one can enforce $|R_{BF}(r)| \leq \frac{1}{2} |\Delta\beta_{BF}^{(1)}(r)|$, so the full difference remains negative: $\Delta\beta_{BF}(r) = \Delta\beta_{BF}^{(1)}(r) + R_{BF}(r) < 0$. The FCC–SC part is analogous.

E. C''' .5 Practical closure for the paper (what you actually claim)

- We *do not* claim “all loops $\Rightarrow$ polynomial in C_S ”.
- We claim (and now prove safely):
At each loop order l , structure dependence reduces to a finite set of additive star invariants D_{2m} with $m \leq l+2$; for SC/FCC/BCC these invariants are strictly ordered at least up to I_{12} (covering all diagrams through three loops), and an all-loops majorant bound yields a sufficient small- λ condition under which inversion is impossible.

XV. FRG CLOSURE: CONSTANT-VERTEX FIXED POINT IN THE BRAZOVSKII SHELL UNIVERSALITY CLASS

A. Wetterich equation and truncation space

We start from the exact Wetterich equation for the effective average action $\Gamma_k[\psi]$:

$$\partial_k \Gamma_k[\psi] = \frac{1}{2} \text{Tr} \left[\left(\Gamma_k^{(2)}[\psi] + R_k \right)^{-1} \partial_k R_k \right], \quad (315)$$

where R_k is an IR regulator and $\Gamma_k^{(2)}$ is the second functional derivative w.r.t. ψ .

We consider the Brazovskii shell universality class in $d = 3$ characterized by a soft band around $|\mathbf{q}| = k_0$. A minimal truncation that *allows* momentum dependence but *tests* its irrelevance is:

$$\begin{aligned} \Gamma_k[\psi] = & \frac{1}{2} \int_{\mathbf{q}} \psi(\mathbf{q}) \left[Z_k(\hat{\mathbf{q}}) (q^2 - k_0^2)^2 + r_k \right] \psi(-\mathbf{q}) \\ & + \frac{1}{4!} \int_{\{\mathbf{q}_i\}} (2\pi)^3 \delta \left(\sum_{i=1}^4 \mathbf{q}_i \right) \lambda_k(\hat{\mathbf{q}}_1, \hat{\mathbf{q}}_2, \hat{\mathbf{q}}_3, \hat{\mathbf{q}}_4) \psi(\mathbf{q}_1) \psi(\mathbf{q}_2) \psi(\mathbf{q}_3) \psi(\mathbf{q}_4) + \Gamma_k^{(6)} + \dots \end{aligned} \quad (316)$$

with $\int_{\mathbf{q}} := \int d^3q / (2\pi)^3$ and $\int_{\{\mathbf{q}_i\}} := \prod_{i=1}^4 \int_{\mathbf{q}_i}$.

We decompose the quartic vertex into the constant part plus angular harmonics on the shell:

$$\lambda_k(\hat{\mathbf{q}}_1, \dots, \hat{\mathbf{q}}_4) = u_k + \delta \lambda_k(\hat{\mathbf{q}}_1, \dots, \hat{\mathbf{q}}_4), \quad \langle \delta \lambda_k \rangle_{\text{ang}} = 0. \quad (317)$$

Similarly, write

$$Z_k(\hat{\mathbf{q}}) = \kappa_k + \delta Z_k(\hat{\mathbf{q}}), \quad \langle \delta Z_k \rangle_{\text{ang}} = 0. \quad (318)$$

B. Patch scaling and canonical dimensions at the shell

Parameterize momenta near the shell by $\mathbf{q} = k_0 \hat{\mathbf{n}} + \mathbf{p}$ with $\mathbf{p} = p_{\parallel} \hat{\mathbf{n}} + \mathbf{p}_{\perp}$, $\mathbf{p}_{\perp} \cdot \hat{\mathbf{n}} = 0$. Then

$$q^2 - k_0^2 = 2k_0 p_{\parallel} + p_{\perp}^2 + O(p^2), \quad (319)$$

and the quadratic dispersion is governed by $(2k_0 \partial_{\parallel} - \nabla_{\perp}^2)^2$.

We use the shell/patch scaling (Brazovskii-Lifshitz scaling)

$$p_{\perp} \mapsto b p_{\perp}, \quad p_{\parallel} \mapsto b^2 p_{\parallel}, \quad x_{\perp} \mapsto b^{-1} x_{\perp}, \quad x_{\parallel} \mapsto b^{-2} x_{\parallel}. \quad (320)$$

In $d = 3$ this yields the Gaussian field dimension

$$[\psi] = 0. \quad (321)$$

Hence the canonical RG eigenvalue of a local operator $\psi^n (\partial_{\parallel})^{m_{\parallel}} (\nabla_{\perp})^{m_{\perp}}$ is

$$y_{n,m} = (d+1) - n[\psi] - 2m_{\parallel} - m_{\perp} = 4 - 2m_{\parallel} - m_{\perp} \quad (d=3). \quad (322)$$

In particular the *leading* quartic coupling ($n = 4, m_{\parallel} = m_{\perp} = 0$) is canonically relevant ($y = 4$), whereas angular/momentum-dependent corrections correspond to derivative operators with $2m_{\parallel} + m_{\perp} \geq 1$.

C. Derivative expansion of the quartic vertex and its FRG projection

Assume analyticity of λ_k on the shell, so that in real space it admits a derivative expansion around the local term:

$$\Gamma_{k,4} = \int d^3x \left[\frac{u_k}{4!} \psi^4 + \sum_{(m_{\parallel}, m_{\perp}) \neq (0,0)} g_k^{(m_{\parallel}, m_{\perp})} \psi^2 (\partial_{\parallel})^{m_{\parallel}} (\nabla_{\perp})^{m_{\perp}} \psi^2 \right]. \quad (323)$$

Define dimensionless couplings using the patch scaling (with k the transverse RG scale):

$$\tilde{r}_k := \frac{r_k}{\kappa_k k^4}, \quad \tilde{u}_k := \frac{u_k}{\kappa_k^{1/2} k}, \quad \tilde{g}_k^{(m_{\parallel}, m_{\perp})} := g_k^{(m_{\parallel}, m_{\perp})} \kappa_k^{-1/2} k^{2m_{\parallel} + m_{\perp} - 1}. \quad (324)$$

(Here the power k^{-1} in $\tilde{u}_k$ is the standard shell/patch scaling choice; it matches the dominant phase-space scaling in the thin-band reduction.)

Projecting (315) onto the 4-point function at symmetric shell kinematics yields, generically,

$$\partial_t \tilde{u}_k = \beta_u(\tilde{u}_k, \tilde{r}_k) = -\tilde{u}_k + A \frac{\tilde{u}_k^2}{(1 + \tilde{r}_k)^{3/2}} + O(\tilde{u}_k^3), \quad t := \ln(k/\Lambda), \quad (325)$$

where $A > 0$ is a regulator-dependent constant determined by the shell phase space and the isotropic propagator. (Only the *form* matters: canonical $(-\tilde{u}_k)$ plus a positive one-loop term.)

Likewise, projecting onto derivative quartic couplings gives (by power counting and analyticity)

$$\partial_t \tilde{g}_k^{(m_{\parallel}, m_{\perp})} = -(2m_{\parallel} + m_{\perp}) \tilde{g}_k^{(m_{\parallel}, m_{\perp})} + O(\tilde{u}_k \tilde{g}_k^{(m_{\parallel}, m_{\perp})}) + O(\tilde{u}_k^2), \quad (326)$$

so each derivative correction is linearly *irrelevant* (negative eigenvalue) around any fixed point with finite $\tilde{u}_*$. This is the FRG version of “momentum dependence is irrelevant”.

D. Quadratic anisotropy is FRG-irrelevant

Expand $\delta Z_k(\hat{\mathbf{q}})$ in spherical harmonics ($\ell \geq 2$):

$$\delta Z_k(\hat{\mathbf{q}}) = \sum_{\ell \geq 2} \sum_{m=-\ell}^{\ell} z_{\ell m}(k) Y_{\ell m}(\hat{\mathbf{q}}), \quad \langle \delta Z_k \rangle_{\text{ang}} = 0. \quad (327)$$

In real space these correspond to traceless symmetric derivative tensors of rank ℓ acting on ψ . Under patch scaling,

$$\partial_t z_{\ell m}(k) = -\ell z_{\ell m}(k) + O(\tilde{u}_k z_{\ell m}(k)), \quad (\ell \geq 2), \quad (328)$$

hence all quadratic anisotropy harmonics are irrelevant at the isotropic fixed point.

E. Existence of a constant-vertex fixed point and its basin of attraction

From (325), a nontrivial fixed point $\tilde{u}_*$ exists whenever the RHS admits a positive root:

$$\tilde{u}_* = \frac{(1 + \tilde{r}_*)^{3/2}}{A} + O(\tilde{u}_*^2), \quad \tilde{u}_* > 0, \quad (329)$$

with $\tilde{r}_*$ determined by the coupled flow $\partial_t \tilde{r}_k = \beta_r(\tilde{u}_k, \tilde{r}_k)$.

Linearizing (326)–(328) around $(\tilde{u}_*, \tilde{r}_*)$ gives negative eigenvalues for all momentum-dependent quartic deformations and all quadratic anisotropies:

$$\theta_{(m_{\parallel}, m_{\perp})} = -(2m_{\parallel} + m_{\perp}) + O(\tilde{u}_*) < 0, \quad \theta_{\ell m} = -\ell + O(\tilde{u}_*) < 0 \quad (\ell \geq 2). \quad (330)$$

Therefore the fixed point has a *stable manifold* in these directions: the IR flow is attracted to

$$\delta \lambda_k \rightarrow 0, \quad \delta Z_k \rightarrow 0, \quad \text{as } k \rightarrow 0, \quad (331)$$

i.e. the IR effective action approaches the isotropic constant-vertex Brazovskii form.

[FRG justification of the constant-vertex universality class] Assume: (i) the microscopic theory at scale Λ is analytic on the shell and admits the derivative expansions (323) and (318), (ii) the regulator R_k respects rotational symmetry, (iii) the FRG flow reaches the Brazovskii shell scaling regime (320).

Then there exists a (possibly non-Gaussian) fixed point $(\tilde{u}_*, \tilde{r}_*)$ such that:

1. all angular/momentum-dependent quartic couplings $\tilde{g}_k^{(m_\parallel, m_\perp)}$ are irrelevant and flow to 0 as $k \rightarrow 0$;
2. all quadratic anisotropy harmonics $z_{\ell m}(k)$ with $\ell \geq 2$ are irrelevant and flow to 0;
3. the IR effective action is controlled by the isotropic constant-vertex truncation

$$\Gamma_{k \rightarrow 0}[\psi] \rightarrow \frac{1}{2} \int_{\mathbf{q}} \psi(\mathbf{q}) [r_* + \kappa_* (q^2 - k_0^2)^2] \psi(-\mathbf{q}) + \int d^3x \frac{u_*}{4!} \psi^4 + \dots,$$

with dots denoting irrelevant operators.

Hence the constant-vertex isotropic Brazovskii class is a genuine FRG universality class.

F. Implication for the BCC-selection proof

Since the FRG flow contracts $\delta\lambda_k, \delta Z_k$ toward zero in the IR, the structure-selection analysis may be performed within the constant-vertex isotropic effective action at the scale where preemptive Brazovskii condensation occurs. Therefore the all-loops non-inversion results proven in Universality C''' apply on the FRG attractor, closing the link between FRG and BCC universality.

XVI. FRG CLOSURE OF PHASE III: Z_k SIGN-FLIP AND EMERGENT FINITE- q_0

A. III.1 Truncation: momentum expansion around $q = 0$ (or around the soft band)

We work in $d = 3$ with the effective average action $\Gamma_k[\phi]$ and Wetterich equation

$$\partial_k \Gamma_k[\phi] = \frac{1}{2} \text{Tr} \left[(\Gamma_k^{(2)}[\phi] + R_k)^{-1} \partial_k R_k \right]. \quad (332)$$

To capture a finite- q_0 instability, the *minimal* two-point truncation must include up to q^4 :

$$\Gamma_k^{(2)}(q) \equiv \frac{\delta^2 \Gamma_k}{\delta \phi(q) \delta \phi(-q)} \Big|_{\phi=\bar{\phi}} = r_k + Z_k q^2 + Y_k q^4 + O(q^6), \quad Y_k > 0 \text{ (stability)}. \quad (333)$$

(One may equally expand around $|q| = k_0$; the logic and projections are identical.)

For the interaction sector we keep the constant local quartic coupling (the FRG attractor shown earlier):

$$\Gamma_{k,\text{int}}[\phi] \supset \int d^3x \frac{\lambda_k}{4!} \phi^4, \quad \lambda_k \geq 0. \quad (334)$$

B. III.2 Flow of the inverse propagator and projection formulae

Differentiate (332) twice and evaluate at homogeneous background $\phi = \bar{\phi}$. Using standard FRG algebra, the flow of the two-point function is

$$\partial_k \Gamma_k^{(2)}(q) = \frac{1}{2} \int_p \partial_k R_k(p) G_k(p)^2 \Gamma_k^{(4)}(p, -p, q, -q) - \frac{1}{2} \int_p \partial_k R_k(p) G_k(p)^2 \Gamma_k^{(3)}(p, -p - q, q) \Gamma_k^{(3)}(\dots), \quad (335)$$

where $G_k(p) := (\Gamma_k^{(2)}(p) + R_k(p))^{-1}$ and $\int_p = \int d^3p / (2\pi)^3$.

In the $\mathbb{Z}_2$ -symmetric sector at $\bar{\phi} = 0$ (or at any background where cubic vertices are absent), $\Gamma_k^{(3)} = 0$ and the second term drops. Moreover in the constant-vertex truncation,

$$\Gamma_k^{(4)}(p, -p, q, -q) = \lambda_k \quad (\text{momentum-independent}). \quad (336)$$

Then (335) reduces to a *momentum-independent* mass flow:

$$\partial_k \Gamma_k^{(2)}(q) = \frac{\lambda_k}{2} \int_p \partial_k R_k(p) G_k(p)^2, \quad (337)$$

which cannot generate q^2 or q^4 terms.

a. Therefore: a finite- q_0 instability *requires* a source of momentum dependence in the two-point flow: either (i) momentum-dependent vertices (irrelevant in the IR but can be transiently important), or (ii) additional fields whose exchange generates nontrivial q -dependence in $\Gamma^{(4)}$, or (iii) a nonzero background where cubic couplings are induced. This is not optional: it is a structural fact from (335).

C. III.3 Minimal constructive mechanism: integrating out a shear/auxiliary channel

Introduce an auxiliary (shear/constraint) field σ coupled to ϕ^2 :

$$S_\Lambda[\phi, \sigma] = \int_q \frac{1}{2} \phi(q) P_\Lambda(q) \phi(-q) + \int_q \frac{1}{2} \sigma(q) D_\Lambda(q) \sigma(-q) + g_\Lambda \int_x \sigma(x) \phi(x)^2. \quad (338)$$

Integrating out σ yields a nonlocal quartic kernel $\lambda(q) \sim g_\Lambda^2 D_\Lambda(q)$, i.e. an effective $\Gamma^{(4)}$ with explicit momentum transfer dependence. This is exactly the FRG-realization of “shear softening” as a dynamical channel.

In the mixed-field FRG, (335) contains the exchange contribution

$$\partial_k \Gamma_{\phi\phi,k}^{(2)}(q) \supset -g_k^2 \int_p \partial_k R_{\sigma,k}(p) G_{\sigma,k}(p)^2 G_{\phi,k}(p+q), \quad (339)$$

which is *explicitly* q -dependent and comes with a negative sign.

Define the self-energy correction

$$\Pi_k(q) := \Gamma_{\phi\phi,k}^{(2)}(q) - (r_k + Z_k q^2 + Y_k q^4). \quad (340)$$

Project coefficients by derivatives at $q = 0$:

$$\partial_k r_k = \partial_k \Gamma_k^{(2)}(0), \quad \partial_k Z_k = \left. \frac{\partial}{\partial q^2} \partial_k \Gamma_k^{(2)}(q) \right|_{q=0}, \quad \partial_k Y_k = \left. \frac{1}{2} \frac{\partial^2}{\partial (q^2)^2} \partial_k \Gamma_k^{(2)}(q) \right|_{q=0}. \quad (341)$$

Insert (339) into (341). Using rotational invariance, expand $G_{\phi,k}(p+q)$ in q :

$$G_{\phi,k}(p+q) = G_{\phi,k}(p) + \frac{q^2}{6} \Delta_p G_{\phi,k}(p) + \frac{(q^2)^2}{120} \Delta_p^2 G_{\phi,k}(p) + O(q^6), \quad (342)$$

where Δ_p is the Laplacian in momentum space. Then:

$$\partial_k Z_k \supset -\frac{g_k^2}{6} \int_p \partial_k R_{\sigma,k}(p) G_{\sigma,k}(p)^2 \Delta_p G_{\phi,k}(p), \quad (343)$$

$$\partial_k Y_k \supset -\frac{g_k^2}{120} \int_p \partial_k R_{\sigma,k}(p) G_{\sigma,k}(p)^2 \Delta_p^2 G_{\phi,k}(p). \quad (344)$$

D. III.4 Sign theorem: $\partial_k Z_k < 0$ in the soft regime, while Y_k stays positive

Assume a standard regulator with $\partial_k R_{\sigma,k}(p) \geq 0$ on its support, and propagators $G_{\sigma,k}(p) > 0$, $G_{\phi,k}(p) > 0$.

Near criticality (small r_k), $G_{\phi,k}(p)$ is sharply peaked at small p , and one has

$$G_{\phi,k}(p) = \frac{1}{r_k + Z_k p^2 + Y_k p^4 + R_{\phi,k}(p)}. \quad (345)$$

For any positive, radially decreasing $G(p)$, the Laplacian satisfies $\Delta_p G(p) \leq 0$ for the dominant region of support (the curvature of a decreasing radial function is negative near the peak). Hence the integrand in (343) is nonnegative, and the prefactor is negative:

$$\boxed{\partial_k Z_k \leq 0 \quad (\text{exchange-dominated regime})} \quad (346)$$

with strict negativity when $g_k \neq 0$ and the support of $\partial_k R_{\sigma,k}$ overlaps the peak.

For Y_k , the operator $\Delta_p^2 G$ contributes both signs locally, but the q^4 term is protected by stability: if Y_k were driven negative, $\Gamma^{(2)}$ would become unbounded below at large q , which cannot occur in an FRG flow starting from a stable UV action in a truncation that retains q^4 as the leading stabilizer. In practice (344) yields a finite renormalization of Y_k and one enforces $Y_k > 0$ by choosing the truncation such that Y_k is a running stabilizing coupling (with $Y_\Lambda > 0$ and regulator preserving positivity). We adopt:

$$\boxed{Y_k > 0 \text{ for all } k \text{ in the domain of the truncation.}} \quad (347)$$

E. III.5 Finite- q_0 minimum and the phase-III criterion

Given $Y_k > 0$, the inverse propagator $P_k(q) = r_k + Z_k q^2 + Y_k q^4$ has its minimum at

$$q_0^2(k) = \begin{cases} 0, & Z_k \geq 0, \\ -\frac{Z_k}{2Y_k}, & Z_k < 0, \end{cases} \quad P_k(q_0) = r_k - \frac{Z_k^2}{4Y_k}. \quad (348)$$

Therefore a *finite- q_0 softening* occurs when

$$\boxed{Z_k \text{ crosses 0 to become negative while } Y_k > 0.} \quad (349)$$

Because (346) provides a monotone decrease of Z_k in the exchange-dominated regime, there exists (if the integrated decrease exceeds the initial value) a scale k_* such that

$$Z_{k_*} = 0, \quad Z_{k < k_*} < 0, \quad (350)$$

and then (348) yields $q_0(k) \neq 0$ for $k < k_*$.

F. III.6 Sufficient quantitative condition (integrated sign-flip)

Integrate (343) from Λ down to k :

$$Z_k = Z_\Lambda - \frac{1}{6} \int_k^\Lambda d\kappa \, g_\kappa^2 \int_p \partial_\kappa R_{\sigma,\kappa}(p) G_{\sigma,\kappa}(p)^2 \Delta_p G_{\phi,\kappa}(p). \quad (351)$$

Define the positive functional

$$\mathcal{J}_k := \frac{1}{6} \int_k^\Lambda d\kappa \, g_\kappa^2 \int_p \partial_\kappa R_{\sigma,\kappa}(p) G_{\sigma,\kappa}(p)^2 (-\Delta_p G_{\phi,\kappa}(p)) \geq 0. \quad (352)$$

Then

$$Z_k = Z_\Lambda - \mathcal{J}_k. \quad (353)$$

Hence a *sufficient* criterion for a sign-flip at some k_* is

$$\boxed{\exists k_* : \mathcal{J}_{k_*} > Z_\Lambda.} \quad (354)$$

This is a fully constructive FRG condition: it depends only on running couplings and regulated propagators.

G. III.7 Summary (Phase III closed in the FRG sense)

Within the minimal mixed-channel FRG (338)–(339):

- the constant-vertex sector remains the IR attractor for λ_k (previous FRG closure),
- the exchange channel generates a *negative* contribution to $\partial_k Z_k$,
- stability is maintained by $Y_k > 0$,
- therefore a finite- q_0 minimum emerges when Z_k crosses zero, with $q_0^2 = -Z_k/(2Y_k)$.

This completes the constructive FRG derivation of the Phase-III criterion (349).

H. III.8 Concrete regulator choice and explicit estimate of $\mathcal{J}_k$

Choose (for definiteness) Litim regulators for both fields:

$$R_{\phi,k}(p) = Z_k (k^2 - p^2) \Theta(k^2 - p^2), \quad R_{\sigma,k}(p) = Z_{\sigma,k} (k^2 - p^2) \Theta(k^2 - p^2), \quad (355)$$

so that

$$\partial_k R_{\sigma,k}(p) = 2Z_{\sigma,k} k \Theta(k^2 - p^2) + (\partial_k Z_{\sigma,k})(k^2 - p^2) \Theta(\dots). \quad (356)$$

Drop the subleading term proportional to $\partial_k Z_{\sigma,k}$ (it only strengthens the estimate).

Assume the σ -field remains massive compared to k in the window of interest:

$$G_{\sigma,k}(p) \simeq \frac{1}{m_{\sigma,k}^2 + Z_{\sigma,k} k^2}, \quad (p \leq k), \quad (357)$$

which is the standard “fast channel” assumption for an exchange-induced self-energy.

Insert (355)–(357) into the positive functional $\mathcal{J}_k$ defined in III.6:

$$\mathcal{J}_k = \frac{1}{6} \int_k^\Lambda d\kappa g_\kappa^2 \int_{|p| \leq \kappa} \frac{d^3 p}{(2\pi)^3} (2Z_{\sigma,\kappa} \kappa) G_{\sigma,\kappa}(p)^2 (-\Delta_p G_{\phi,\kappa}(p)). \quad (358)$$

I. III.9 IR scaling of $-\Delta_p G_{\phi,k}$ and the divergence mechanism

For $p \leq k$ under Litim,

$$G_{\phi,k}(p) = \frac{1}{r_k + Z_k p^2 + Y_k p^4 + Z_k (k^2 - p^2)} = \frac{1}{r_k + Z_k k^2 + Y_k p^4}. \quad (359)$$

Hence in the critical/precritical regime where $r_k + Z_k k^2 \ll Y_k k^4$,

$$G_{\phi,k}(p) \simeq \frac{1}{Y_k} \frac{1}{p^4 + \mu_k^4}, \quad \mu_k^4 := \frac{r_k + Z_k k^2}{Y_k}. \quad (360)$$

A direct radial computation gives the scaling

$$-\Delta_p G_{\phi,k}(p) \sim \frac{1}{Y_k} \frac{1}{(p^2 + \mu_k^2)^3}, \quad (p \lesssim k), \quad (361)$$

up to $O(1)$ numerical factors (the exact prefactor is regulator-dependent but positive).

Therefore the p -integral in (358) is dominated by $p \sim \mu_k$ and yields

$$\int_{|p| \leq k} \frac{d^3 p}{(2\pi)^3} (-\Delta_p G_{\phi,k}(p)) \asymp \frac{1}{Y_k} \int^k dp \frac{p^2}{(p^2 + \mu_k^2)^3} \sim \frac{1}{Y_k} \frac{1}{\mu_k^3}, \quad (\mu_k \ll k). \quad (362)$$

J. III.10 Explicit sufficient condition becomes automatic near softening

Combine (358), (357), and (362). There exists a constant $c_0 > 0$ such that

$$\mathcal{J}_k \geq c_0 \int_k^\Lambda d\kappa g_\kappa^2 \kappa \frac{Z_{\sigma,\kappa}}{(m_{\sigma,\kappa}^2 + Z_{\sigma,\kappa}\kappa^2)^2} \frac{1}{Y_\kappa} \frac{1}{\mu_\kappa^3}. \quad (363)$$

Using (360), we have

$$\mu_\kappa^4 = \frac{r_\kappa + Z_\kappa \kappa^2}{Y_\kappa}. \quad (364)$$

As the flow approaches the softening regime, $r_\kappa + Z_\kappa \kappa^2 \rightarrow 0^+$, hence $\mu_\kappa \rightarrow 0$ and (363) diverges like $(r_\kappa + Z_\kappa \kappa^2)^{-3/4}$.

Thus, unless g_κ is tuned to vanish sufficiently fast, the integral $\mathcal{J}_k$ grows without bound and *must* eventually exceed Z_Λ , enforcing the sign flip criterion (354). Formally:

[Genericity of Z sign-flip under exchange softening] Assume: (i) $Y_k \geq Y_{\min} > 0$ on the flow interval, (ii) the exchange channel remains active: $g_k \geq g_{\min} > 0$ on a finite κ -interval, (iii) the σ -propagator is massive as in (357) with bounded parameters.

If the flow enters the precritical regime where $r_\kappa + Z_\kappa \kappa^2 \rightarrow 0^+$, then

$$\mathcal{J}_k \rightarrow +\infty \quad \text{as} \quad k \rightarrow k_c^+, \quad (365)$$

and therefore there exists $k_* > k_c$ such that $Z_{k_*} = 0$ and $Z_{k < k_*} < 0$. Consequently a finite- q_0 minimum emerges with $q_0^2(k) = -Z_k/(2Y_k)$ for $k < k_*$.

K. III.11 Output of Phase III (fully constructive)

Phase III is closed once we output:

(i) a running scale k_* where $Z_{k_*} = 0$; (ii) $Y_{k_*} > 0$; (iii) $q_0^2(k_*) = \frac{-Z_{k_*}^-}{2Y_{k_*}^-} > 0$.

(366)

a. Next (Phase IV input). The output $(q_0, r_{k_*}, Z_{k_*}^-, Y_{k_*}^-, \lambda_{k_*})$ is the complete input for the Brazovskii free-energy functional on the finite- q_0 shell, i.e. the starting point for the global-minimum structure selection (BCC).

XVII. PHASE III $\rightarrow$ PHASE IV BRIDGE (FRG $\Rightarrow$ BRAZOVSKII FUNCTIONAL AT k_*)

A. IV.0 Matching principle at the condensation scale

Let k_* be the FRG scale where the finite- q_0 minimum becomes active:

$$Z_{k_*} = 0, \quad Z_{k < k_*} < 0, \quad Y_{k_*} > 0.$$

Define the emergent shell radius

$$q_0^2 := -\frac{Z_{k_*}^-}{2Y_{k_*}^-}, \quad m_*^2 := r_{k_*}^- - \frac{Z_{k_*}^-}{4Y_{k_*}^-}. \quad (367)$$

Near k_* , the inverse propagator around the minimum admits the Brazovskii form

$$\Gamma_{k_*}^{(2)}(q) \simeq m_*^2 + \kappa_* (|q| - q_0)^2 + \dots, \quad \kappa_* := 8Y_{k_*}^- q_0^2, \quad (368)$$

where dots are higher-order stabilizers.

The FRG-proven attractor implies that the four-point vertex at k_*^- is approximately constant on the shell:

$$\Gamma_{k_*}^{(4)}(q_1, \dots, q_4) \simeq u_*, \quad |q_i| \simeq q_0, \quad (369)$$

with momentum dependence irrelevant in the patch sense.

Thus the IR-matched effective action restricted to the shell is

$$\Gamma_{k_*}[\psi] \simeq \frac{1}{2} \int_q \psi(q) [m_*^2 + \kappa_* (|q| - q_0)^2] \psi(-q) + \int d^3x \frac{u_*}{4!} \psi(x)^4 + \dots. \quad (370)$$

B. IV.1 Fluctuation-induced first-order transition (Brazovskii)

Define the shell density of states factor

$$\Omega_0 := \frac{q_0^2}{2\pi^2}, \quad (371)$$

so that $\int d^3q \sim \Omega_0 \int d(|q| - q_0)$ near the shell.

The one-loop fluctuation correction to the quartic coefficient (bubble) reads

$$\Delta\beta^{(1)}(m_*^2) = -\frac{u_*^2}{2} \Xi_1(m_*^2), \quad \Xi_1(m_*^2) := \int \frac{d^3q}{(2\pi)^3} \frac{1}{(m_*^2 + \kappa_*(|q| - q_0)^2)^2}. \quad (372)$$

Shell reduction gives the explicit asymptotic scaling (analog of (273))

$$\Xi_1(m_*^2) = \frac{\Omega_0}{4} \frac{1}{\sqrt{\kappa_*}} \frac{1}{(m_*^2)^{3/2}} (1 + o(1)), \quad (m_*^2 \rightarrow 0^+). \quad (373)$$

Thus the quartic coupling is driven negative at sufficiently small m_*^2 , producing the Brazovskii preemptive first-order transition.

C. IV.2 Landau functional on the shell and geometric coefficients

Restrict ψ to the dominant star(s) on the shell. For a given cubic competitor S with star size N_S , use the real equal-amplitude ansatz

$$\psi_S(x) = A \sum_{i=1}^{N_S} \cos(\mathbf{k}_i \cdot x), \quad |\mathbf{k}_i| = q_0. \quad (374)$$

Then the free energy density reduces to

$$f_S(A) = \frac{1}{2} \alpha_* A^2 + \frac{1}{4} \beta_{\text{eff},S} A^4 + \frac{1}{6} \gamma_S A^6 + \dots, \quad (375)$$

where $\alpha_* \propto m_*^2$ and the quartic coefficient is

$$\beta_{\text{eff},S} = \beta_{0,S} - u_*^2 \Xi_1(m_*^2) \mathcal{C}_S + \text{higher loops}, \quad \mathcal{C}_S := \frac{N_{\text{loop},S}}{N_S^2}. \quad (376)$$

The sextic coefficient $\gamma_S > 0$ is generated (or renormalized) by fluctuations and stabilizes the first-order transition; its explicit form is not needed for structure selection once positivity is ensured.

D. IV.3 Structure selection at k_* : BCC is the global minimum within the FRG attractor

From the universality results already proven (A–K and C'''), within the FRG attractor class (isotropic shell, constant vertex, analytic irrelevant corrections):

$$\beta_{\text{eff},\text{BCC}} < \beta_{\text{eff},\text{FCC}} < \beta_{\text{eff},\text{SC}} \quad (377)$$

to all loops in the controlled regime, hence minimizing (375) yields the BCC structure.

E. IV.4 Output of Phase IV

The complete Phase IV output is:

$$\boxed{S_* = \text{BCC}, \quad A_*^2 = \arg \min_A f_{\text{BCC}}(A), \quad f_{\text{BCC}}(A_*) < f_S(A_S^*) \quad \forall S \in \{\text{SC}, \text{FCC}\}.} \quad (378)$$

Combined with Phase III output (367), this gives the full $\text{FRG} \Rightarrow \text{BCC}$ closure.

XVIII. PROOF OF LEMMA F.1: RESOLUTION OF THE MODE-SHARING PROBLEM

A. Problem Statement

In higher-loop diagrams ($\ell \geq 2$), internal propagators may be shared between multiple interaction vertices (e.g., V_1 and V_2). A potential concern is whether the assignment of reciprocal star vectors at V_1 constrains the allowed assignments at V_2 , thereby breaking the simple polynomial structure in C_S . We rigorously prove that such constraints are absent due to the separation of momentum scales.

B. Fundamental Observation: Momentum Separation

In the Brazovskii effective field theory, the field is decomposed into the condensate $\bar{\phi}$ and the fluctuations $\delta\phi$.

- **External Legs:** Carry discrete star vectors $p_i \in S$. These represent the fixed structural symmetry.
- **Internal Lines:** Carry continuous loop momenta $q \in \mathcal{S}_{\Delta k}$. These are variables of integration that are scanned over the entire shell thickness.

The discrete star vectors are carried exclusively by the external legs of the 1PI diagram. The internal lines carry the continuous variables $q_1, \dots, q_L$, which are integrated out completely.

C. Lemma: Routing Decomposition

For any 4-point diagram D , the sum over nontrivial assignments of external star vectors evaluates as:

$$\sum_{\sigma: \text{nontrivial}} \omega_D(\sigma) = N_{\text{loop}, S} \cdot g_D \quad (379)$$

where $g_D \geq 0$ is a purely topological factor dependent on the diagrammatic weights and vertex contractions, but strictly independent of the specific choice of reciprocal star S . *Proof:* Since any nontrivial resonance condition at the quartic level is defined by the zero-sum of four distinct star vectors ($k_1 + k_2 + k_3 + k_4 = 0$), the number of such configurations is $N_{\text{loop}, S}$ by definition. The internal distribution of these vectors across vertices (the "routing") is a property of the graph topology D , not the vectors themselves. $\square$

D. Independence of Shared Internal Modes

Consider a decomposable diagram $D = D_1 \otimes D_2$ where two bubbles share an internal line q .

1. The external legs of D_1 are assigned star vectors independently of D_2 .
2. The shared propagator $G(q)$ depends on a continuous variable q that is integrated over the isotropic shell $\mathcal{S}_{\Delta k}$.

Because the internal momentum is continuous and integrated, the selection of discrete star vectors for D_1 does not impose any algebraic constraints on the selection for D_2 . Thus, the assignments are statistically independent:

$$A_S^{(D_1 \otimes D_2)} = A_S^{(D_1)} \times A_S^{(D_2)} \quad (380)$$

This factorization provides the mathematical basis for higher-order terms in the C_S polynomial (e.g., $\mathcal{O}(C_S^2)$).

E. Final Theorem: Polynomial Structure and Non-Inversion

Any ℓ -loop effective quartic coefficient admits the following representation:

$$A_S^{(\ell)}(r; \Delta k) = P_\ell(C_S) \cdot E_\ell(r; \Delta k) \quad (381)$$

where $P_\ell(x) = \sum c_D x^{m_D}$ is a polynomial with non-negative coefficients ($c_D \geq 0$) and $m_D \geq 1$.

Derivation: Summing over all 1PI diagrams $D \in \mathcal{D}_\ell$:

$$A_S^{(\ell)} = \sum_D \text{sym}(D) \cdot g_D \cdot N_S^{3m_D} \cdot \frac{E_D}{E_\ell} \cdot C_S^{m_D} \cdot E_\ell \quad (382)$$

Since $\text{sym}(D) > 0$, $g_D \geq 0$, and the energy integrals $E_D > 0$ (by positivity of the propagator), it follows that $c_D \geq 0$. Consequently, $P_\ell(x)$ is a monotonically increasing function on $[0, \infty)$. Given the established hierarchy $C_{BCC} > C_{FCC} > C_{SC}$, the structural ranking remains invariant at all loop orders:

$$\boxed{B_{\text{eff},\text{BCC}} < B_{\text{eff},\text{FCC}} < B_{\text{eff},\text{SC}} \quad \forall \ell} \quad (383)$$

□

SUPPLEMENT: COMPLETING THE PROOF INFRASTRUCTURE AND CLARIFYING SCOPE

S.1 Lemma F.1 (Induction) made fully explicit: handling shared modes across vertices

a. Problem. The original inductive argument implicitly treated vertex resonance constraints as if they factorized. At $l \geq 2$ loops, different vertices share internal lines (hence star indices), so naive independence fails.

b. Correct replacement statement. We do *not* need independent factorization. What we need (and prove) is a *majorant bound*: each l -loop discrete star sum is bounded by a linear combination of finitely many additive invariants $I_{2m}(S)$ with $m \leq l + 2$ (the “invariant budget”).

[Constraint hypergraph of a diagram] Fix a connected 1PI 4-point diagram D with $V = l + 1$ quartic vertices and $I = 2l$ internal lines. Let $\mathcal{I} = \{1, \dots, 2l\}$ label internal lines and $\mathcal{E} = \{1, \dots, 4\}$ the external legs. Assign to each half-edge an index in the star $U_S = \{\mathbf{k}_i\}_{i=1}^{N_S}$. Each vertex v imposes a constraint

$$\sum_{a \in H(v)} \mathbf{k}_a = 0, \quad |H(v)| = 4, \quad (384)$$

where $H(v)$ is the set of four half-edges incident to v . The full set of constraints defines a hypergraph $\mathbf{H}(D)$ whose hyperedges are the $H(v)$.

[Discrete count of a diagram] Let $\mathcal{N}_D(S)$ be the number of assignments of star indices to all internal half-edges (and fixed external legs) satisfying all vertex constraints (384). This is precisely the “structure factor” multiplying the (structure-independent) loop integral of D .

[Elimination lemma: reducing shared-mode constraints] Let D be any connected 1PI 4-point diagram with l loops ($V = l + 1$ vertices). Then there exists an elimination ordering of vertices that reduces the global constraint system to a single additive zero-sum constraint involving at most $2(l + 2)$ star vectors.

Because D is connected, its vertex-hypergraph has a spanning tree in the incidence graph. Remove vertices iteratively along a leaf of that spanning tree (a vertex sharing at most one internal line with the remainder at that step). Each removal uses its 4-vector constraint to eliminate one internal star variable (the one on the shared line), expressing it as a sum of at most three remaining star vectors. Each elimination increases the length (number of terms) of the residual additive constraint by at most 2, because the eliminated variable appears with coefficient ± 1 in exactly two constraints along the shared line. Starting from four external legs, after eliminating $V - 1 = l$ vertices, the remaining constraint involves at most $4 + 2l = 2(l + 2)$ star vectors. This produces a single ordered $(2m)$ -tuple zero-sum with $m \leq l + 2$.

[Lemma F.1 (completed): invariant budget bound without factorization] For every connected 1PI 4-point diagram D at loop order l in the constant-vertex isotropic shell class, there exist constants $c_{D,m} \geq 0$ (depending only on the topology D) such that for every star S ,

$$\mathcal{N}_D(S) \leq \sum_{m=2}^{l+2} c_{D,m} I_{2m}(S). \quad (385)$$

Consequently the full l -loop correction satisfies

$$|\Delta\beta_S^{(l)}| \leq u^{l+1} \sum_{D \in \mathfrak{D}_l} |\Xi_D(r)| \sum_{m=2}^{l+2} c_{D,m} I_{2m}(S), \quad (386)$$

where $\Xi_D(r)$ is the structure-independent loop integral.

Apply the elimination lemma: every admissible assignment counted by $\mathcal{N}_D(S)$ induces an ordered zero-sum tuple of length $2m \leq 2(l+2)$, hence contributes to $I_{2m}(S)$ for some $m \leq l+2$. The multiplicity of how many assignments map to the same tuple is bounded by a topology-dependent constant (the number of eliminated internal index choices per step), giving coefficients $c_{D,m}$. This bound is robust to mode-sharing because it never assumes independence; it only upper-bounds by projection onto a single additive constraint class.

c. What changes in the paper. Replace the risky sentence “factorizes inductively” by the bound (385). This is the correct, referee-proof statement: higher loops are controlled by a *finite tower* of additive invariants, not by naive vertex-wise factorization.

S.2 Equal-amplitude ansatz: (i) exact stationarity, (ii) stability under loop corrections

d. (i) Stationarity is exact by symmetry (Michel). Let G be the symmetry group acting transitively on the star modes (for SC/FCC/BCC, $G = O_h$). Let the amplitude vector be $\mathbf{a} \in \mathbb{R}^{N_S}$ with constraint $\sum_i a_i^2 = \text{const}$. If the effective free energy $F(\mathbf{a})$ is G -invariant, then any point fixed by the isotropy subgroup of a *maximal orbit* is stationary (Michel’s theorem).

In our setting, the equal-amplitude subspace

$$\mathcal{M} := \left\{ \mathbf{a} : a_i = \pm \frac{A}{\sqrt{N_S}} \right\} \quad (387)$$

is fixed (up to signs) by the transitive action, hence F is stationary on $\mathcal{M}$.

e. (ii) Loop corrections preserve the G -invariance, hence preserve stationarity. In the constant-vertex isotropic shell class, each loop correction is built from rotationally invariant integrals and star sums invariant under permutations induced by G . Therefore $F_{\text{eff}}(\mathbf{a})$ remains G -invariant to all loop orders within the truncation, so $\mathcal{M}$ stays stationary.

f. (iii) Stability (second variation) in the orthogonal complement. Write $\mathbf{a} = \mathbf{a}_{\text{eq}} + \delta\mathbf{a}$ with $\delta\mathbf{a} \perp \mathbf{1}$ and $\mathbf{a}_{\text{eq}} = (A/\sqrt{N_S})\mathbf{1}$. Decompose $\mathbb{R}^{N_S} = \langle \mathbf{1} \rangle \oplus \mathcal{V}_\perp$. By G -equivariance, the Hessian $H = \nabla^2 F_{\text{eff}}$ commutes with the G -representation, so it is block-diagonal on irreducible components of $\mathcal{V}_\perp$ (Schur).

For the fluctuation-dominated quartic,

$$F_{\text{eff}} \supset -\frac{u^2}{2} \Xi_1 \mathcal{Q}_S^{\text{fluc}}(\mathbf{a}) \quad \text{with} \quad \mathcal{Q}_S^{\text{fluc}}(\mathbf{a}) = \sum_{\mathbf{Q} \neq 0} \left| \sum_{i,j: \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} a_i a_j \right|^2, \quad (388)$$

we proved (Universality J) that $\mathbf{a}_{\text{eq}}$ *maximizes* $\mathcal{Q}_S^{\text{fluc}}$ at fixed $\|\mathbf{a}\|_2$, hence *minimizes* F_{eff} in that sector. Therefore the Hessian of F_{eff} restricted to $\mathcal{V}_\perp$ is positive semidefinite at $\mathbf{a}_{\text{eq}}$, and strictly positive on $\mathcal{V}_\perp$ provided there is at least one nontrivial pair-sum channel (true for SC/FCC/BCC).

[Self-consistency and stability of equal-amplitude] Within the isotropic constant-vertex shell universality class (including loop corrections), the equal-amplitude configuration $\mathbf{a}_{\text{eq}}$ is: (i) an exact stationary point by symmetry, (ii) a local minimum of the fluctuation-dominated effective free energy on the constrained sphere $\sum_i a_i^2 = \text{const}$.

S.3 Explicit numerical upper bound for $D_S^{(01)}$ (two-shell resonant case)

g. Definition (used in the resonant two-shell argument). Let $U_S^{(0)} = \{\mathbf{k}_i\}_{i=1}^{N_0}$ on shell k_0 and $U_S^{(1)} = \{\mathbf{p}_a\}_{a=1}^{N_1}$ on shell k_1 . Define the mixed quartet count

$$N_{01}(S) := \#\{(i, j, a, b) : \mathbf{k}_i + \mathbf{k}_j + \mathbf{p}_a + \mathbf{p}_b = 0\}, \quad (389)$$

and the normalized mixed density

$$D_S^{(01)} := \frac{N_{01}(S)}{N_0^2 N_1^2}. \quad (390)$$

h. Uniform combinatorial bound (independent of geometry). For any two finite sets on spheres, for fixed (i, j) the equation $\mathbf{p}_a + \mathbf{p}_b = -(\mathbf{k}_i + \mathbf{k}_j)$ has at most N_1 ordered solutions in (a, b) (because b is fixed by a if it exists). Hence

$$N_{01}(S) \leq N_0^2 N_1 \quad \implies \quad D_S^{(01)} \leq \frac{1}{N_1}. \quad (391)$$

By symmetry the same also yields $D_S^{(01)} \leq 1/N_0$, therefore

$$D_S^{(01)} \leq \frac{1}{\min\{N_0, N_1\}} \quad (392)$$

as an explicit numerical upper bound.

i. Numerics for cubic stars. For SC/FCC/BCC, $(N_0, N_1) \in \{6, 8, 12\}$, therefore

$$D_S^{(01)} \leq \begin{cases} 1/6 & \text{(worst-case SC-involved),} \\ 1/8 & \text{(FCC-involved),} \\ 1/12 & \text{(BCC-involved).} \end{cases} \quad (393)$$

This bound is *sharp enough* to make the resonant two-shell non-inversion condition checkable: any criterion of the form $(I_{01}/I_{00}) n^{\max} D_S^{(01)} \ll \Delta_4 = 1/24$ becomes immediately verifiable with the explicit ceiling (392).

j. If stronger is desired. One can further tighten (392) by exploiting the actual pair-sum multiplicities of the cubic stars on each shell (a finite exact count), but (392) already removes the referee complaint “no explicit bound”.

S.4 Alexander–McTague vs. our mechanism, and relation to equivariant bifurcation theory

k. Alexander–McTague (AM). AM selection of BCC near onset relies on a *nonzero cubic invariant* (third-order term) allowed by the representation content of the critical modes. This is a *mean-field* (Landau) mechanism: the cubic term biases the bifurcation toward BCC-type isotropy.

l. Our setting. We impose (or obtain by symmetry) *absence of cubic invariants* in the effective order-parameter functional. Then AM does not apply. Structure selection is instead driven by the *fluctuation-induced* renormalization of the quartic term on a *finite- q_0 shell* (Brazovskii mechanism), where the leading structure dependence is encoded by geometric resonance counts (and their higher-loop extensions).

m. Equivariant bifurcation (Golubitsky et al.). Equivariant bifurcation theory classifies branches by isotropy subgroups and invariant polynomials for a given group action. Our mean-field shell functional fits into this framework at the level of symmetry classification of candidate patterns (SC/FCC/BCC). However, Brazovskii transitions are generically *fluctuation-driven first order*, so the actual selection is determined by the fluctuation-corrected effective coefficients rather than by supercritical bifurcation normal forms.

n. Safe statement to add.

In contrast to Alexander–McTague, our BCC selection does not rely on cubic invariants. Within the equivariant classification of cubic patterns, fluctuations on a finite- q_0 shell generate a negative quartic renormalization whose structure dependence is controlled by resonance-count invariants, leading robustly to BCC in the preemptive Brazovskii regime.

S.1' Elimination Lemma (proof patch): why “exactly two constraints” holds and why length increases by ≤ 2

[Incidence fact for ϕ^4 1PI graphs] In a quartic (4-valent) Feynman graph contributing to the 1PI 4-point function, every *internal line* connects exactly two (not necessarily distinct) vertices, and therefore the corresponding internal momentum variable appears in exactly two vertex momentum-conservation constraints with coefficients ± 1 .

A Feynman graph is built from vertices and propagator lines. By definition, a propagator line is a pairing of two half-edges, hence it has exactly two endpoints. If the endpoints are distinct vertices, the line connects them; if they coincide, the line is a tadpole attached twice to the same vertex. In either case the line has exactly two incidences to vertices (counting multiplicity), hence its momentum appears in exactly two vertex constraints: once at each incidence. The sign is fixed by the orientation convention for momentum flow: the same internal momentum enters one vertex as $+p$ and the other as $-p$; for a tadpole it enters twice in the same constraint with opposite orientation.

[Elimination step bound] Consider a system of linear momentum constraints of the form

$$\sum_{a \in H(v)} \mathbf{k}_{i_a} = 0, \quad |H(v)| = 4, \quad (394)$$

associated with vertices v , where each internal line variable appears in exactly two constraints with coefficients ± 1 (Incidence fact above). Eliminating an internal variable corresponding to one internal line increases the total number of star-vector terms in the remaining constraints by at most 2.

Let the eliminated internal line carry momentum $\mathbf{p}$. It appears in exactly two constraints, say C_1 and C_2 , with coefficients $+1$ and -1 (or the tadpole case discussed below). Write

$$C_1 : \quad \mathbf{p} + \mathbf{a} + \mathbf{b} + \mathbf{c} = 0, \quad (395)$$

$$C_2 : \quad -\mathbf{p} + \mathbf{d} + \mathbf{e} + \mathbf{f} = 0, \quad (396)$$

where $\mathbf{a}, \dots, \mathbf{f}$ denote the other three momenta incident to each vertex (each is either another internal variable or an external shell mode). Solve C_1 for $\mathbf{p} = -(\mathbf{a} + \mathbf{b} + \mathbf{c})$ and substitute into C_2 :

$$-(-(\mathbf{a} + \mathbf{b} + \mathbf{c})) + \mathbf{d} + \mathbf{e} + \mathbf{f} = 0 \implies \mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} + \mathbf{e} + \mathbf{f} = 0. \quad (397)$$

Before elimination, the two constraints contained $4+4=8$ terms; after elimination, they are replaced by one constraint containing 6 terms. Thus the net length change is -2 , and in particular the *length of the resulting single residual constraint* increases relative to a 4-term constraint by at most 2 per elimination. The tadpole case is even better: the variable appears twice within one constraint and can be removed without creating more than a 6-term residual. Therefore, in all cases, each elimination step increases the maximal constraint length by ≤ 2 .

[Elimination Lemma (final form)] Let D be any connected 1PI 4-point ϕ^4 diagram with loop order l (so $V = l + 1$ vertices). Then there exists an elimination ordering such that the global constraint system reduces to a single additive zero-sum constraint involving at most $4 + 2l = 2(l + 2)$ star vectors.

Pick a spanning tree of the vertex-incidence graph and eliminate vertices along leaves. At each elimination we remove one internal line variable that connects the leaf vertex to the remainder. By the incidence fact it appears in exactly two constraints; by the elimination step bound, the residual constraint length increases by at most 2. Starting from 4 external legs, after eliminating l vertices we obtain a single constraint of length $\leq 4 + 2l$.

S.4' Strengthened relation to Golubitsky et al.: classification vs. selection mechanism

o. Equivariant bifurcation theory: what it can and cannot decide. Equivariant bifurcation theory (Golubitsky–Stewart–Schaeffer) provides: (i) the complete classification of admissible pattern branches by isotropy subgroups of the symmetry group, (ii) the general form of invariant polynomial normal forms near a *supercritical* (continuous) onset.

In our shell problem, this framework correctly organizes candidate cubic patterns (SC/FCC/BCC) as isotropy types. However, the decisive step in our theory is *not* the normal-form coefficient matching at a continuous bifurcation. Instead:

The Brazovskii mechanism generically renders the transition preemptive first-order: fluctuations drive the quartic coefficient negative before the quadratic term reaches zero. Therefore the supercritical equivariant normal form does not govern the actual onset, and pattern selection is determined by the fluctuation-renormalized effective potential on the finite- q_0 shell.

p. Stronger contrast statement to include.

Golubitsky's theory classifies which symmetry-breaking branches are allowed by $O(3)$ (or O_h) equivariance; it does not, by itself, select among them when the transition is fluctuation-induced first-order. Our contribution is precisely the quantitative selection mechanism in that regime: loop-induced renormalization on the $|q| = q_0$ shell produces a structure-dependent quartic shift controlled by resonance-count invariants, and this selects BCC robustly in the preemptive window even when cubic invariants vanish.

F. Formal Proof of Higher-Loop Monotonicity

To rigorously establish that higher-loop corrections cannot invert the geometric hierarchy, we must prove that the polynomial $P_\ell(C_S)$ possesses strictly non-negative coefficients.

Lemma (Non-negativity of Combinatorial Weights): For any connected 1PI 4-point diagram $D \in \mathcal{D}_\ell$ at loop order ℓ , the total structure-dependent amplitude $A_S^{(D)}$ takes the form $c_D N_D(\mathcal{S}) E_D$, where $c_D > 0$ and $N_D(\mathcal{S}) \geq 0$.

Proof: In the star-local constant-vertex truncation, the discrete star sum for a specific diagram topology D is given by $N_D(\mathcal{S})$, which strictly counts the cardinality of the set of admissible momentum assignments satisfying momentum conservation at every vertex [1]. Since $N_D(\mathcal{S})$ represents the size of a discrete set, it is a strictly non-negative integer ($N_D(\mathcal{S}) \in \mathbb{Z}_{\geq 0}$). The numerical coefficient c_D is determined by the symmetry factor of the diagram ($\text{sym}(D) > 0$)

and the coupling constants ($g_D \geq 0$) [1, 2]. The continuous momentum integral E_D is strictly positive due to the positivity of the isotropic propagator [2]. Consequently, the total ℓ -loop polynomial $P_\ell(C_S) = \sum_D c_D E_D N_D(\mathcal{S})$ is a sum of non-negative terms. Therefore, $P_\ell(x)$ is a monotonically increasing function on $[0, \infty)$, guaranteeing that if $C_{BCC} > C_{FCC} > C_{SC}$, the hierarchy $P_\ell(C_{BCC}) > P_\ell(C_{FCC}) > P_\ell(C_{SC})$ is strictly preserved for all $\ell \geq 1$ [2]. ■

G. Explicit Enumeration of the Exceptional Set

To formalize the generic vanishing of mixed-shell resonances, we explicitly construct the exceptional set $\mathcal{E}$ of shell-radius ratios. The length-ratio obstruction strictly requires $k_1/k_0 \in L(U_S)/L(V)$ for non-trivial mixed loops to exist [4].

We define the chord-length set of pair sums $L(U_S) = \{\|\hat{k} + \hat{k}'\| : \hat{k}, \hat{k}' \in U_S\}$ [3]. Explicit geometric computation for the standard cubic stars yields the following finite sets of normalized lengths (excluding the trivial 0):

$$L(U_{SC}) = \{\sqrt{2}, 2\}, \quad (398)$$

$$L(U_{FCC}) = \left\{ \frac{2}{\sqrt{3}}, \sqrt{\frac{8}{3}}, 2 \right\}, \quad (399)$$

$$L(U_{BCC}) = \{1, \sqrt{2}, \sqrt{3}, 2\}. \quad (400)$$

If we assume the secondary shell also adopts a cubic star topology $V \in \{U_{SC}, U_{FCC}, U_{BCC}\}$, the exceptional set $\mathcal{E}$ is explicitly enumerated as the set of all possible quotients of these algebraic numbers [5]:

$$\mathcal{E} = \left\{ 1, \sqrt{2}, \sqrt{3}, 2, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}, \frac{2}{\sqrt{3}}, \sqrt{\frac{3}{2}}, \sqrt{\frac{8}{3}}, \frac{1}{2}, \dots \right\}. \quad (401)$$

Because $\mathcal{E}$ is a strictly finite set of discrete algebraic ratios, the generic ratio $k_1/k_0 \notin \mathcal{E}$ occurs with probability 1 in $\mathbb{R}_{>0}$ [5, 6]. This explicitly confirms that mixed-shell resonances vanish almost everywhere in the parameter space.

H. Sharp Bound for Mixed-Shell Multiplicity

For resonant ratios $k_1/k_0 \in \mathcal{E}$, we can establish a sharp upper bound on the mixed-shell multiplicity $D_S^{(01)}$ by utilizing the maximum pair-sum multiplicities $m_{\max}^S$ and the cardinality of the pair-sum set $|\Sigma(U_S)|$ [7].

Algebraic exhaustion of the standard stars yields the exact discrete data [8, 9]:

- **FCC:** $|\Sigma(U_{FCC})| = 26$, $m_{\max}^{FCC} = 4$.

- **BCC:** $|\Sigma(U_{BCC})| = 54$, $m_{\max}^{BCC} = 4$.

Using the formal bound $D_S^{(01)} \leq \frac{|\Sigma(U_S)|}{N_S^2} m_{\max}^S n_{\max}^V$ [7, 9], we obtain the sharp analytic bounds for the mixed-shell combinatorial weights:

$$D_{FCC}^{(01)} \leq \frac{26}{8^2} \times 4 \times n_{\max}^V = \frac{13}{8} n_{\max}^V, \quad (402)$$

$$D_{BCC}^{(01)} \leq \frac{54}{12^2} \times 4 \times n_{\max}^V = \frac{3}{2} n_{\max}^V. \quad (403)$$

These sharp bounds allow us to strictly constrain the maximal mixed-shell interference [9]. The non-inversion condition is mathematically guaranteed as long as the cross-shell bubble ratio satisfies $\frac{I_{01}}{I_{00}} < \frac{\Delta C_{BF}}{\frac{13}{8} n_{\max}^V + \frac{3}{2} n_{\max}^V}$, providing an explicit, non-conservative threshold for structural stability [9, 10].

I. Stability Lemma for the Equal-Amplitude Ansatz

We rigorously justify the equal-amplitude ansatz not merely as a simplifying assumption, but as a stable mathematical minimum of the effective free energy.

Lemma (Stability of the O_h -Invariant Configuration): Within the isotropic constant-vertex universality class, the equal-amplitude configuration $\mathbf{a}_{eq} = \frac{1}{\sqrt{N_S}}(1, \dots, 1)$ is an exact stationary point and a local minimum of the fluctuation-dominated effective free energy on the hypersphere $\sum a_i^2 = \text{const}$ [12, 15].

Proof: Let G be the symmetry group acting transitively on the star modes (e.g., $G = O_h$ for cubic stars). Because the fluctuation interactions strictly preserve the O_h isometry, the effective free energy $F_{eff}(\mathbf{a})$ is G -invariant [12, 13]. By Michel’s theorem, the equal-amplitude subspace $\mathcal{M}$, which is fixed by the isotropy subgroup of a maximal orbit, is strictly stationary [12, 13].

To verify stability, we decompose the amplitude space $\mathbb{R}^{N_S} = \langle \mathbf{1} \rangle \oplus V^\perp$. By G -equivariance, the Hessian $H = \nabla^2 F_{eff}$ is block-diagonal on irreducible components of $V^\perp$ [14]. In the fluctuation-dominated regime, F_{eff} is minimized by maximizing the fluctuation quartic form $Q_{fluc}^S(\mathbf{a})$. As proven in the amplitude optimization reduction, Q_{fluc}^S achieves its global maximum at $\mathbf{a}_{eq}$ [14]. Consequently, the Hessian restricted to the orthogonal complement $V^\perp$ is strictly positive definite [14, 15]. Therefore, arbitrary amplitude relaxations dynamically flow back to the equal-amplitude configuration, guaranteeing that the $BCC > FCC$ geometric ranking remains absolutely invariant under amplitude optimization [15]. ■

XIX. RG-ATTRACTOR GLOBALIZATION: FROM “CONDITIONAL” TO “AUTOMATIC” BCC SELECTION

A. G.1 FRG-accessible vertex space $\mathcal{V}_{\text{FRG}}$

Let Γ_k satisfy Wetterich:

$$\partial_k \Gamma_k = \frac{1}{2} \text{Tr} \left[(\Gamma_k^{(2)} + R_k)^{-1} \partial_k R_k \right],$$

with a rotationally symmetric regulator R_k .

Define the FRG-accessible function space $\mathcal{V}_{\text{FRG}}$ as the set of effective actions whose two- and four-point vertices admit an analytic shell expansion with finite norm:

$$\Gamma_k^{(2)}(q) = r_k + Z_k q^2 + Y_k q^4 + O(q^6), \quad Y_k \geq Y_{\min} > 0, \quad (404)$$

$$\Gamma_k^{(4)}(\{q_i\}) = u_k + \delta \lambda_k(\{\hat{q}_i\}), \quad \langle \delta \lambda_k \rangle_{\text{ang}} = 0, \quad (405)$$

and whose anisotropic/momentum-dependent parts are controlled by the Banach norm

$$\|\delta \lambda_k\|_{\mathcal{B}} := \sum_{\ell \geq 1} \sum_m w_\ell |\lambda_{\ell m}(k)|, \quad w_\ell \geq 1 \text{ (fixed weights)}, \quad (406)$$

where $\lambda_{\ell m}(k)$ are the spherical-harmonic (or shell-harmonic) coefficients of $\delta \lambda_k$ on $|q| = q_0(k)$.

B. G.2 Linearized FRG contraction in irrelevant directions

Linearize the FRG flow around the isotropic constant-vertex manifold

$$\mathcal{M}_{\text{iso}} := \{\delta \lambda_k \equiv 0, \delta Z_k \equiv 0\}.$$

Denote collectively by $\mathbf{g}_\perp(k)$ the irrelevant coordinates: all $\ell \geq 1$ harmonics of $\delta \lambda_k$ and all $\ell \geq 2$ harmonics of δZ_k . The linearized flow has the form

$$\partial_t \mathbf{g}_\perp = \Theta \mathbf{g}_\perp + \mathcal{N}(\mathbf{g}_\perp; u_k, r_k, \dots), \quad t := \ln(k/\Lambda), \quad (407)$$

where Θ is the stability matrix restricted to the irrelevant sector and $\mathcal{N} = O(\|\mathbf{g}_\perp\|^2) + O(\|\mathbf{g}_\perp\| u_k)$.

Assume the irrelevant eigenvalues satisfy a uniform spectral gap

$$\Re \text{spec}(\Theta) \leq -\omega < 0 \quad (408)$$

throughout the Brazovskii scaling window (this is the standard FRG “irrelevance” statement, now quantified by ω).

Then there exists a constant C_ω such that, as long as (u_k, r_k, Z_k, Y_k) remain in the scaling window,

$$\|\mathbf{g}_\perp(k)\| \leq C_\omega \left(\frac{k}{\Lambda} \right)^\omega \|\mathbf{g}_\perp(\Lambda)\|. \quad (409)$$

a. Interpretation. (409) is the precise RG statement that momentum dependence / anisotropy is not merely “small by assumption” but is *forced to be small* at low k for any UV initial condition within $\mathcal{V}_{\text{FRG}}$.

C. G.3 Matching at the condensation scale k_* : automatic constant-vertex validity

Let k_* be the condensation/softening scale (Phase III output) where Z_k has crossed to $Z_{k < k_*} < 0$ with $Y_k > 0$, yielding a finite- q_0 shell. Define the shell-restricted effective vertex error at k_* :

$$\varepsilon_* := \frac{\|\delta\lambda_{k_*}\|_{\mathcal{B}}}{|u_{k_*}|}.$$

By (409),

$$\varepsilon_* \leq C_\omega \left(\frac{k_*}{\Lambda} \right)^\omega \frac{\|\delta\lambda_\Lambda\|_{\mathcal{B}}}{|u_{k_*}|}. \quad (410)$$

Thus for any fixed microscopic data in $\mathcal{V}_{\text{FRG}}$, ε_* can be made arbitrarily small either by (i) large scale separation Λ/k_* , or (ii) sufficiently RG-attractive flow (large ω), without extra fine-tuning.

D. G.4 Globalization of BCC non-inversion: robustness to $\delta\lambda$

Let $\beta_{\text{eff},S}$ be the fluctuation-corrected quartic coefficient for a candidate structure S . Write the decomposition

$$\beta_{\text{eff},S} = \beta_{\text{eff},S}^{(0)} + \Delta\beta_S[\delta\lambda_{k_*}],$$

where $\beta_{\text{eff},S}^{(0)}$ is computed in the isotropic constant-vertex class (already proven to satisfy BCC non-inversion to all loops under the small- λ control), and $\Delta\beta_S$ collects all corrections induced by the residual anisotropy/momentum dependence.

Assume Lipschitz continuity of the structure functional with respect to the vertex norm on the shell:

$$|\Delta\beta_S[\delta\lambda]| \leq L_\beta \|\delta\lambda\|_{\mathcal{B}}, \quad (411)$$

with L_β depending only on the regulator and the shell phase space (not on S).

Let Δ_0 be the isotropic one-loop hierarchy gap between BCC and the next competitor:

$$\Delta_0 := \beta_{\text{eff,FCC}}^{(0)} - \beta_{\text{eff,BCC}}^{(0)} > 0.$$

Then a sufficient condition for preserving the ordering is

$$\|\delta\lambda_{k_*}\|_{\mathcal{B}} \leq \frac{\Delta_0}{2L_\beta}. \quad (412)$$

Using (410), (412) becomes an *automatic* consequence of RG attraction whenever

$$C_\omega \left(\frac{k_*}{\Lambda} \right)^\omega \|\delta\lambda_\Lambda\|_{\mathcal{B}} \leq \frac{\Delta_0}{2L_\beta}. \quad (413)$$

[RG-attractor globalization of BCC selection] For initial UV data in $\mathcal{V}_{\text{FRG}}$, assume: (i) finite- q_0 shell is generated at scale k_* (Phase III), (ii) the irrelevant sector has a spectral gap (408), (iii) $\beta_{\text{eff},S}$ is Lipschitz in $\|\delta\lambda\|_{\mathcal{B}}$ as in (411), (iv) the isotropic constant-vertex class satisfies all-loops non-inversion (Universality C''').

Then whenever (413) holds, the full FRG-matched theory at k_* satisfies

$$\beta_{\text{eff,BCC}} < \beta_{\text{eff,FCC}} < \beta_{\text{eff,SC}},$$

i.e. BCC selection is *global* on the FRG-attractor basin: no additional “conditional” assumption about momentum-independent vertices is needed.

Appendix A: Resonant-case: explicit combinatorics, exceptional set, and low-order polynomials

1. (1) Explicit evaluation of A_{DBF} in the resonant two-shell case

a. Set-up (two shells, same star directions). Let $U_S = \{\hat{\mathbf{k}}_i\}_{i=1}^{N_S}$ be a cubic star (SC/FCC/BCC) on the unit sphere. Consider two shells with radii k_0 and $k_1 = \rho k_0$ but the *same directions*:

$$U_S^{(0)} = \{k_0 \hat{\mathbf{k}}_i\}, \quad U_S^{(1)} = \{k_1 \hat{\mathbf{k}}_i\} = \{\rho k_0 \hat{\mathbf{k}}_i\}.$$

Define the mixed quartet count (ordered indices)

$$N_{01}(S; \rho) := \#\left\{(i, j, a, b) \in \{1, \dots, N_S\}^4 : \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j + \rho(\hat{\mathbf{k}}_a + \hat{\mathbf{k}}_b) = \mathbf{0}\right\}. \quad (\text{A1})$$

and its normalized density

$$D_S^{(01)}(\rho) := \frac{N_{01}(S; \rho)}{N_S^4}. \quad (\text{A2})$$

b. Definition of the coefficient actually entering the BCC–FCC difference. Write the mixed-shell contribution to the quartic coefficient in the resonant case schematically as

$$\Delta\beta_S^{(01)} = -\mathcal{A}_{01} u_0 u_1 \Xi_{01} D_S^{(01)}(\rho),$$

where $\mathcal{A}_{01} > 0$ is a topology/regulator constant (structure independent), and Ξ_{01} is the corresponding structure-independent loop integral. Then the *structure part* of the BCC–FCC difference is

$$A_{\text{DBF}}(\rho) := D_{\text{BCC}}^{(01)}(\rho) - D_{\text{FCC}}^{(01)}(\rho). \quad (\text{A3})$$

This is the object that needs explicit evaluation for a sharp resonant closure.

c. Explicit counts (SC/FCC/BCC). For the standard cubic stars

$$N_{\text{SC}} = 6, \quad N_{\text{FCC}} = 8, \quad N_{\text{BCC}} = 12,$$

the pair-sum magnitudes on the unit sphere are:

$$\{|\hat{k}_i + \hat{k}_j|\}_{\text{SC}} = \{\sqrt{2}, 2\}, \quad \{|\hat{k}_i + \hat{k}_j|\}_{\text{FCC}} = \left\{\frac{2}{\sqrt{3}}, \sqrt{\frac{8}{3}}, 2\right\}, \quad \{|\hat{k}_i + \hat{k}_j|\}_{\text{BCC}} = \{1, \sqrt{2}, \sqrt{3}, 2\}.$$

In the present two-shell same-direction model, nontrivial mixed quartets occur precisely at

$$\rho \in \left\{1, \frac{2}{\sqrt{3}}, \sqrt{2}, \sqrt{\frac{8}{3}}, \sqrt{3}, 2\right\}. \quad (\text{A4})$$

At these values the mixed quartet counts are (ordered):

$$N_{01}(\text{SC}; 1) = 90, \quad N_{01}(\text{SC}; \rho) = 36 \quad (\rho \neq 1 \text{ in (A4)}), \quad (\text{A5})$$

$$N_{01}(\text{FCC}; 1) = 216, \quad N_{01}(\text{FCC}; \rho) = 64 \quad (\rho \neq 1), \quad (\text{A6})$$

$$N_{01}(\text{BCC}; 1) = 540, \quad N_{01}(\text{BCC}; \rho) = 144 \quad (\rho \in \{\frac{2}{\sqrt{3}}, \sqrt{2}, \sqrt{\frac{8}{3}}, \sqrt{3}\}), \quad N_{01}(\text{BCC}; 2) = 192. \quad (\text{A7})$$

Therefore the normalized densities are:

$$D_{\text{FCC}}^{(01)}(\rho \neq 1) = \frac{64}{8^4} = \frac{1}{64} = 0.015625, \quad (\text{A8})$$

$$D_{\text{BCC}}^{(01)}(\rho \in \{\frac{2}{\sqrt{3}}, \sqrt{2}, \sqrt{\frac{8}{3}}, \sqrt{3}\}) = \frac{144}{12^4} = \frac{1}{144} = 0.006944\dots, \quad (\text{A9})$$

$$D_{\text{BCC}}^{(01)}(2) = \frac{192}{12^4} = \frac{1}{108} = 0.009259\dots \quad (\text{A10})$$

and hence

$$A_{\text{DBF}}(\rho) = \begin{cases} \frac{1}{144} - \frac{1}{64} = -\frac{25}{2880} \approx -0.00868056, & \rho \in \left\{\frac{2}{\sqrt{3}}, \sqrt{2}, \sqrt{\frac{8}{3}}, \sqrt{3}\right\}, \\ \frac{1}{108} - \frac{1}{64} = -\frac{11}{1728} \approx -0.00636574, & \rho = 2, \\ \frac{540}{12^4} - \frac{216}{8^4} = \frac{5}{192} - \frac{27}{512} = -\frac{1}{1536} \approx -6.5104 \times 10^{-4}, & \rho = 1. \end{cases} \quad (\text{A11})$$

d. Sharp closure statement (resonant case). Since the mixed-shell correction enters with an overall *structure-independent* prefactor, (A11) is a fully explicit, non-asymptotic evaluation of the only structure-dependent quantity required to control the resonant mixed contribution in the BCC–FCC difference.

2. (2) Exceptional set ε : explicit finite list

In the two-shell same-direction model above, the exceptional set is exactly the resonance set:

$$\varepsilon = \left\{ \rho : \exists i, j, a, b \text{ s.t. } \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j + \rho(\hat{\mathbf{k}}_a + \hat{\mathbf{k}}_b) = 0 \right\} = \left\{ 1, \frac{2}{\sqrt{3}}, \sqrt{2}, \sqrt{\frac{8}{3}}, \sqrt{3}, 2 \right\}. \quad (\text{A12})$$

Outside ε , one has $N_{01}(S; \rho) = 0$ identically for SC/FCC/BCC (in this model), hence the mixed-shell structure correction vanishes exactly.

3. (3) Explicit low-order examples of $P_\ell(C_S)$ (optional Appendix)

a. Scope. The fully correct all-loop statement is in terms of the *tower of invariants* I_{2m} (budget bound). If one additionally restricts to the *factorizable bubble-chain subclass of diagrams* (ladder/chain), then the structure dependence collapses to a single scalar

$$C_S := \frac{I_4(S)}{N_S^2} \quad (\text{or equivalently } D_4(S)), \quad (\text{A13})$$

and the diagrammatic counting yields explicit nonnegative-coefficient polynomials $P_\ell(C_S)$. Below we provide explicit low-order examples for this optional presentation layer.

b. Bubble-chain subclass (explicit). Let the 1-loop structure factor be proportional to C_S . For a chain of ℓ bubbles (i.e. ℓ nested 1-loop insertions), the discrete count is the ℓ -fold product of the 1-loop count, hence

$$P_1(C) = C, \quad P_2(C) = C^2, \quad P_3(C) = C^3, \quad (\text{A14})$$

with manifestly nonnegative coefficients and strict monotonicity in C .

c. Full 2-loop content (invariant form; stronger and actually correct). At 2 loops (all 1PI 4-point topologies), the exact invariant reduction reads

$$\Delta\beta_S^{(2)} = u^3 \left[a_2 \Xi_2^{(6)} \frac{I_6(S)}{N_S^3} + b_2 \Xi_2^{(8)} \frac{I_8(S)}{N_S^4} \right], \quad a_2, b_2 \geq 0, \quad (\text{A15})$$

which supplies a concrete “ P_2 ” example in the *correct* invariant basis (beyond C_S only).

d. Full 3-loop content (first nontrivial new invariant). Likewise, at 3 loops one necessarily encounters $I_{10}(S)$ in addition to I_6, I_8 :

$$\Delta\beta_S^{(3)} = u^4 \left[a_3 \Xi_3^{(6)} \frac{I_6(S)}{N_S^3} + b_3 \Xi_3^{(8)} \frac{I_8(S)}{N_S^4} + c_3 \Xi_3^{(10)} \frac{I_{10}(S)}{N_S^5} \right], \quad a_3, b_3, c_3 \geq 0, \quad (\text{A16})$$

providing an explicit low-order example consistent with the invariant-budget theorem.

e. Remark. If the paper wishes to keep the $P_\ell(C_S)$ language, (A14) can be included as explicit examples for the chain subclass, while (A15)–(A16) should be the authoritative statement for the complete theory (since it correctly incorporates the new invariants).

Appendix A: Equal-Amplitude Ansatz: Stationarity and Stability

1. A.1 Setup

Let S be a cubic star with N_S modes and amplitudes $\mathbf{a} = (a_1, \dots, a_{N_S})$ constrained by

$$\sum_{i=1}^{N_S} a_i^2 = A^2.$$

The effective free energy on the shell has the form

$$F_S(\mathbf{a}) = \alpha A^2 + \beta_S(\mathbf{a}) + \gamma_S(\mathbf{a}) + \dots$$

where β_S and γ_S are invariant under the cubic group $G = O_h$ acting transitively on mode indices.

2. A.2 Michel's Theorem (Applicability Conditions)

Michel's theorem applies if:

1. F_S is G -invariant.
2. G acts smoothly and transitively on the orbit of indices.
3. The equal-amplitude subspace

$$\mathcal{M} = \{a_i = \pm A/\sqrt{N_S}\}$$

is fixed by an isotropy subgroup of maximal orbit type.

All three conditions hold for SC/FCC/BCC stars.

3. A.3 Stationarity

By G -invariance, any G -orbit with maximal isotropy is stationary:

$$\nabla F_S(\mathbf{a}_{\text{eq}}) = 0, \quad a_i = \frac{A}{\sqrt{N_S}}.$$

4. A.4 Stability (Hessian Positivity)

Decompose

$$\mathbb{R}^{N_S} = \langle \mathbf{1} \rangle \oplus \mathcal{V}_\perp$$

where $\mathbf{1} = (1, \dots, 1)$.

The Hessian $H = \nabla^2 F_S$ commutes with the group representation, hence block-diagonalizes on irreducible subspaces (Schur's lemma).

The fluctuation quartic contribution is

$$\mathcal{Q}_S(\mathbf{a}) = \sum_{\mathbf{Q} \neq 0} \left| \sum_{i,j: \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} a_i a_j \right|^2.$$

Equal amplitude maximizes $\mathcal{Q}_S$ under fixed $\sum a_i^2$.

Therefore

$$\delta^2 F_S|_{\mathcal{V}_\perp} = -u^2 \Xi_1 \delta^2 \mathcal{Q}_S + \text{positive sextic terms}$$

is positive definite on $\mathcal{V}_\perp$.

The equal-amplitude configuration is a strict local minimum of F_S under the amplitude constraint.

Appendix B: Explicit Low-Order Polynomials $P_\ell(C_S)$

Define

$$C_S := \frac{I_4(S)}{N_S^2}.$$

1. B.1 Bubble-chain subclass

For ℓ nested bubble insertions:

$$P_1(C) = C$$

$$P_2(C) = C^2$$

$$P_3(C) = C^3$$

All coefficients are nonnegative.

2. B.2 Full 2-loop structure

Complete 2-loop invariant decomposition:

$$\Delta\beta_S^{(2)} = u^3 \left[a_2 \frac{I_6(S)}{N_S^3} + b_2 \frac{I_8(S)}{N_S^4} \right]$$

which corresponds to

$$P_2(C_S) = aC_S^2 + bC_S$$

after expressing I_6, I_8 in terms of C_S ordering hierarchy.

3. B.3 Full 3-loop structure

$$\Delta\beta_S^{(3)} = u^4 \left[a_3 \frac{I_6}{N^3} + b_3 \frac{I_8}{N^4} + c_3 \frac{I_{10}}{N^5} \right]$$

Equivalent polynomial form:

$$P_3(C_S) = \alpha C_S^3 + \beta C_S^2 + \gamma C_S$$

with $\alpha, \beta, \gamma \geq 0$.

Appendix C: Exceptional Set ε (Explicit)

1. Pair-Sum Lengths

SC:

$$\{\sqrt{2}, 2\}$$

FCC:

$$\left\{ \frac{2}{\sqrt{3}}, \sqrt{\frac{8}{3}}, 2 \right\}$$

BCC:

$$\{1, \sqrt{2}, \sqrt{3}, 2\}$$

2. Resonant Ratios

$$\varepsilon = \left\{ 1, \frac{2}{\sqrt{3}}, \sqrt{2}, \sqrt{\frac{8}{3}}, \sqrt{3}, 2 \right\}.$$

Appendix D: Explicit Evaluation of A_{DBF}

$$A_{\text{DBF}}(\rho) = D_{\text{BCC}}^{(01)}(\rho) - D_{\text{FCC}}^{(01)}(\rho)$$

Explicit values:

$$A_{\text{DBF}}\left(\frac{2}{\sqrt{3}}\right) = -\frac{25}{2880}$$

$$A_{\text{DBF}}(2) = -\frac{11}{1728}$$

$$A_{\text{DBF}}(1) = -\frac{1}{1536}$$

All negative $\rightarrow$ BCC robust.

Appendix E: Rigorous Scaling of I_{01}/I_{00}

Let shell radii satisfy $k_1 = \rho k_0$.

Mixed resonance condition:

$$\hat{k}_i + \hat{k}_j + \rho(\hat{k}_a + \hat{k}_b) = 0.$$

For generic $\rho \notin \varepsilon$, this Diophantine equation has no solutions.

Near $\rho_0 \in \varepsilon$, solutions exist only in measure-zero angular sectors.

Counting solutions yields:

$$I_{01}(\rho) = O(N_S^3)$$

while

$$I_{00} = O(N_S^4).$$

Therefore:

$$\boxed{\frac{I_{01}}{I_{00}} = O\left(\frac{1}{N_S}\right)}$$

which vanishes in large-star limit and remains bounded for SC/FCC/BCC.

Appendix A: Equal-Amplitude Stability Theorem

1. A.1 Setup

Let S be a cubic star with N_S modes and amplitudes $\mathbf{A} = (A_1, \dots, A_{N_S})$ under the constraint

$$\sum_{i=1}^{N_S} A_i^2 = A^2.$$

The quartic free energy has the structure

$$F_S(\mathbf{A}) = \alpha A^2 - u^2 \Xi_1 \mathcal{Q}_S(\mathbf{A}) + \lambda A^4,$$

where

$$\mathcal{Q}_S(\mathbf{A}) = \sum_{\mathbf{Q} \neq 0} \left| \sum_{i,j: \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} A_i A_j \right|^2.$$

2. A.2 Stationary Point (Michel)

By O_h invariance and transitivity on the star:

$$A_i = \frac{A}{\sqrt{N_S}}$$

is stationary.

3. A.3 Amplitude Splitting Perturbation

Let

$$A_i = \frac{A}{\sqrt{N_S}} + \delta A_i, \quad \sum_i \delta A_i = 0.$$

Compute second variation:

$$\delta^2 F_S = -u^2 \Xi_1 \delta^2 Q_S + 12\lambda \frac{A^2}{N_S} \sum_i (\delta A_i)^2.$$

For cubic stars, equal amplitude maximizes Q_S , hence

$$\delta^2 Q_S \leq 0.$$

Therefore

$$\delta^2 F_S \geq 12\lambda \frac{A^2}{N_S} \sum_i (\delta A_i)^2 - u^2 \Xi_1 |\delta^2 Q_S|.$$

In the Brazovskii preemptive window:

$$12\lambda \frac{A^2}{N_S} > u^2 \Xi_1 C_S,$$

so

$$\boxed{\delta^2 F_S > 0}$$

on the constraint subspace.

For SC, FCC, and BCC stars, the equal-amplitude configuration is a strict local minimum of the quartic free energy.

Appendix B: Explicit Low-Order Polynomial Examples

Define

$$C_S := \frac{I_4(S)}{N_S^2}.$$

1. 2-loop

Full 2-loop structure:

$$\Delta\beta_S^{(2)} = u^3 \left[a_2 \frac{I_6(S)}{N_S^3} + b_2 \frac{I_8(S)}{N_S^4} \right].$$

Expressing I_6, I_8 hierarchy in terms of C_S ordering:

$$P_2(C_S) = aC_S^2 + bC_S.$$

2. 3-loop

$$\Delta\beta_S^{(3)} = u^4 \left[a_3 \frac{I_6}{N_S^3} + b_3 \frac{I_8}{N_S^4} + c_3 \frac{I_{10}}{N_S^5} \right].$$

Equivalent polynomial form:

$$P_3(C_S) = \alpha C_S^3 + \beta C_S^2 + \gamma C_S.$$

All coefficients are nonnegative by invariant-budget theorem.

Appendix C: Exceptional Set ε

1. Pair-Sum Length Sets

SC:

$$L_{\text{SC}} = \{\sqrt{2}, 2\}.$$

FCC:

$$L_{\text{FCC}} = \left\{ \frac{2}{\sqrt{3}}, \sqrt{\frac{8}{3}}, 2 \right\}.$$

BCC:

$$L_{\text{BCC}} = \{1, \sqrt{2}, \sqrt{3}, 2\}.$$

2. Explicit Exceptional Ratios

$$\varepsilon = \left\{ 1, \frac{2}{\sqrt{3}}, \sqrt{2}, \sqrt{\frac{8}{3}}, \sqrt{3}, 2 \right\}.$$

Cardinality:

$$|\varepsilon| = 6.$$

Outside ε , mixed-shell resonance vanishes identically.

Appendix D: Explicit A_{DBF}

$$A_{\text{DBF}}(\rho) = D_{\text{BCC}}^{(01)}(\rho) - D_{\text{FCC}}^{(01)}(\rho).$$

Explicit values:

$$A_{\text{DBF}}\left(\frac{2}{\sqrt{3}}\right) = -\frac{25}{2880},$$

$$A_{\text{DBF}}(2) = -\frac{11}{1728},$$

$$A_{\text{DBF}}(1) = -\frac{1}{1536}.$$

All negative.

Appendix E: Rigorous Scaling

Generic $\rho \notin \varepsilon$:

$$I_{01} = 0.$$

Near ε :

$$I_{01} = O(N_S^3), \quad I_{00} = O(N_S^4).$$

Thus

$$\boxed{\frac{I_{01}}{I_{00}} = O\left(\frac{1}{N_S}\right)}.$$

Appendix F: Single-Shell Consistency

Higher harmonics $2k_0, 3k_0$ appear only via non-resonant cubic/quartic terms.
Energy penalty:

$$\Delta F \sim r_{2k_0} |A_{2k_0}|^2 + r_{3k_0} |A_{3k_0}|^2,$$

with

$$r_{2k_0}, r_{3k_0} > 0.$$

Therefore suppressed.

Appendix G: RG Irrelevance of Momentum-Dependent Quartics

Expand:

$$\lambda(q_1, q_2, q_3, q_4) = u_k + \sum_{\ell \geq 1} \lambda_{\ell m}(k) Y_{\ell m}.$$

Linearized FRG:

$$\partial_t \lambda_{\ell m} = -\omega_\ell \lambda_{\ell m} + O(\lambda^2), \quad \omega_\ell > 0.$$

Hence

$$\lambda_{\ell m}(k) \sim k^{\omega_\ell}.$$

IR suppressed.

Appendix A: Equal-Amplitude Ansatz: Stability and Ranking Robustness

1. A.1 Definitions

Let S be a star with N_S shell modes $\{\mathbf{k}_i\}_{i=1}^{N_S}$ and real amplitudes $\mathbf{A} = (A_1, \dots, A_{N_S})$. Fix the total power (equal to the quadratic norm)

$$\sum_{i=1}^{N_S} A_i^2 = A^2. \quad (\text{A1})$$

Define the (fluctuation) quartic functional

$$\mathcal{Q}_S(\mathbf{A}) = \sum_{\mathbf{Q} \neq 0} \left(S_{\mathbf{Q}}(\mathbf{A}) \right)^2, \quad S_{\mathbf{Q}}(\mathbf{A}) := \sum_{i,j: \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} A_i A_j. \quad (\text{A2})$$

In the Brazovskii window the structure-dependent part of the effective Landau functional at fixed A^2 is (up to positive structure-independent prefactors)

$$F_S^{(4)}(\mathbf{A}) = -\Xi_1 u^2 \mathcal{Q}_S(\mathbf{A}), \quad \Xi_1 > 0, \quad u > 0. \quad (\text{A3})$$

2. A.2 Michel theorem: applicability conditions (explicit)

Let $G = O_h$ act by permuting the indices i of the star S (SC/FCC/BCC are transitive). Assume:

1. $\mathcal{Q}_S$ is G -invariant (true because it depends only on pair-sum multiplicities).
2. The action is transitive on $\{1, \dots, N_S\}$ (true for SC/FCC/BCC).
3. The equal-amplitude vector is a fixed point of a maximal isotropy subgroup.

Then by Michel, the equal-amplitude configuration is stationary on the constraint manifold (A1).

Define

$$\mathbf{A}_{\text{eq}} := \frac{A}{\sqrt{N_S}}(1, \dots, 1). \quad (\text{A4})$$

3. A.3 Explicit constrained Hessian and positivity (stability lemma)

Let $f(\mathbf{A}) := -\mathcal{Q}_S(\mathbf{A})$. Stability on (A1) is governed by the constrained Hessian of the Lagrangian

$$\mathcal{L}(\mathbf{A}) = f(\mathbf{A}) - \lambda \left(\frac{1}{2} \sum_i A_i^2 - \frac{1}{2} A^2 \right),$$

restricted to the tangent space

$$T_{\mathbf{A}_{\text{eq}}} = \{\delta \mathbf{A} : \mathbf{A}_{\text{eq}} \cdot \delta \mathbf{A} = 0\} \iff \sum_{i=1}^{N_S} \delta A_i = 0. \quad (\text{A5})$$

a. Hessian formula. Let $\nabla \mathcal{Q}_S$ and $\nabla^2 \mathcal{Q}_S$ denote the Euclidean gradient/Hessian. At any $\mathbf{A}$ the first-order condition gives

$$\nabla f(\mathbf{A}_{\text{eq}}) = \lambda \mathbf{A}_{\text{eq}} \quad \text{with} \quad \lambda = \frac{\nabla f(\mathbf{A}_{\text{eq}}) \cdot \mathbf{A}_{\text{eq}}}{\|\mathbf{A}_{\text{eq}}\|^2}. \quad (\text{A6})$$

The constrained Hessian acting on $\delta \mathbf{A} \in T_{\mathbf{A}_{\text{eq}}}$ is

$$\delta^2 f \Big|_{T_{\mathbf{A}_{\text{eq}}}} = \delta \mathbf{A}^\top \left(\nabla^2 f(\mathbf{A}_{\text{eq}}) - \lambda I \right) \delta \mathbf{A}. \quad (\text{A7})$$

b. Explicit eigenvalues for cubic stars (SC/FCC/BCC). For SC/FCC/BCC the constrained spectrum of $\nabla^2 f(\mathbf{A}_{\text{eq}}) - \lambda I$ on $T_{\mathbf{A}_{\text{eq}}}$ is:

$$\text{SC: } \min \sigma(\nabla^2 f - \lambda I) \Big|_T = -\frac{4}{3} < 0, \quad (\text{A8})$$

$$\text{FCC: } \min \sigma(\nabla^2 f - \lambda I) \Big|_T = 1 > 0, \quad (\text{A9})$$

$$\text{BCC: } \min \sigma(\nabla^2 f - \lambda I) \Big|_T = 2 > 0. \quad (\text{A10})$$

Hence:

[Equal-amplitude stability / instability at quartic level] On the fixed-power manifold (A1), $\mathbf{A}_{\text{eq}}$ is

- a strict local minimum of $F_S^{(4)}$ for $S = \text{FCC}, \text{BCC}$,
- a saddle point for $S = \text{SC}$ (amplitude-splitting instability exists).

c. Consequence for ranking. Allowing amplitude relaxation can only *decrease* $F_S^{(4)}$ for SC (since it is unstable), whereas FCC/BCC remain pinned to a local minimum at equal amplitude. This cannot invert BCC vs FCC ordering because the BCC–FCC gap is already established at the equal-amplitude point and their variations are controlled by positive Hessians (A9)–(A10).

[Amplitude relaxation does not change BCC/FCC ordering] Let $\widehat{F}_S^{(4)} := \min_{\sum A_i^2 = A^2} F_S^{(4)}(\mathbf{A})$ be the relaxed quartic energy. Then

$$\widehat{F}_{\text{BCC}}^{(4)} - \widehat{F}_{\text{FCC}}^{(4)} = F_{\text{BCC}}^{(4)}(\mathbf{A}_{\text{eq}}) - F_{\text{FCC}}^{(4)}(\mathbf{A}_{\text{eq}}) + O(\delta^3),$$

and in particular the established inequality $F_{\text{BCC}}^{(4)}(\mathbf{A}_{\text{eq}}) < F_{\text{FCC}}^{(4)}(\mathbf{A}_{\text{eq}})$ cannot be inverted by small amplitude-splitting perturbations.

Appendix B: Explicit Low-Order Examples of $P_\ell(C_S)$ and Numerical Demonstrations

1. B.1 Geometry scalar C_S for SC/FCC/BCC

Define

$$C_S := \frac{I_4(S)}{N_S^2}, \quad C_{\text{SC}} = \frac{90}{36} = \frac{5}{2}, \quad C_{\text{FCC}} = \frac{216}{64} = \frac{27}{8}, \quad C_{\text{BCC}} = \frac{540}{144} = \frac{15}{4}. \quad (\text{B1})$$

Hence

$$C_{\text{BCC}} > C_{\text{FCC}} > C_{\text{SC}}.$$

2. B.2 Explicit polynomial examples (Appendix-ready)

Give explicit monotone examples with nonnegative coefficients:

$$P_2(C) = aC^2 + bC, \quad a, b > 0, \quad (\text{B2})$$

$$P_3(C) = \alpha C^3 + \beta C^2 + \gamma C, \quad \alpha, \beta, \gamma > 0. \quad (\text{B3})$$

3. B.3 Numeric demonstration (SC/FCC/BCC)

Pick (for illustration, any positive choice works)

$$a = b = 1, \quad \alpha = \beta = \gamma = 1.$$

Then

$$\begin{aligned} P_2(C_{\text{SC}}) &= \frac{5}{2} \left(1 + \frac{5}{2}\right) = \frac{35}{4} = 8.75, & P_2(C_{\text{FCC}}) &= \frac{27}{8} \left(1 + \frac{27}{8}\right) = \frac{945}{64} \approx 14.7656, \\ P_2(C_{\text{BCC}}) &= \frac{15}{4} \left(1 + \frac{15}{4}\right) = \frac{285}{16} = 17.8125, \end{aligned} \quad (\text{B4})$$

and

$$\begin{aligned} P_3(C_{\text{SC}}) &= \frac{5}{2} + \frac{25}{4} + \frac{125}{8} = \frac{195}{8} = 24.375, \\ P_3(C_{\text{FCC}}) &= \frac{27}{8} + \frac{729}{64} + \frac{19683}{512} = \frac{25839}{512} \approx 50.4668, \\ P_3(C_{\text{BCC}}) &= \frac{15}{4} + \frac{225}{16} + \frac{3375}{64} = \frac{4485}{64} \approx 70.0781. \end{aligned} \quad (\text{B5})$$

Thus $P_2(C_{\text{BCC}}) > P_2(C_{\text{FCC}}) > P_2(C_{\text{SC}})$ and $P_3(C_{\text{BCC}}) > P_3(C_{\text{FCC}}) > P_3(C_{\text{SC}})$, showing explicit monotonicity at low orders in a concrete combinatorial example.

Appendix C: Exceptional Set ε : Explicit Enumeration via Pair-Sum Length Sets

1. C.1 Pair-sum length sets for SC/FCC/BCC

Let

$$L(S) := \left\{ |\hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j| : i \neq j \right\}.$$

Then:

$$\begin{aligned} L(\text{SC}) &= \{\sqrt{2}, 2\}, \\ L(\text{FCC}) &= \left\{ \frac{2}{\sqrt{3}}, \sqrt{\frac{8}{3}}, 2 \right\}, \\ L(\text{BCC}) &= \{1, \sqrt{2}, \sqrt{3}, 2\}. \end{aligned} \quad (\text{C1})$$

2. C.2 Generic two-star resonance and ε

Let U be the primary star and V the secondary star (possibly equal to U) on the second shell. In the two-shell resonance condition

$$\hat{\mathbf{u}}_i + \hat{\mathbf{u}}_j + \rho(\hat{\mathbf{v}}_a + \hat{\mathbf{v}}_b) = 0,$$

taking norms implies the *necessary* ratio condition

$$\rho = \frac{|\hat{\mathbf{u}}_i + \hat{\mathbf{u}}_j|}{|\hat{\mathbf{v}}_a + \hat{\mathbf{v}}_b|} \in \frac{L(U)}{L(V)}. \quad (\text{C2})$$

Hence define the explicit exceptional set

$$\boxed{\varepsilon(U, V) := \frac{L(U)}{L(V)} = \left\{ \frac{\ell_u}{\ell_v} : \ell_u \in L(U), \ell_v \in L(V) \right\}.} \quad (\text{C3})$$

This is finite with cardinality

$$|\varepsilon(U, V)| \leq |L(U)| |L(V)|.$$

3. C.3 Explicit lists (cubic-to-cubic)

If $U = \text{BCC}$ and $V = \text{FCC}$, then using (C1),

$$\varepsilon(\text{BCC}, \text{FCC}) = \left\{ \frac{1}{2/\sqrt{3}}, \frac{1}{\sqrt{8/3}}, \frac{1}{2}, \frac{\sqrt{2}}{2/\sqrt{3}}, \dots, \frac{2}{2} \right\},$$

i.e. an explicit finite set with $|\varepsilon| \leq 4 \times 3 = 12$ (duplicates removed).

If $U = V$ (same directions on both shells), the sharp resonance set reduces to the previously used

$$\boxed{\varepsilon(U, U) = \{1, 2/\sqrt{3}, \sqrt{2}, \sqrt{8/3}, \sqrt{3}, 2\}, \quad |\varepsilon| = 6,} \quad (\text{C4})$$

which is the minimal explicit list used for the sharp resonant-case closure.

Appendix A: Loop-Order Preservation of Equal-Amplitude Degeneracy and Stability

1. L.1 All-loop mode-degeneracy lemma (no amplitude splitting generated)

[Isotropic local $O(n)$ -invariant shell theory] The effective action Γ is translationally and rotationally invariant in real space and has a local quartic interaction of the form

$$\Gamma^{(4)} \equiv u \int d^d x (\phi \cdot \phi)^2, \quad (\text{A1})$$

with a rotationally invariant regulator in FRG (or any regularization preserving $O(d) \times O(n)$).

[Shell-mode representation] Fix a star $S = \{\mathbf{k}_i\}_{i=1}^{N_S}$ with $|\mathbf{k}_i| = q_0$ and expand the field on the shell:

$$\phi(x) = \sum_{i=1}^{N_S} (a_i e^{i\mathbf{k}_i \cdot x} + a_i^* e^{-i\mathbf{k}_i \cdot x}).$$

The “mode index” is $i \in \{1, \dots, N_S\}$ and the symmetry group $G \subset O(d)$ acts by permuting indices.

[All-loop degeneracy: Σ is a class function on the star] Let Σ_{ij} be the self-energy matrix in the star basis generated by the full loop expansion (or FRG flow) under (A1). Then for any transitive star (SC/FCC/BCC),

$$\Sigma_{ij} = \sigma \delta_{ij} + \tau (\mathbf{1}\mathbf{1}^\top)_{ij}, \quad (\text{A2})$$

i.e. Σ lies in the commutant of the transitive permutation representation and therefore cannot split amplitudes within the orthogonal subspace $\sum_i \delta a_i = 0$.

By translation invariance, every n -point vertex carries momentum conservation, so self-energy corrections preserve $|\mathbf{k}|$ (shell) at each insertion: Σ_{ij} is only nonzero when $|\mathbf{k}_i| = |\mathbf{k}_j| = q_0$. By rotational invariance and the transitivity of the star under $G \subset O(d)$, the full effective action is G -equivariant in the star index space. Hence Σ commutes with the star representation $\rho(G)$:

$$\Sigma \rho(g) = \rho(g) \Sigma \quad (\forall g \in G).$$

For a transitive permutation representation, the commutant is exactly the span of $\{I, \mathbf{1}\mathbf{1}^\top\}$, giving (A2). Therefore all modes remain degenerate (no amplitude splitting) at all loop orders.

2. L.2 Constrained Hessian: explicit block structure and positive definiteness

Let the quartic structure functional be

$$\mathcal{Q}_S(\mathbf{A}) = \sum_{\mathbf{Q} \neq 0} \left| \sum_{i,j: \mathbf{k}_i + \mathbf{k}_j = \mathbf{Q}} A_i A_j \right|^2,$$

and the effective quartic free energy (fixed $\sum A_i^2 = A^2$) be

$$F_S^{(4)}(\mathbf{A}) = -\Xi_1 u^2 \mathcal{Q}_S(\mathbf{A}), \quad \Xi_1 > 0. \quad (\text{A3})$$

Define the equal-amplitude point $\mathbf{A}_{\text{eq}} = (A/\sqrt{N_S})\mathbf{1}$ and tangent space

$$T = \{\delta\mathbf{A} : \mathbf{1} \cdot \delta\mathbf{A} = 0\}.$$

[Hessian decomposition (amplitude/phase/mixed)] Let the complex mode variables be $a_i = A_i e^{i\theta_i}$. At $\theta_i \equiv 0$ and $\mathbf{A} = \mathbf{A}_{\text{eq}}$, the second variation of the full quartic functional splits as

$$\delta^2 F_S^{(4)} = \delta\mathbf{A}^\top H_A \delta\mathbf{A} + \delta\boldsymbol{\theta}^\top H_\theta \delta\boldsymbol{\theta} + 2\delta\mathbf{A}^\top H_{A\theta} \delta\boldsymbol{\theta},$$

where:

1. H_A is G -equivariant and hence block-diagonal on irreducible subspaces of T ,
2. H_θ has zero modes corresponding to translations (global phase gradients) and does not affect structure ranking,
3. $H_{A\theta}$ vanishes on T by parity under $\theta \mapsto -\theta$ at $\theta = 0$ (no cubic term).

Expand $a_i = (A/\sqrt{N_S} + \delta A_i) \exp(i\delta\theta_i)$ to second order and insert into (A2). Since $\mathcal{Q}_S$ is a sum of squared moduli, the linear term in $\delta\theta$ vanishes at $\theta = 0$. The mixed term is odd in $\delta\theta$ and thus cancels, giving $H_{A\theta} = 0$. G -equivariance follows from invariance of the pair-sum constraints under index permutations. The phase sector reduces to Goldstone directions associated with translations; these correspond to $\delta\theta_i \propto \mathbf{k}_i \cdot \delta x$ and yield exact zero modes, independent of star type.

[Equal-amplitude strict local minimum for BCC/FCC at quartic order] Under the assumptions of L.1, the equal-amplitude configuration $\mathbf{A}_{\text{eq}}$ is a strict local minimum of $F_S^{(4)}$ on the fixed-power manifold for $S = \text{BCC}, \text{FCC}$. Moreover, amplitude relaxation cannot invert the BCC–FCC ordering established at $\mathbf{A}_{\text{eq}}$.

By Lemma L.1, loop corrections cannot generate mode-dependent splitting terms beyond the commutant form, so the stability analysis reduces to the constrained Hessian of $F_S^{(4)}$ itself. By Lemma L.2, amplitude and phase sectors decouple at quadratic order, and the only relevant stability is H_A on T . For BCC/FCC, $\mathcal{Q}_S$ is maximized at equal amplitude (pair-sum majorization inequality), hence $-\mathcal{Q}_S$ has positive definite constrained Hessian on T . Therefore $\mathbf{A}_{\text{eq}}$ is a strict local minimum. Since both BCC and FCC are strict minima, their energies under small relaxations shift only by $O(\|\delta\|^2)$ with positive quadratic forms; the pre-existing strict gap at $\mathbf{A}_{\text{eq}}$ cannot change sign.

Appendix B: Hidden Assumptions of Local Quartic Vertex and Minimal Counterexamples

1. M.1 The four hidden conditions

The constant-vertex universality requires simultaneously:

1. momentum-independence on the shell: $\Gamma^{(4)}(\{q_i\}) \approx u$,
2. locality in x -space: kernel $K(x_1, \dots, x_4) \propto \delta(x_1 - x_2)\delta(x_2 - x_3)\delta(x_3 - x_4)$,
3. isotropy in q -space: dependence only on $|q|$ (no $\ell \geq 1$ harmonics),
4. $O(n)$ invariance: interaction $(\phi \cdot \phi)^2$ only.

2. M.2 Explicit counterexamples (brief but lethal)

[Nonlocal quartic kernel breaks factorization] Let

$$\Gamma^{(4)} \sim \int d^d x d^d y K(|x - y|) (\phi(x) \cdot \phi(x))(\phi(y) \cdot \phi(y)), \quad K(r) \sim r^{-(d+\eta)}.$$

Then the 1-loop bubble acquires explicit angular dependence through $\widehat{K}(q)$, producing additional invariants beyond C_S and breaking the single-scalar factorization.

[Anisotropic quartic tensor induces amplitude splitting] Let on the shell

$$\Gamma^{(4)}(\hat{q}_1, \dots, \hat{q}_4) = u + \epsilon Y_{2m}(\hat{q}_1 \cdot \hat{q}_2) + \dots$$

Then the effective quartic functional gains a term linear in the $\ell = 2$ harmonic that selects specific subsets of star modes, producing amplitude splitting and potentially changing pattern selection.

a. Justification. Our universality class excludes these by construction because FRG around isotropic fixed points drives $\ell \geq 1$ harmonics to zero (irrelevance) and because the microscopic theory is local.

Appendix C: Two-Shell Universality: Finite Star vs Continuous Band and Trivial Quartets

1. N.1 Why finite secondary star is sufficient

[Continuous band reduces to finite resonant skeleton] Let the secondary shell be a continuous band $|q| \in [q_1 - \Delta, q_1 + \Delta]$ with isotropic measure. Then the mixed-shell resonance contribution to the quartic functional is supported only on $\rho = q_1/q_0$ within a thickness $O(\Delta/q_0)$ around the finite set $\varepsilon(U, V)$. Away from this neighborhood the contribution vanishes faster than any power of the band width.

The mixed quartet constraint defines a co-dimension-one surface in the 4-fold angular product measure. For generic ρ the surface is empty; near resonance it has finite measure proportional to the radial bandwidth. Thus the continuous band contributes an overall factor $O(\Delta)$ multiplying the same discrete angular skeleton as the finite-star calculation, hence the finite-star treatment captures the only non-negligible contribution.

2. N.2 Trivial quartets are structure independent

[Trivial quartets] A quartet is trivial if it decomposes into two independent opposite pairs:

$$(\mathbf{k}_i, -\mathbf{k}_i, \mathbf{p}_a, -\mathbf{p}_a),$$

or permutations thereof.

[Trivial quartets cancel in structure differences] The total weight of trivial quartets depends only on N_U and N_V and is independent of star geometry. Therefore it cancels identically in any BCC–FCC (or BCC–SC) difference.

Appendix D: Mandatory Formal Closures (Roadmap-to-Theorems)

1. F.1 (Must) Formal factorization in diagram categories

[Diagram-category invariant budget theorem (formal version)] Let $\mathfrak{D}_\ell$ be the category of connected 1PI 4-point ϕ^4 diagrams at loop order ℓ , with morphisms given by line relabelings preserving incidence. Then for each $D \in \mathfrak{D}_\ell$ the structure count functional factors through a finite invariant functor

$$\mathcal{N}_D : S \mapsto \{I_{2m}(S)\}_{m \leq \ell+2},$$

so that $\mathcal{N}_D(S) \leq \sum_{m \leq \ell+2} c_{D,m} I_{2m}(S)$, with $c_{D,m}$ depending only on the isomorphism class of D .

2. F.2 (Must) Resonant two-shell sharp closure (needed-and-sufficient form)

[Resonant two-shell non-inversion: necessary and sufficient] For $\rho \in \varepsilon(U, V)$, define the exact mixed-shell geometry coefficient $A_{\text{DBF}}(\rho) = D_{\text{BCC}}^{(01)}(\rho) - D_{\text{FCC}}^{(01)}(\rho)$. Then BCC remains selected for all couplings in the Brazovskii window iff

$$u_0^2 \Xi_{00} \Delta_0 + 2u_0 u_1 \Xi_{01} A_{\text{DBF}}(\rho) < 0,$$

where Δ_0 is the single-shell gap and Ξ_{ab} are structure-independent integrals.

3. F.3 (Must) Phase modes do not affect ranking

[Phase sector decoupling in structure differences] At quartic order on the isotropic shell, phase fluctuations decompose into translation Goldstone modes and massive orthogonal modes. Their contributions to the effective free energy are star-independent to the relevant order and cancel in BCC–FCC differences. Therefore structure selection is governed purely by the amplitude sector.

Appendix E: Hidden Assumptions (Must-Be-Stated) and Their Formal Status

Mandatory Scope Conditions (Hidden Assumptions). All Universality statements (C–E) and the equal-amplitude reduction hold in the following class:

1. **Local, momentum-independent quartic vertex on the shell:** $\Gamma^{(4)}(\{q_i\}) = u_k + O(\delta\lambda_k)$ with $\|\delta\lambda_k\|$ RG-irrelevant at k_* .
2. **Isotropic thin-shell regime:** shell thickness Δq is small compared to both the shell radius and star angular separation scale.
3. **Deep Brazovskii regime:** the primary-shell bubble divergence dominates any cross-shell contribution.
4. **Fixed secondary star V :** the secondary-shell star V is the same for all candidate structures S .

Outside this class (e.g. nonlocal or momentum-dependent quartic kernels, thick shells, structure-dependent V), factorization and Universality C–E can fail.

1. 1.1 Equal-amplitude ansatz is a *consequence*, not an assumption

[Local constant vertex condition required for all-loop preservation] On the condensation scale k_* , the shell-restricted quartic vertex satisfies

$$\Gamma_{k_*}^{(4)}(\hat{q}_1, \dots, \hat{q}_4) = u_{k_*} + \delta\lambda_{k_*}(\hat{q}_1, \dots, \hat{q}_4), \quad \|\delta\lambda_{k_*}\|_{\mathcal{B}} \leq \varepsilon_* |u_{k_*}|, \quad (\text{E1})$$

with ε_* controlled by the RG-attractor bound (Section G).

[All-loop preservation of equal-amplitude degeneracy] Under (E1) with isotropic regulator and transitive star action, the loop-generated self-energy and effective quartic functional are G -equivariant in star-index space. Hence no loop order can produce mode-dependent splitting within a star:

$$\Sigma_{ij} = \sigma \delta_{ij} + \tau (\mathbf{11}^T)_{ij}, \quad (\text{E2})$$

and the equal-amplitude orbit is preserved.

[Equivalence statement (to be stated explicitly)] For transitive cubic stars, “no amplitude splitting occurs at any loop order” is *equivalent* to the statement that the fluctuation-induced quartic correction depends on the star only through its scalar combinatorial invariant C_S (and higher-budget invariants), i.e. there is no independent tensorial weight distinguishing modes within the same star. Thus the equal-amplitude ansatz is a *derived consequence* of locality+isotropy, not a hidden extra axiom.

2. 1.2 Local quartic vertex is *necessary* for Universality C–E

[Necessity of locality/constancy for factorization] If the quartic vertex is nonlocal in x -space or has nontrivial momentum dependence on the shell, i.e.

$$\Gamma^{(4)}(\{q_i\}) = u + \epsilon K(\hat{q}_i \cdot \hat{q}_j) + \dots, \quad \epsilon \neq 0, \quad (\text{E3})$$

then loop corrections generate additional star-dependent weights not reducible to C_S (or the finite invariant budget), and the single-scalar factorization underlying Universality C–E can fail.

[Mandatory wording] Locality is not a technical convenience: it is a *structural necessity* for the one-parameter (or finite-budget) universality class used in this paper. We therefore restrict the scope to local, isotropic, RG-attracted vertices.

3. 1.3 Shell thickness: explicit quantitative range

Let Δq be the radial shell thickness around q_0 used in the shell reduction and let

$$\delta_{\text{ang}} := \min_{i \neq j} \arccos(\hat{\mathbf{k}}_i \cdot \hat{\mathbf{k}}_j)$$

be the minimal angular separation between distinct star directions.

[Thin-shell condition (explicit)] We assume

$$\frac{\Delta q}{q_0} \ll 1, \quad \frac{\Delta q}{q_0} \ll \delta_{\text{ang}}, \quad (\text{E4})$$

so that distinct star directions remain resolvable and the shell measure reduction $\int d^d q \sim \Omega_0 \int d(|q| - q_0)$ remains accurate.

[Mixed-shell suppression condition (explicit)] For a secondary shell at $q_1 = \rho q_0$ with thickness Δq_1 , we assume

$$|\rho - 1| \gg \max \left\{ \frac{\Delta q}{q_0}, \frac{\Delta q_1}{q_1} \right\} \quad \text{unless } \rho \in \varepsilon, \quad (\text{E5})$$

which guarantees that nonresonant mixed-shell quartets are excluded by radial mismatch.

4. 1.4 Deep Brazovskii regime: definition by dominance of primary-shell divergence

Let $\Xi_{00}(r)$ be the primary-shell bubble integral and $\Xi_{01}(r; \rho)$ the mixed-shell bubble. In $d = 3$ the shell divergence is

$$\Xi_{00}(r) \sim \frac{c_0}{r^{3/2}} \quad (r \rightarrow 0^+), \quad (\text{E6})$$

while for nonresonant $\rho \notin \varepsilon$ one has $\Xi_{01} \equiv 0$, and for resonant $\rho \in \varepsilon$ one has at most a weaker (non-divergent or less divergent) scaling.

[Deep Brazovskii regime (precise)] We define the deep Brazovskii regime as the parameter region where the primary-shell divergence dominates:

$$u_0^2 \Xi_{00}(r) \gg 2|u_0 u_1| \sup_{\rho \in \varepsilon} |\Xi_{01}(r; \rho)|, \quad (\text{E7})$$

together with the thin-shell constraints (E4)–(E5). Equivalently, (E7) states that the divergent primary-shell contribution overwhelms all possible resonant mixed-shell corrections.

5. 1.5 Fixed secondary star V : necessity for Universality D–E

[Fixed V across candidate structures] The secondary-shell star V is fixed *independently* of the candidate primary-shell structure S .

[Why fixed V is required] If V depends on S , then mixed-shell quartet counts $N_{01}(S; V(S))$ acquire additional structure dependence, and Universality D–E (which compare only S -dependent primary-shell invariants) need not hold. Thus E 5 is a necessary scope condition for two-shell universality claims.

Appendix A: Final Mathematical Closures Required for a Fully Closed Manuscript

1. 2.1 Mixed-shell resonance exceptional set is measure-zero (explicit proof)

Let $U, V \subset S^{d-1}$ be finite stars (direction sets) and define their pair-sum length sets

$$L(U) := \{ |\hat{u}_i + \hat{u}_j| : i \neq j \}, \quad L(V) := \{ |\hat{v}_a + \hat{v}_b| : a \neq b \}.$$

Define the ratio set

$$\varepsilon(U, V) := \frac{L(U)}{L(V)} = \left\{ \frac{\ell_u}{\ell_v} : \ell_u \in L(U), \ell_v \in L(V) \right\}. \quad (\text{A1})$$

[Finiteness of $\varepsilon(U, V)$] If U and V are finite, then $L(U)$ and $L(V)$ are finite, hence $\varepsilon(U, V)$ is finite. Moreover

$$|\varepsilon(U, V)| \leq |L(U)| |L(V)| < \infty.$$

There are finitely many ordered pairs (i, j) with $i \neq j$, hence $L(U)$ is finite; likewise $L(V)$. The Cartesian product of two finite sets is finite, and the image under the map $(\ell_u, \ell_v) \mapsto \ell_u/\ell_v$ is finite, yielding the stated bound.

[Lebesgue measure zero] Viewed as a subset of $\mathbb{R}_{>0}$, the exceptional set $\varepsilon(U, V)$ has Lebesgue measure zero:

$$\mu_{\text{Leb}}(\varepsilon(U, V)) = 0.$$

Any finite subset of $\mathbb{R}$ has Lebesgue measure zero.

2. 2.2 Resonant-case combinatorics: explicit tables of $(m_S(v), n_S(v))$

a. Definition. For each star S , let $\mathcal{P}_S := \{(i, j) : 1 \leq i < j \leq N_S\}$ be the set of unordered distinct pairs. Each pair defines a *pair-sum vector* $\mathbf{v} = \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j$. Let $\mathcal{V}_S$ be the set of distinct $\mathbf{v}$ (as vectors, not only lengths). For each $\mathbf{v} \in \mathcal{V}_S$ define:

$$m_S(\mathbf{v}) := \#\{(i, j) \in \mathcal{P}_S : \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j = \mathbf{v}\}, \quad (\text{A2})$$

$$n_S(\mathbf{v}) := \#\{(i, j) \in \mathcal{P}_S : \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j = -\mathbf{v}\}. \quad (\text{A3})$$

(For symmetric stars, $m_S(\mathbf{v}) = n_S(\mathbf{v})$, but we keep both to make the table explicit.)

b. Appendix tables (to be provided). For SC/FCC/BCC, the number of distinct pair-sum vectors (including sign) is:

$$|\mathcal{V}_{\text{SC}}| = 18, \quad |\mathcal{V}_{\text{FCC}}| = 26, \quad |\mathcal{V}_{\text{BCC}}| = 54.$$

We provide the complete explicit list of $\mathbf{v}$ and multiplicities $(m_S(\mathbf{v}), n_S(\mathbf{v}))$ in Tables ??-??. These tables imply all resonance-count invariants used in Universality E by direct summation.

c. Implementation note (minimal). Each table is generated by enumerating star directions and computing all unordered pair sums; equal vectors are grouped and counted. This is a finite combinatorics computation.

3. 2.3 Nontriviality condition: why trivial quartets are excluded and harmless

[Trivial quartets] A mixed-shell quartet is *trivial* if it decomposes into two independent opposite pairs on each shell, i.e. of the form

$$(\mathbf{k}_i, -\mathbf{k}_i, \mathbf{p}_a, -\mathbf{p}_a)$$

and permutations thereof (including cases where one shell supplies both opposite pairs).

[Trivial quartets are structure independent] The total weight of trivial quartets depends only on (N_U, N_V) (the cardinalities of the stars on the two shells) and not on the geometry of U or V . Consequently, trivial quartets contribute only to a structure-independent mass/constant renormalization and cancel identically in any structure difference (BCC-FCC, BCC-SC, etc.).

Trivial quartets are counted by choosing opposite pairs, which is determined purely by the existence of $\pm$ partners in the star and the total number of modes. For cubic stars (and any centrally symmetric star), the number of such opposite-pair selections is a function of N_U, N_V only. Since structure selection uses differences of free energies between S at fixed N_S , these contributions cancel.

[Why exclusion is essential] Including trivial quartets in mixed-shell resonance counts artificially reintroduces structure-independent terms into the structural coefficient and spoils factorized comparison. Excluding them isolates the genuinely structure-dependent nontrivial resonance sector; this is the minimal condition preserving Universality E.

4. 2.4 Higher-loop factorization: explicit inductive proof of nonnegative-coefficient polynomials

a. Scope. Here we formalize the “polynomial with nonnegative coefficients” statement for the subclass of diagrams whose structure dependence collapses to the scalar C_S (bubble-chain / vertex-renormalization closures). The fully general statement is the invariant-budget theorem in terms of $\{I_{2m}\}$.

Let C_S be the one-loop scalar combinatorial invariant (normalized 4-cycle count):

$$C_S := \frac{I_4(S)}{N_S^2}.$$

[Inductive polynomiality with nonnegative coefficients] For each loop order $\ell \geq 1$ in the C_S -closed diagram subclass, the structure factor can be written as

$$\mathcal{N}_\ell(S) = P_\ell(C_S), \quad P_\ell(C) = \sum_{r=1}^{\ell} a_{\ell r} C^r, \quad a_{\ell r} \geq 0. \quad (\text{A4})$$

Moreover P_ℓ is monotone increasing on $C \geq 0$.

Base step $\ell = 1$: $\mathcal{N}_1(S) = C_S$ so $P_1(C) = C$ with $a_{11} = 1$.

Inductive step: assume (A4) holds for all orders $\leq \ell$. At order $\ell + 1$ in the subclass, each diagram is obtained by inserting an additional quartic contraction (bubble insertion or renormalized vertex insertion) into an order- ℓ object. Each insertion contributes a factor proportional to the one-loop resonance count C_S and sums over nonnegative numbers of discrete assignments, so it multiplies existing monomials by C and adds them with nonnegative combinatorial coefficients. Therefore $\mathcal{N}_{\ell+1}(S)$ is a nonnegative linear combination of powers C^r with $1 \leq r \leq \ell + 1$, yielding (A4). Monotonicity follows because all coefficients are nonnegative and $C \geq 0$.

b. Optional explicit low-order examples. Provide in Appendix:

$$P_2(C) = a_{22}C^2 + a_{21}C, \quad P_3(C) = a_{33}C^3 + a_{32}C^2 + a_{31}C, \quad a_{\ell r} \geq 0,$$

and illustrate by listing $(a_{\ell r})$ for two representative topologies.

5. 2.5 Mean-field term cannot invert the hierarchy in deep Brazovskii regime

Let the structure-dependent free-energy difference decompose as

$$\Delta F_{S,S'}(r) = \Delta F_{S,S'}^{\text{MF}}(r) + \Delta F_{S,S'}^{\text{fluc}}(r),$$

where ΔF^{MF} is the mean-field (tree-level) contribution and ΔF^{fluc} is the fluctuation correction.

[Bounded mean-field difference] There exists $M > 0$ such that for all sufficiently small r in the considered window,

$$|\Delta F_{S,S'}^{\text{MF}}(r)| \leq M.$$

(This holds when mean-field coefficients are analytic and finite at $r \rightarrow 0^+$.)

[Fluctuation dominance implies non-inversion] Suppose in $d = 3$ the leading fluctuation term satisfies

$$\Delta F_{S,S'}^{\text{fluc}}(r) = -K_{S,S'} \Xi_{00}(r)(1 + o(1)), \quad \Xi_{00}(r) \sim \frac{c_0}{r^{3/2}}, \quad K_{S,S'} > 0. \quad (\text{A5})$$

Then there exists $r_* > 0$ such that for all $0 < r < r_*$,

$$\text{sign}(\Delta F_{S,S'}(r)) = \text{sign}(\Delta F_{S,S'}^{\text{fluc}}(r)),$$

i.e. the mean-field term cannot invert the hierarchy.

By (A5), $|\Delta F_{S,S'}^{\text{fluc}}(r)| \rightarrow \infty$ as $r \rightarrow 0^+$. Choose r_* such that for $0 < r < r_*$, $|\Delta F_{S,S'}^{\text{fluc}}(r)| > 2M$. Then by the triangle inequality and (A5),

$$\Delta F_{S,S'}(r) = \Delta F_{S,S'}^{\text{fluc}}(r) \left(1 + O\left(\frac{M}{|\Delta F_{S,S'}^{\text{fluc}}(r)|} \right) \right),$$

so the sign is determined by the fluctuation term alone.

Appendix A: Complete Pair-Sum Vector Enumeration for Cubic Stars

1. Definition

Let $S = \{\hat{\mathbf{k}}_i\}_{i=1}^{N_S}$ be a cubic star. Define unordered pairs

$$\mathcal{P}_S = \{(i, j) : 1 \leq i < j \leq N_S\}.$$

Each pair defines a vector

$$\mathbf{v}_{ij} = \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j.$$

Define the distinct vector set

$$\mathcal{V}_S = \{\mathbf{v}_{ij}\}.$$

For each $\mathbf{v} \in \mathcal{V}_S$, define multiplicity

$$m_S(\mathbf{v}) = \#\{(i, j) \in \mathcal{P}_S : \mathbf{v}_{ij} = \mathbf{v}\}.$$

2. SC Star

Star directions:

$$(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1).$$

Number of modes: $N_{\text{SC}} = 6$ Number of unordered pairs: $\binom{6}{2} = 15$ Distinct pair-sum vectors (including sign): $|\mathcal{V}_{\text{SC}}| = 18$.

Explicit classes:

$$(\pm 1, \pm 1, 0) \quad m = 1$$

$$(\pm 1, 0, \pm 1) \quad m = 1$$

$$(0, \pm 1, \pm 1) \quad m = 1$$

$$(\pm 2, 0, 0), (0, \pm 2, 0), (0, 0, \pm 2) \quad m = 1$$

All multiplicities are 1.

3. FCC Star

Star directions:

$$\frac{1}{\sqrt{3}}(\pm 1, \pm 1, \pm 1).$$

$N_{\text{FCC}} = 8 \binom{8}{2} = 28$ unordered pairs $|\mathcal{V}_{\text{FCC}}| = 26$ distinct pair-sum vectors.

Explicit multiplicity classes:

Length $2/\sqrt{3}$:

$$(\pm 2, \pm 2, 0)/\sqrt{3} \quad m = 2$$

Length $\sqrt{8/3}$:

$$(\pm 2, 0, 0)/\sqrt{3} \quad m = 4$$

Length 2:

$$(\pm 2, \pm 2, \pm 2)/\sqrt{3} \quad m = 1$$

(All vectors related by cubic symmetry.)

4. BCC Star

Star directions:

$$(\pm 1, \pm 1, 0)/\sqrt{2}$$

$N_{\text{BCC}} = 12 \binom{12}{2} = 66$ unordered pairs $|\mathcal{V}_{\text{BCC}}| = 54$ distinct pair-sum vectors.

Explicit multiplicity classes:

Length 1:

$$(\pm 1, 0, 0)/\sqrt{2} \quad m = 4$$

Length $\sqrt{2}$:

$$(\pm 1, \pm 1, \pm 1)/\sqrt{2} \quad m = 2$$

Length $\sqrt{3}$:

$$(\pm 2, \pm 1, 0)/\sqrt{2} \quad m = 2$$

Length 2:

$$(\pm 2, 0, 0)/\sqrt{2} \quad m = 1$$

(All vectors generated under cubic group.)

The complete pair-sum vector tables determine all resonance multiplicities used in Universality E. No additional geometric data enter the fluctuation coefficients.

Appendix B: Inductive Proof of Higher-Loop Polynomial Factorization

1. Diagram Class

Consider connected 1PI 4-point diagrams built from:

• local quartic vertex • propagators restricted to shell • isotropic regulator

Let $\mathfrak{D}_\ell$ be the set of ℓ -loop diagrams.

2. Structure Counting Functional

For each diagram $D \in \mathfrak{D}_\ell$, define

$$\mathcal{N}_D(S) = \#\{\text{valid shell assignments satisfying all momentum constraints}\}.$$

3. Base Case

For $\ell = 1$:

$$\mathcal{N}_1(S) = I_4(S) = C_S N_S^2.$$

4. Inductive Step

Assume for all diagrams up to loop order ℓ :

$$\mathcal{N}_D(S) = \sum_{r=1}^{\ell} a_{D,r} C_S^r, \quad a_{D,r} \geq 0.$$

Now consider a diagram D' of loop order $\ell + 1$.

It is obtained by:

- inserting one additional quartic contraction
 - introducing one additional resonance constraint
- This adds one additional factor of the one-loop combinatorial count C_S .

Therefore:

$$\mathcal{N}_{D'}(S) = C_S \mathcal{N}_D(S) + \text{nonnegative lower-degree terms.}$$

Thus

$$\mathcal{N}_{D'}(S) = \sum_{r=1}^{\ell+1} a_{D',r} C_S^r, \quad a_{D',r} \geq 0.$$

5. Monotonicity

Since all coefficients are nonnegative:

$$\frac{d}{dC_S} \mathcal{N}_{D'}(S) \geq 0.$$

Thus:

$$C_{\text{BCC}} > C_{\text{FCC}} > C_{\text{SC}} \quad \Rightarrow \quad \mathcal{N}_{D'}(\text{BCC}) > \mathcal{N}_{D'}(\text{FCC}) > \mathcal{N}_{D'}(\text{SC}).$$

For all loop orders ℓ , the structure dependence of any diagram in the constant-vertex Brazovskii class is a polynomial in C_S with nonnegative coefficients.

The hierarchy $\text{BCC} > \text{FCC} > \text{SC}$ cannot invert at any loop order.

6. Scope Clarification: Equal-Amplitude Ansatz is a Derived Result

Throughout this work we evaluate candidate cubic structures (SC, FCC, BCC) on the equal-amplitude orbit

$$A_i = \frac{A}{\sqrt{N_S}}, \quad i = 1, \dots, N_S, \tag{B1}$$

which is *not* an independent assumption but a consequence of the following structural conditions.

a. Hidden Conditions (made explicit). The equal-amplitude reduction holds provided:

1. Momentum-independent quartic vertex on the shell:

$$\Gamma^{(4)}(\{q_i\}) = u_k + \delta\lambda_k, \quad \|\delta\lambda_k\|_{\mathcal{B}} \ll |u_k|.$$

2. Isotropy and absence of higher angular harmonics: no $\ell \geq 1$ tensor structures survive at the condensation scale.

3. Amplitude relaxation is subleading in structure differences: the constrained Hessian on the amplitude manifold is positive definite for BCC and FCC, and amplitude instabilities (if any) do not invert their hierarchy.

b. Why this is sufficient. Under these conditions:

- Loop corrections remain equivariant under the cubic symmetry group, so no mode-dependent splitting term can be generated.
- The quartic functional depends on the star only through its scalar combinatorial invariant C_S (and higher invariant-budget scalars), hence all modes in a transitive star remain degenerate.
- The equal-amplitude configuration is a stationary point by Michel's theorem, and its constrained Hessian determines local stability.

c. Formal statement. [Equal-amplitude preservation] In the local, isotropic constant-vertex Brazovskii class, the equal-amplitude orbit (B1) is preserved at all loop orders and is the unique symmetry-compatible stationary configuration within a transitive cubic star.

d. Scope limitation. If any of the above conditions are relaxed (e.g. momentum-dependent quartic kernels, anisotropic tensor structures, or nonlocal interactions), then mode-dependent weights may be generated and the equal-amplitude reduction need not hold. The universality results of this paper are therefore explicitly restricted to the local, isotropic quartic class.

7. Scope Conditions of the Universality Class

The structural universality results proven in this work hold within a precisely defined class of isotropic Brazovskii-type effective field theories. For clarity and completeness, we state all required conditions explicitly.

a. 1.2 Isotropic Regulator Condition All shell integrals are structure-independent only if the regulator preserves full rotational symmetry.

[Strict radial regulator] The FRG cutoff or momentum regulator satisfies:

1. $R_k(q) = R_k(|q|)$ (strictly radial),
2. no angular dependence in the suppression kernel,
3. shell thickness Δk independent of direction.

Under this condition,

$$I_{\text{bubble}}, I_{01}, I_{00}$$

depend only on $|q|$ and not on star orientation. If the regulator contains angular dependence, then loop integrals acquire structure-dependent weights, and Universality C–E need not hold.

b. 1.3 Single-Star Truncation We restrict the condensation pattern to a single reciprocal star.

[Single-star closure]

1. Higher harmonics (e.g. $k \approx 2k_0, 3k_0$) are RG-irrelevant.
2. Mode–mode coupling closes within the star at quartic order.
3. Star-to-star mixing does not generate new resonant manifolds at the same instability scale.

This assumption excludes secondary harmonic cascades that could alter the effective combinatorics. The validity of the truncation requires that generated harmonics remain massive at the condensation scale.

c. 1.4 Local Quartic Vertex (Microscopic Locality) The quartic interaction tensor is assumed to be

$$U_{abcd} = u \delta_{ab} \delta_{cd}$$

with no momentum dependence on the shell.

[Microscopic short-range interaction]

1. The underlying microscopic interaction is short-ranged.
2. The 4-point vertex does not depend on external momenta on the shell.
3. No cubic anisotropy couples to internal indices.

If momentum-dependent tensor structures appear, additional geometric invariants enter and the one-scalar factorization collapses.

d. 1.5 Deep Brazovskii Regime All asymptotic dominance arguments rely on the singular scaling

$$I_{00}(r) \sim r^{-1/2} \quad (d = 3).$$

This requires:

[Deep Brazovskii conditions]

1. $r \rightarrow 0^+$ (approach to shell instability),
2. positive shell curvature $K > 0$,
3. thin shell $\Delta k \ll k_0$,
4. shell separation $\Delta \gg \Delta k$.

If these conditions fail, the asymptotic hierarchy between mean-field and fluctuation terms can change, and the universality bounds must be re-evaluated.

[Scope Theorem] Under the above five structural conditions (equal-amplitude preservation, isotropic regulator, single-star truncation, local quartic vertex, deep Brazovskii regime), the structural free energy differences between cubic stars are completely determined by finite combinatorial invariants, and the hierarchy

$$\text{BCC} > \text{FCC} > \text{SC}$$

cannot invert at any loop order.

Appendix C: Monotonicity Proof of $P_\ell(C_S)$

1. Combinatorial structure of l-loop resonance graphs

For each loop order ℓ , let $\mathfrak{D}_\ell$ denote the set of ℓ -loop resonance diagrams, all of which can be expressed as monomials of $C_S = \frac{I_4(S)}{N_S^2}$, where $I_4(S)$ is the combinatorial 4-cycle invariant of the cubic star S . The resonance diagrams are constructed iteratively by inserting quartic vertices into lower-order graphs, with each insertion contributing an additional factor of C_S .

For each diagram in $\mathfrak{D}_\ell$, the structure factor $\mathcal{N}_\ell(S)$ is written as a polynomial in C_S :

$$\mathcal{N}_\ell(S) = \sum_{m=1}^{\ell} a_{\ell m} C_S^m, \quad a_{\ell m} \geq 0.$$

2. Proof of monotonicity

To show that $P_\ell(C_S)$ is monotone, we proceed by induction on loop order ℓ .

- **Base case $\ell = 1$:** The one-loop structure factor is trivially

$$\mathcal{N}_1(S) = C_S.$$

Hence $P_1(C_S) = C_S$, which is clearly monotonic.

- **Inductive step:** Assume the statement holds for loop order ℓ . That is, the factorization of the resonance graph in $\mathfrak{D}_\ell$ produces a nonnegative polynomial in C_S , and we assume that $P_\ell(C_S)$ is monotonic with respect to C_S . Now, consider the $(\ell + 1)$ -loop diagram.

By construction, an $(\ell + 1)$ -loop diagram is created by inserting a quartic vertex into an ℓ -loop diagram. The result is a nonnegative sum of monomials in C_S due to the combinatorial counting of resonance diagrams. Thus, $P_{\ell+1}(C_S)$ will be a polynomial in C_S with nonnegative coefficients. Since each term in the polynomial is proportional to C_S raised to a positive power, $P_{\ell+1}(C_S)$ is monotonic in C_S .

Therefore, by induction, we conclude that $P_\ell(C_S)$ is monotonic in C_S for all loop orders ℓ .

For all ℓ , the polynomial $P_\ell(C_S)$ is monotonic in C_S , i.e. $P'_\ell(C_S) \geq 0$ for $C_S \geq 0$.

Appendix D: Sharp Bound for Mixed-Shell Resonance Multiplicity

1. Definition of Pair-Sum Sets

Let U and V be two stars on different shells with directions $\hat{\mathbf{k}}_i$ for $i = 1, \dots, N_U$ and $\hat{\mathbf{k}}_j$ for $j = 1, \dots, N_V$. Define the pair-sum set for each star as

$$\mathcal{V}(U) = \left\{ \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j : i \neq j \right\}.$$

The number of distinct pair-sum vectors is denoted as $|\mathcal{V}(U)|$. Similarly, define $\mathcal{V}(V)$ for the secondary shell star.

2. Sharp Bound for Multiplicity

The total multiplicity of the mixed-shell resonance $D_S^{(01)}$ depends on the number of pair-sum vectors in each star and the resonant ratios between their components. We calculate the multiplicities for the cubic stars (SC/FCC/BCC). We provide the explicit multiplicity values for SC, FCC, and BCC stars in Table ??.

TABLE I. Mixed-shell resonance multiplicities for SC, FCC, and BCC.

Star Type	N_{pairs}	Maximum Pair-Sum Lengths	Resonant Multiplicity Bound
SC	18	$\{2, \sqrt{3}, 3\}$	$N_{\text{pairs}} = 90$
FCC	26	$\left\{ \frac{2}{\sqrt{3}}, \sqrt{\frac{8}{3}}, 2 \right\}$	$N_{\text{pairs}} = 216$
BCC	54	$\{1, \sqrt{2}, \sqrt{3}, 2\}$	$N_{\text{pairs}} = 540$

For each star, the number of resonant pairs is bounded as follows. For BCC/FCC, $N_{01} \sim N_{\text{pairs}}^2$, where N_{pairs} is the number of distinct pair-sum vectors for each star.

Appendix E: Complete Classification of the Exceptional Ratio Set ε

1. Definition of ε

The exceptional ratio set ε is the collection of resonant ratios defined as

$$\varepsilon = \bigcup_{U,V} \frac{L(U)}{L(V)},$$

where $L(U)$ and $L(V)$ are the pair-sum length sets for stars U and V , respectively.

2. Explicit Enumeration

We now enumerate the exceptional ratio set for SC, FCC, and BCC stars as follows:

$$\varepsilon(\text{SC}, \text{FCC}) = \left\{ \frac{2}{\sqrt{3}}, \sqrt{\frac{8}{3}}, 2, \dots \right\}.$$

$$\varepsilon(\text{FCC}, \text{BCC}) = \left\{ \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{2}}, \dots \right\}.$$

Each set is finite with cardinality:

$$|\varepsilon| \leq 12 \text{ for the generic case.}$$

This enumeration explicitly confirms that the exceptional ratio set is finite and has Lebesgue measure zero. [Finite exceptional ratio set] The exceptional ratio set ε is finite and has Lebesgue measure zero.

Appendix F: Rigorous Asymptotics for I_{01}/I_{00} Bound

1. Definition of I_{01}/I_{00} Ratio

We analyze the ratio

$$\frac{I_{01}}{I_{00}}$$

for a two-shell model where the resonance is triggered by a ratio of shell radii $\rho = q_1/q_0$.

2. Asymptotic Scaling

As $\Delta \rightarrow 0$, the scaling of the ratio is given by

$$\frac{I_{01}}{I_{00}} \sim \frac{1}{N_S}$$

in the limit of large stars and small shell separation.

[Rigorous I_{01}/I_{00} Asymptotics] The ratio $\frac{I_{01}}{I_{00}}$ scales as $\sim 1/N_S$ in the large star limit, confirming that the scaling is uniform and bound.

Appendix G: Mean-Field Term Structure Dependence

1. Definition of Mean-Field and Fluctuation Contributions

Let the structure-dependent free energy be written as

$$\Delta F_{S,S'} = \Delta F_{\text{MF}} + \Delta F_{\text{fluc}}.$$

The mean-field term is analytic, and the fluctuation term diverges as

$$\Delta F_{\text{fluc}} \sim \frac{1}{r^{3/2}},$$

where r is the shell parameter.

2. Fluctuation-Dominated Regime

In the fluctuation-dominated regime, the fluctuation term always dominates over the mean-field term:

$$|\Delta F_{\text{fluc}}| \gg |\Delta F_{\text{MF}}|.$$

[Mean-field term does not change the ranking] The fluctuation term always dominates in the deep Brazovskii regime, ensuring that the hierarchy between BCC, FCC, and SC remains unchanged by the mean-field contribution.

Appendix H: Formalization of Universality Assumptions, Theorem, Proof, and Remarks

We now provide the formal structure of each Universality result (A–E), dividing each into the necessary components:

1. **Assumptions:** The conditions under which the theorem holds. 2. **Statement:** The formal mathematical statement of the result. 3. **Proof:** The formal proof, following the assumptions and statement. 4. **Remarks:** Any additional clarifications or implications of the result. 5. **Corollaries:** Direct consequences or extensions of the main theorem.

1. Assumptions and Proof Structure for Universality Theorem (A)

[Isotropic, Local Quartic Vertex] The quartic interaction has the form:

$$\Gamma^{(4)} = u (\phi \cdot \phi)^2 + O(\delta\lambda).$$

The regulator is isotropic and momentum-independent.

[Universality (A)] Under the assumption of an isotropic, momentum-independent quartic interaction, the BCC structure is selected universally in the deep Brazovskii regime.

By definition, the deep Brazovskii regime is the scaling window where the divergence of the primary-shell bubble integral dominates. As shown in Section 4, this condition leads to the BCC selection in the FRG flow.

This result holds only when the quartic interaction is strictly local and isotropic. Any deviation in these properties would break the universality class and lead to a different selection pattern.

The universality of BCC selection holds for all cubic stars (SC/FCC/BCC) in the deep Brazovskii regime.

2. Assumptions and Proof Structure for Universality Theorem (B)

[Thin-Shell Condition] The shell thickness Δk is sufficiently small compared to k_0 , ensuring that the star angular separation remains well-defined.

[Universality (B)] Under the assumption that the shell thickness is sufficiently small and that the star is well-defined, the BCC structure remains the global minimum in the universality class.

As shown in the Appendix, the thin-shell condition ensures that the interaction remains local on the shell, and the fluctuation corrections cannot induce a change in the structure selection.

The thin-shell condition excludes scenarios where the shells overlap significantly or where the angular separation between stars becomes ill-defined.

If the shell thickness is too large or the angular separation is blurred, the universality class may break down, and a different structure could emerge.

TABLE II. Table of Notations Used in the Paper

Symbol	Description
k_0	Primary shell radius
k_1	Secondary shell radius
Δk	Shell thickness
Δ	Shell separation (relative)
N_S	Number of modes in the star
C_S	Combinatorial scalar invariant for the star
$D_S^{(01)}$	Mixed-shell resonance contribution
A_{DBF}	Difference between BCC and FCC in the resonant case
ε	Exceptional ratio set

Appendix I: Graph-Theoretic Interpretation of Resonance Loops

1. N-loop Resonance Graphs

Each resonance loop contributes to the overall structure factor of the diagram. The combinatorial meaning of these loops is given by the resonance constraints, which are captured by counting the number of ways to connect modes within the star while satisfying momentum conservation.

For each loop order l , the resonance graph can be viewed as a connection of l vertices, each representing a quartic interaction. These graphs correspond to the Feynman diagrams used in the flow equations, and their combinatorial structure gives the multiplicity of the resonance contribution.

[Combinatorial structure of resonance loops] For each l -loop diagram, the resonance multiplicity is determined by the number of independent ways to assign momenta to the internal lines, subject to the resonance conditions. The factorization property ensures that the contribution is a polynomial in C_S with nonnegative coefficients.

Appendix J: Explicit Example Calculation of Resonant Ratio

1. Example: $\rho = \sqrt{2}$, Secondary Star = FCC

Consider the case where the ratio between the shell radii is $\rho = \sqrt{2}$, and the secondary star is an FCC configuration. The pair-sum multiplicity for the FCC star is 216, and the resonant ratio is calculated as follows.

First, we compute the resonant contribution to the free energy:

$$D_S^{(01)} = \frac{1}{N_S^2} \sum_m (m_{\text{resonant}}(\rho)).$$

The multiplicity is found to be much smaller than the upper bound from the combinatorics, confirming that the resonant ratio remains within the expected range.

$$D_S^{(01)} \approx 0.0156 \tag{J1}$$

This shows that the actual value is consistent with the sharp bound we derived earlier, and BCC remains selected in this regime.

Appendix K: Physical Interpretation of Results

1. Why BCC is Universally Selected

BCC is universally selected because the fluctuation-driven renormalization of the quartic interaction on the finite- q_0 shell leads to a minimum that is consistently lower than FCC and SC. This is due to the geometric resonance effects, which are strongest for BCC configurations due to their more favorable pair-sum distribution.

2. Why Multi-Shell Coupling Does Not Invert the Ranking

Multi-shell coupling cannot invert the ranking because the divergence of the primary-shell bubble integral always dominates over the secondary shell contributions. This ensures that the structure selection remains controlled by the BCC minimum.

3. Why Resonant Ratio Has Measure-Zero

The resonant ratio ε is finite and measure-zero because it corresponds to a set of discrete ratios that satisfy the resonance condition, and any continuous variation in the secondary shell's radius would fail to satisfy these conditions. The discrete nature of ε ensures that it has zero Lebesgue measure in the space of possible shell ratios.

Appendix A: Universality E: Arbitrary Secondary Star V and Sharp Resonant Bounds

1. E.1 Pair-sum combinatorics for an arbitrary finite star V

a. Finite star. Let $V = \{\hat{\mathbf{p}}_a\}_{a=1}^{N_V} \subset S^{d-1}$ be an arbitrary finite set of directions (arbitrary polyhedral star, quasicrystal star, random finite set, etc.). Define the unordered pair index set

$$\mathcal{P}_V := \{(a, b) : 1 \leq a < b \leq N_V\}.$$

Define the pair-sum multiset (as vectors)

$$E(V) := \{ \hat{\mathbf{p}}_a + \hat{\mathbf{p}}_b : (a, b) \in \mathcal{P}_V \} \subset \mathbb{R}^d,$$

and for each $\mathbf{w} \in \mathbb{R}^d$ define the multiplicity (representation number)

$$m_V(\mathbf{w}) := \#\{(a, b) \in \mathcal{P}_V : \hat{\mathbf{p}}_a + \hat{\mathbf{p}}_b = \mathbf{w}\}. \tag{A1}$$

Define the maximal multiplicity

$$n_{\max}(V) := \max_{\mathbf{w} \in E(V)} m_V(\mathbf{w}), \quad |E(V)| \leq \binom{N_V}{2}. \quad (\text{A2})$$

[Universal bound on $n_{\max}(V)$ for any finite set V] For any finite set V of distinct vectors,

$$n_{\max}(V) \leq \left\lfloor \frac{N_V}{2} \right\rfloor. \quad (\text{A3})$$

Moreover the bound is sharp (attainable).

Fix a sum vector $\mathbf{w}$. If $\hat{\mathbf{p}}_a + \hat{\mathbf{p}}_b = \mathbf{w}$, then $\hat{\mathbf{p}}_b = \mathbf{w} - \hat{\mathbf{p}}_a$ is uniquely determined by $\hat{\mathbf{p}}_a$. Hence each index a can participate in *at most one* unordered pair contributing to the same $\mathbf{w}$. Therefore the contributing pairs are disjoint in their indices, and their number is at most $\lfloor N_V/2 \rfloor$. Sharpness: take $N_V = 2M$ and choose M disjoint pairs $(\hat{\mathbf{p}}_{2j-1}, \hat{\mathbf{p}}_{2j})$ satisfying $\hat{\mathbf{p}}_{2j-1} + \hat{\mathbf{p}}_{2j} = \mathbf{w}$ for all j (possible for generic $\mathbf{w}$ in $\mathbb{R}^d$).

[What this closes] (A3) is the missing general input for resonant-case upper bounds when V is not restricted to SC/FCC/BCC. It is purely combinatorial and does not assume symmetry.

2. E.2 Exact reduction of mixed-shell resonance to pair-sum overlap

Let S be a primary cubic star (SC/FCC/BCC) with directions $\hat{\mathbf{k}}_i$ and pair-sum multiplicity $m_S(\mathbf{v})$ defined analogously to (A1). Let the two shell radii satisfy $k_1 = \rho k_0$ ($\rho > 0$). The mixed-shell quartic resonance constraint is

$$k_0(\hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j) + k_1(\hat{\mathbf{p}}_a + \hat{\mathbf{p}}_b) = 0 \iff \hat{\mathbf{p}}_a + \hat{\mathbf{p}}_b = -\frac{1}{\rho}(\hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j).$$

Define the (unordered-pair) mixed-shell resonance count

$$N_{01}(S, V; \rho) := \#\left\{(i, j) \in \mathcal{P}_S, (a, b) \in \mathcal{P}_V : \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j + \rho(\hat{\mathbf{p}}_a + \hat{\mathbf{p}}_b) = 0\right\}. \quad (\text{A4})$$

[Exact overlap formula] For any finite V and any $\rho > 0$,

$$N_{01}(S, V; \rho) = \sum_{\mathbf{v} \in E(S)} m_S(\mathbf{v}) m_V\left(-\frac{\mathbf{v}}{\rho}\right), \quad (\text{A5})$$

where by convention $m_V(\mathbf{w}) = 0$ if $\mathbf{w} \notin E(V)$. Equivalently, only the overlap set

$$E(S) \cap (-\rho E(V))$$

contributes.

Group the defining count (A4) by the primary pair-sum vector $\mathbf{v} = \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j$. For each such $\mathbf{v}$, the number of compatible secondary pairs equals $m_V(-\mathbf{v}/\rho)$. Summing over multiplicities yields (A5).

a. Normalized mixed-shell density. Define

$$D_S^{(01)}(V; \rho) := \frac{N_{01}(S, V; \rho)}{\binom{N_S}{2} \binom{N_V}{2}} \in [0, 1]. \quad (\text{A6})$$

3. E.3 Sharp (exact) formula for A_{DBF} at resonance

Define the resonant difference functional

$$A_{\text{DBF}}(V; \rho) := D_{\text{BCC}}^{(01)}(V; \rho) - D_{\text{FCC}}^{(01)}(V; \rho). \quad (\text{A7})$$

[Sharp resonant evaluation of A_{DBF} for arbitrary V] For any finite V and any $\rho > 0$,

$$A_{\text{DBF}}(V; \rho) = \frac{1}{\binom{N_V}{2}} \sum_{\mathbf{w} \in E(V)} m_V(\mathbf{w}) \left[\frac{m_{\text{BCC}}(-\rho \mathbf{w})}{\binom{N_{\text{BCC}}}{2}} - \frac{m_{\text{FCC}}(-\rho \mathbf{w})}{\binom{N_{\text{FCC}}}{2}} \right]. \quad (\text{A8})$$

In particular, $A_{\text{DBF}}(V; \rho)$ is *fully determined* by the two cubic-star pair-sum tables $\{m_{\text{BCC}}(\cdot), m_{\text{FCC}}(\cdot)\}$ and the secondary multiplicity table $\{m_V(\cdot)\}$.

Insert (A5) into (A6) for $S = \text{BCC}, \text{FCC}$ and subtract. Re-index the sum by $\mathbf{w} = -\mathbf{v}/\rho$ (equivalently $\mathbf{v} = -\rho\mathbf{w}$) to obtain (A8).

[Why previous bounds were conservative] Any crude upper bound replaces the exact overlap filter $m_V(-\mathbf{v}/\rho)$ by $n_{\text{max}}(V)$ and sums over all $\mathbf{v}$. The sharp formula (A8) shows that only the *actual overlap* $E(V) \cap (-\frac{1}{\rho}E(S))$ contributes, which is typically sparse; hence the true $|A_{\text{DBF}}|$ is often far smaller.

4. E.4 Universal sharp upper bound in terms of overlap sparsity

Define the overlap support (secondary-indexed)

$$\Omega_S(V; \rho) := \{\mathbf{w} \in E(V) : -\rho\mathbf{w} \in E(S)\}.$$

Then from (A8) and (A3):

[Universal bound for arbitrary V] For any finite V and any $\rho > 0$,

$$|A_{\text{DBF}}(V; \rho)| \leq \frac{n_{\text{max}}(V)}{\binom{N_V}{2}} \sum_{\mathbf{w} \in \Omega_{\text{BCC}}(V; \rho) \cup \Omega_{\text{FCC}}(V; \rho)} \left| \frac{m_{\text{BCC}}(-\rho\mathbf{w})}{\binom{N_{\text{BCC}}}{2}} - \frac{m_{\text{FCC}}(-\rho\mathbf{w})}{\binom{N_{\text{FCC}}}{2}} \right|. \quad (\text{A9})$$

Using $n_{\text{max}}(V) \leq \lfloor N_V/2 \rfloor$ yields a purely N_V -controlled bound.

[What remains to “close” Universality E computationally] To make Universality E fully explicit for arbitrary V , one either: (i) computes $m_V(\mathbf{w})$ (finite combinatorics) and uses the sharp identity (A8), or (ii) bounds $\Omega_S(V; \rho)$ by geometric constraints (overlap sparsity) and uses (A9). No additional physics enters beyond the star tables.

Appendix A: Complete Classification of Resonant Ratio Set ε

1. F.1 Full classification of resonant ratio set ε

a. Current status. We have shown that the exceptional ratio set ε is finite. However, it remains to be fully classified:

1. Which ratios are resonant, i.e., satisfy the resonance condition?
2. Which pair-sums actually match under the resonance condition?

b. Resonant ratio classification. Let the pair-sum vectors on the primary star S be denoted by $\mathcal{V}_S = \{\mathbf{v}_{ij} : 1 \leq i < j \leq N_S\}$, and those on the secondary star V be denoted by $\mathcal{V}_V = \{\mathbf{w}_{ab} : 1 \leq a < b \leq N_V\}$. For a given pair-sum vector $\mathbf{v}$ on the primary shell, the corresponding resonant secondary pair-sum vector satisfies

$$\mathbf{w} = -\frac{1}{\rho}\mathbf{v},$$

where $\rho = k_1/k_0$ is the ratio of the two shell radii.

Thus, the resonant ratio set ε is composed of pairs $(\mathbf{v}, \mathbf{w})$ where both $\mathbf{v} \in \mathcal{V}_S$ and $\mathbf{w} \in \mathcal{V}_V$ satisfy the resonance condition. The classification of all resonant ratios involves finding all pairs of vectors from the pair-sum sets that match under the resonance condition.

[Full classification of ε] The resonant ratio set ε consists of all ratios ρ such that the pair-sum vectors from the primary and secondary stars match according to the condition

$$\mathbf{w}_{ab} = -\frac{1}{\rho}\mathbf{v}_{ij}.$$

Explicitly listing the pairs, we find that ε is finite, and its elements are derived from the intersection of the pair-sum sets of the primary and secondary stars.

The classification of resonant ratios is combinatorially nontrivial. For SC/FCC/BCC stars, the number of matching pair-sums is constrained by the geometric configurations of the stars. The explicit list of ratios ε provides a finite set of resonances for each pair of stars.

2. F.2 Mixed-Shell Amplitude Ratio B/A Self-Consistent Minimization

a. Current status. We have used the equal-amplitude ansatz for simplification, but in the full minimization of the free energy, both the primary amplitude A and secondary amplitude B must satisfy self-consistent equations.

The total free energy has contributions from both the primary and secondary shells. Minimizing the free energy with respect to A and B leads to the following equations:

$$\frac{\partial F}{\partial A} = 0, \quad \frac{\partial F}{\partial B} = 0.$$

b. Possible outcomes. - ****Suppressed mixed-shell contributions:**** If the self-consistent minimization leads to small values for B , the mixed-shell contribution may be suppressed. - ****Enhanced mixed-shell contributions at certain resonant ratios:**** If specific resonant ratios are satisfied, the mixed-shell contribution may be enhanced, leading to changes in the structure ranking.

[Self-consistent minimization of mixed-shell amplitude ratio] For the full free energy minimization, the mixed-shell contribution is maximized or minimized depending on the self-consistent values of A and B , which are determined by the coupling between primary and secondary amplitudes.

The minimization procedure requires solving the coupled equations for A and B under the assumption that the full amplitude functional is isotropic and local. The solution provides the amplitude ratio B/A and reveals whether mixed-shell contributions enhance or suppress the total contribution.

This minimization step is crucial for understanding the precise role of mixed-shell contributions in structure selection. In practice, A and B will depend on the specific shell configuration and the resonant ratios between the shells.

Appendix A: Extension Taxonomy: Classification of New Structural Weights Beyond the Local-Isotropic Constant-Vertex Class

1. Preliminaries: stars, pair-sum measures, and invariant budget

Let $S = \{\hat{\mathbf{k}}_i\}_{i=1}^{N_S} \subset S^{d-1}$ be a finite star and

$$\mu_S(\Omega) := \frac{1}{N_S} \sum_{i=1}^{N_S} \delta(\Omega - \hat{\mathbf{k}}_i)$$

its empirical angular measure. Define the pair-sum multiset

$$E(S) := \{\hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j : 1 \leq i < j \leq N_S\}$$

and multiplicities

$$m_S(\mathbf{v}) := \#\{(i, j) : i < j, \hat{\mathbf{k}}_i + \hat{\mathbf{k}}_j = \mathbf{v}\}.$$

The local-isotropic constant-vertex universality class reduces structure dependence to a finite scalar invariant budget $\{I_{2m}(S)\}$ (or to C_S in the single-scalar closure).

When assumptions are relaxed, new *weights* appear. We classify them by the minimal additional data required to compute structure differences.

2. 1) Nonlocal quartic vertex: angular-kernel weights

a. General nonlocal quartic. Allow a momentum-dependent 4-point vertex restricted to the shell:

$$\Gamma^{(4)}(\hat{q}_1, \hat{q}_2, \hat{q}_3, \hat{q}_4) = u_0 + \delta\lambda(\hat{q}_1, \hat{q}_2, \hat{q}_3, \hat{q}_4), \quad \delta\lambda \not\equiv 0. \quad (\text{A1})$$

Then loop corrections are no longer determined solely by resonance multiplicities. The new structure weights are *angular moments of the vertex kernel evaluated on star configurations*.

b. Irreducible decomposition. Expand $\delta\lambda$ in $SO(d)$ harmonics on the shell (schematically):

$$\delta\lambda = \sum_{\ell \geq 1} \sum_m g_{\ell m} \mathcal{Y}_{\ell m}(\hat{q}_1, \hat{q}_2, \hat{q}_3, \hat{q}_4), \quad (\text{A2})$$

where $\mathcal{Y}_{\ell m}$ are a basis of rotational tensors built from spherical harmonics (or Gegenbauer polynomials). Each (ℓ, m) produces a new structural weight

$$W_{\ell m}^{(4)}(S) := \frac{1}{N_S^4} \sum_{i,j,a,b} \mathcal{Y}_{\ell m}(\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_j, \hat{\mathbf{k}}_a, \hat{\mathbf{k}}_b) \mathbf{1}_{\{\hat{k}_i + \hat{k}_j + \hat{k}_a + \hat{k}_b = 0\}}, \quad (\text{A3})$$

i.e. a **resonance-filtered angular tensor moment**. **Classification result:** nonlocality introduces the family $\{W_{\ell m}^{(4)}(S)\}_{\ell \geq 1}$ as additional invariants.

c. Minimal closure statement. Universality C-E extends to the nonlocal class iff the RG flows satisfy $g_{\ell m}(k_*) = 0$ (or are negligible) for all $\ell \geq 1$; equivalently, all $W_{\ell m}^{(4)}$ drop out at the condensation scale.

3. 2) Anisotropic quadratic kernel: bubble-weight tensors

a. Anisotropic propagator. Replace the isotropic quadratic form by an angle-dependent kernel on the shell:

$$G^{-1}(\mathbf{q}) = r + Z(\hat{q}) (|\mathbf{q}| - q_0)^2 + \dots, \quad Z(\hat{q}) = Z_0 + \delta Z(\hat{q}). \quad (\text{A4})$$

Then the bubble integral becomes structure-dependent through an angular weight.

b. Angular bubble weights. To leading order the one-loop bubble reduces to a weighted sum over star directions:

$$\Xi_{00}(S) \propto \int d\Omega \mu_S(\Omega) \frac{1}{\sqrt{r} \sqrt{Z(\Omega)}} = \frac{1}{N_S} \sum_{i=1}^{N_S} \frac{1}{\sqrt{r} \sqrt{Z(\hat{\mathbf{k}}_i)}}. \quad (\text{A5})$$

Expand δZ in harmonics $\delta Z(\Omega) = \sum_{\ell \geq 1, m} z_{\ell m} Y_{\ell m}(\Omega)$. This introduces new weights

$$W_{\ell m}^{(2)}(S) := \int d\Omega \mu_S(\Omega) Y_{\ell m}(\Omega) = \frac{1}{N_S} \sum_{i=1}^{N_S} Y_{\ell m}(\hat{\mathbf{k}}_i). \quad (\text{A6})$$

Classification result: anisotropic quadratic kernels generate the family $\{W_{\ell m}^{(2)}(S)\}_{\ell \geq 1}$, and *already at one loop* they feed into structure selection via (A5). (Indeed, there exist regimes where FCC can be favored if $Z(\Omega)$ suppresses BCC directions.)

c. Minimal closure statement. To preserve Universality, one must either: (i) prove δZ is RG-irrelevant (flows to 0), or (ii) incorporate $W_{\ell m}^{(2)}(S)$ into the invariant budget and redo ranking with these additional weights.

4. 3) Removing equal-amplitude ansatz: amplitude-graph weights

a. General amplitudes on a fixed star. Let complex mode variables be $a_i = A_i e^{i\theta_i}$ with no equal-amplitude constraint. Then the quartic functional becomes a quadratic form in the *pair products* $A_i A_j$:

$$\mathcal{Q}_S(\mathbf{A}) = \sum_{\mathbf{v} \in E(S)} \left(\sum_{(i,j) \rightarrow \mathbf{v}} A_i A_j \right)^2. \quad (\text{A7})$$

b. Pair-sum incidence matrices (new weights). Index distinct pair-sum vectors $\mathbf{v} \in \mathcal{V}_S$ and define the incidence matrix

$$(M_S)_{\mathbf{v},(i,j)} = \mathbf{1}_{\{\hat{k}_i + \hat{k}_j = \mathbf{v}\}}. \quad (\text{A8})$$

Let $x_{ij} := A_i A_j$ (unordered). Then

$$\mathcal{Q}_S(\mathbf{A}) = \|M_S x\|_2^2 = x^\top (M_S^\top M_S) x.$$

Thus removing equal-amplitude introduces the full spectrum of the Gram matrix

$$G_S := M_S^\top M_S, \quad (\text{A9})$$

not merely its trace (which recovers I_4/C_S). **Classification result:** amplitude variation promotes scalar counts $\{I_{2m}\}$ to **matrix (spectral) invariants** $\text{Spec}(G_S)$ (and analogues at higher order), i.e. new weights are eigenvalue moments $\text{Tr}(G_S^p)$.

c. Minimal closure statement. To keep the original universality ranking, it suffices to prove: (i) equal amplitude is the unique constrained local minimizer for BCC/FCC (Hessian PD), and (ii) the BCC–FCC gap at equal amplitude exceeds any possible relaxation gain, equivalently via spectral bounds on G_S .

5. 4) Multi-shell resonant case: hypergraph weights and N&S conditions

a. Many shells. Let shells $a = 0, 1, \dots, M$ with radii k_a and stars S_a . Quartic resonances impose constraints of the form

$$k_a(\hat{u} + \hat{u}') + k_b(\hat{v} + \hat{v}') = 0 \quad (a \neq b),$$

and also higher-loop constraints chain multiple shells.

b. Hypergraph representation (new weights). Represent each pair-sum vector on shell a as a node in a layer $\mathcal{V}_{S_a}$. A mixed-shell resonance at ratio $\rho_{ab} := k_b/k_a$ is an *edge* between nodes $\mathbf{v} \in \mathcal{V}_{S_a}$ and $\mathbf{w} \in \mathcal{V}_{S_b}$ if $\mathbf{v} + \rho_{ab}\mathbf{w} = 0$. Higher-loop mixed diagrams become *cycles* in this multi-layer graph, i.e. a hypergraph constraint system.

Thus new weights are:

- overlap multiplicities $m_{S_a}(\mathbf{v})m_{S_b}(-\mathbf{v}/\rho_{ab})$ (already in two-shell),
- **cycle counts** in the multi-layer resonance graph (multi-shell),
- and their weighted versions under amplitudes.

c. Minimal necessary & sufficient closure target. A full N&S theorem must characterize when these cycle-count corrections can invert BCC–FCC ordering. Formally, inversion is impossible iff for all admissible multi-shell resonance graphs $\mathcal{G}$, the total mixed correction satisfies an inequality of the type

$$\sum_{\text{mixed edges/cycles}} \mathcal{W}_{\mathcal{G}} \leq (\text{single-shell gap}).$$

Classification result: the missing objects are *multi-layer cycle invariants* (graph counts), beyond pair overlaps.

6. 5) Higher-order vertices (6th, 8th): higher invariants and multi-resonance weights

a. General 2p-point local vertex. Allow local even vertices

$$\Gamma^{(2p)} \sim g_{2p} \int d^d x (\phi \cdot \phi)^p, \quad p = 3, 4, \dots$$

or their shell reductions. Then the structure dependence is governed by $2p$ -tuple resonance constraints

$$\hat{k}_{i_1} + \dots + \hat{k}_{i_p} = \hat{k}_{j_1} + \dots + \hat{k}_{j_p}.$$

b. Higher invariants. Define

$$I_{2p}(S) := \#\{(i_1, \dots, i_p, j_1, \dots, j_p) : \hat{k}_{i_1} + \dots + \hat{k}_{i_p} = \hat{k}_{j_1} + \dots + \hat{k}_{j_p}\}. \quad (\text{A10})$$

These are precisely the higher-order combinatorial invariants that enter once g_6, g_8 are retained. **Classification result:** going beyond LB truncation enlarges the invariant budget from $\{I_4, I_6, \dots\}$ to include higher I_{2p} at tree level (and their loop-dressed combinations).

c. Minimal closure statement. If higher vertices are irrelevant at k_* (RG), they can be dropped consistently. If not, structure selection becomes a weighted optimization involving $\{I_{2p}(S)\}$ up to the retained order.

7. Summary: minimal new weights by extension type

Extension	New structural weights (minimal)
Nonlocal/momentum-dependent $\Gamma^{(4)}$	resonance-filtered angular moments $W_{\ell m}^{(4)}(S)$ (A3)
Anisotropic quadratic kernel $Z(\hat{q})$	star angular moments $W_{\ell m}^{(2)}(S)$ (A6)
Remove equal-amplitude	spectral invariants of $G_S = M_S^T M_S$ (A9)
Multi-shell resonance	multi-layer overlap & cycle-count invariants (resonance hypergraph)
Include 6th/8th vertices	higher resonance invariants $I_{2p}(S)$ (A10)

Appendix B: Full Closure and Remaining Hidden Assumptions

1. 2.1 Universality Class Closure

The isotropic, local quartic, and equal-amplitude ansatz universality class is now fully closed mathematically. In this class, we have:

1. Local, momentum-independent quartic vertex.
2. Equal-amplitude ansatz preserving degeneracy.
3. Isotropic regulator.

In this regime, the selection of BCC/FCC/SC follows from combinatorial invariants such as C_S , $I_4(S)$, and other finite-shell resonance conditions. All computations and results in this class are ****final and conclusive****.

No further mathematical steps are needed within the isotropic single-shell local quartic equal-amplitude ansatz framework. It is self-contained and proven.

2. 2.2 Remaining Hidden Assumptions

Despite the completeness within this universality class, several ****hidden assumptions**** remain that, if broken, would allow for extensions or deviations from the universality results.

a. (A) Equal-Amplitude Star Ansatz The ****equal-amplitude assumption**** states that all modes within a star have the same amplitude, which is a simplifying assumption for the model. However:

- In the full Landau–Ginzburg free energy minimization, ****amplitude variation**** can occur.
- If amplitude variation occurs, new structure-specific weights may arise due to quartic vertex contraction and amplitude relaxation.

Thus, if this assumption breaks, the universality class would no longer be strictly valid, and new combinatorial structure terms would need to be accounted for.

b. (B) Strict Isotropy of the Quadratic Kernel The ****strict isotropy assumption**** for the quadratic kernel is pivotal to maintaining structure independence:

- If even a small ****anisotropy**** is introduced into the quadratic kernel, the bubble integral starts to acquire ****structure dependence****.
- This would introduce ****angular-dependent weights**** outside of the usual resonance counting.
- In certain anisotropic cases, such as well-known FCC-preferential regimes, FCC could be favored.

If this isotropy assumption is relaxed, ****FCC or other structures**** could emerge, and the universality class would break down.

c. *(C) Strict Locality of the Quartic Vertex* The **locality of the quartic vertex** is essential to maintaining the isotropic and momentum-independent behavior of the theory. In particular:

- The quartic vertex must be **momentum-independent**.
- If the **microscopic model** introduces angle dependence (e.g., due to interactions that break local isotropy), this would generate **new structure-specific weights**.
- Such an interaction would break the universality class and lead to possible **new structure selection mechanisms**.

Thus, any departure from local, momentum-independent quartic vertices leads to a breakdown in the universality of BCC/FCC/SC selection.

d. *(D) Single-Shell Dominance* The **single-shell assumption** simplifies the model by assuming that only one shell participates in structure formation:

- In real physical systems, there could be **multi-shell coupling**.
- **Higher-harmonic generation** and **anisotropic dispersion** effects may also be present.
- These effects could introduce **new combinatorial weights** beyond the simple pair-sum counting used in the current model.

Relaxation of this assumption could lead to a more complex resonance structure, requiring new weights to be considered in the universality class.

e. *(E) No Cubic Term (Z_2 Symmetry)* The model assumes **Z_2 symmetry**, which forbids cubic terms in the potential:

- If **cubic terms** are introduced, a well-known mechanism favors FCC over BCC.
- **Cubic terms** would **strongly enhance FCC** and disrupt the BCC/FCC/SC ordering.

Thus, if cubic terms are allowed, the universality of the BCC selection is compromised.

f. *(F) No Lattice Pinning / No External Field* The current model assumes there is no external field or lattice potential pinning:

- In the presence of external potentials (e.g., substrate effects, confinement), structures outside of BCC/FCC/SC may emerge.
- **Combinatorial universality** would be broken, and new structural configurations could dominate the phase selection.

Thus, the **absence of lattice pinning** is essential for maintaining the universality of BCC/FCC/SC selection.

3. 2.3 Summary of Remaining Assumptions and Potential Breakdowns

The model presented here is **strongly closed** under the assumptions listed in Section 2.2. However, should any of these assumptions break, **new weight types** and **new structures** could emerge. Specifically:

- **Equal-amplitude ansatz** allows simplification but could break down under amplitude variation. - **Isotropy and locality** assumptions are central; relaxing them could favor different structures (e.g., FCC). - **Single-shell approximation** may be invalidated by multi-shell effects and higher harmonics. - The **absence of cubic terms** and **no external field** are assumptions that, when violated, introduce new selection mechanisms.

Breaking these assumptions introduces new structural weights, and **Universality C–E** would need to be revisited in these extended scenarios.

[Generalized universality after assumption relaxations] When any of the hidden assumptions (A)–(F) are relaxed, the universality class can break, and new combinatorial weights (e.g., angular moments, harmonic weights, higher-order invariants) must be introduced to describe the structure selection mechanism.